ALAMO/ALTO X-Message Format Specification

XMTPS

Generated from `spec/xmessage-format.md` by `spec/build-formats.py`

This document specifies the binary X-message format that MRV ALAMO and ALTO profiling floats use to send their data over Iridium Short Burst Data (SBD). It also says how XMTPS decodes each message and which listing-file lines each message becomes.

Status: compiled on 2026-10-25. It brings together everything known about the format:

- the SOLO-II X-format specification it derives from;
- the ALAMO X-format spreadsheet (v2.4);
- MRV's firmware message definitions (7.3.5, 10.5.0, 10.12.0);
- the XMTPS decoder source;
- hand-decoded production packets.

No single earlier document was complete or current, and they disagree in places. Where they disagree, this document says so, and gives the layout that the decoder and real packets agree on.

Decoder fixes in build 3309 (2026-10-25). Writing this document turned up several decoder bugs, which build 3309 fixes. The parts below describe these bugs as found and mark each one fixed:

- Argo `nSeek` is read as two bytes.
- `APS` SNROut shows the output SNR.
- Mission `DRT` and `PPC0`/`PPC1` list their own values.
- 0xEA versions 0x13/0x14 read raw samples before empty bins.
- The 0xEA version 0x10 error heading formats.
- Mission and BIT blobs too short for their layout are rejected.

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How to read this document

Structure. A float's data for one dive cycle is split into *packets*. Each packet travels as one Iridium SBD message, a binary e-mail attachment of at most 340 bytes. A packet has the form `X nn mm dd p <data> $cc`, and its `<data>` is a sequence of *blobs* (the SOLO-II documents call them "sensor messages"). Each blob has the form `ID jj <data> ;`. The 1-byte ID says what the blob holds; `jj` is the blob length. Part A.1 gives the details.

Offsets in every table count from the blob's ID byte, which is offset 0. All multi-byte fields are big-endian, including the IEEE-754 floats.

Times are seconds since 2000-01-01 00:00 UTC unless a table says UNIX time (seconds since 1970-01-01).

Sensor values sent as counts are converted with the gains and offsets in the same dive's Argo message 0xF0: `value = counts / gain − offset` (A.1.5).

Source tags. The parts mark where each statement comes from:

Tag (part A)	Tag (parts B and C)	Source
[D]	[decoder]	The XMTPS decoder source (`src/data/xmsg/`, `src/data/XFloat.cpp`, `src/data/XFloatMailMessage.*`) as of 2026-10-25. It is the reference for what is implemented.
[O]	Production	Production packets, decoded by hand from the hex dumps (`SHX`/`shx` lines) in XMTPS listings.
[S]	SOLO2	*Format of SOLOII X messages*, Argo version manual/decoder v2.0, 31 Mar 2014 (`SOLO2_Xformat_v2.0_20Mar2014.pdf`).
[X]	XF2.4	ALAMO X-format specification v2.4 spreadsheet (`xfmt-format-spec-2.4-alamo_2018-05-15.xlsx`).
[M]	NG10.5, NG10.12, DF7.3.5	MRV firmware definitions: `nextgen_v10.5.0_blobs.txt`, `nextgen-10.12.0-xmessage.txt`, `DataFormat-v7.3.5.yaml`, plus `xfs.txt` (on-float storage) and `system-error-codes.txt`.
[I]	[inferred]	Not documented anywhere. Worked out from production data or decoder behavior.

The MRV documents and the SOLO-II PDF are in the Float Group's `mrv/doc` folder. The ALAMO spreadsheets, v2.2 to v2.4 (2015–2018), are in this repository's `doc/` folder, which git doesn't track.

Firmware versions are written as 32-bit words `0xMMmmppbb`: major, minor, patch, then build/product byte. For example, `0x0A0C0001` is firmware 10.12.0.1 and `0x0A0C0032` is 10.12.0.32. In the last byte, 0x31 marks WIMS (wave) floats and 0x32/0x33 mark MM2/Micro-ALTO floats (C.1.4).

Message type index

ID	Message	Section	Decoded by XMTPS	Listing lines
0x00	GPS fix after the diagnostic dive	A.2	yes	`GPS`
0x01	GPS fix at the start of a dive	A.2	yes	`GPS`
0x02	GPS fix at the end of a dive	A.2	yes	`GPS`
0x03	GPS fix after a mission abort	A.2	yes	`GPS`
0x05	GPS fix during the built-in test	A.2	yes	`GPS`
0x10–0x1F	Binned profile pressure	A.3	yes	`PRO/pro`, `PRD/prd`
0x20–0x2F	Binned profile temperature (format 3: auxiliary temperature channel)	A.3, A.4.3	yes	`PRO/pro`, `PRD/prd`
0x30–0x3F	Binned profile salinity	A.3	yes	`PRO/pro`, `PRD/prd`
0x40–0x4F	Fall (descent) pressure time series	A.5.1	yes	`F/f`
0x50–0x5F	Rise (ascent) pressure time series	A.5.1	yes	`R/r`
0x60–0x6F	Pump operations	A.5.3	yes	`PMP/pmp`
0x78–0x7F	Drift sample times	A.4.1	yes	`D/d`
0x80–0x8F	Surface time series	A.5.1	yes (formats 0–2)	`SFC/sfc`
0x90–0x97	High-resolution profile pressure	A.3	yes	`PRU/pru`
0x98–0x9F	Drift pressure	A.3	yes	`D/d`
0xA0–0xA7	High-resolution profile temperature	A.3	yes	`PRU/pru`

ID	Message	Section	Decoded by XMTPS	Listing lines
0xA8–0xAF	Drift temperature	A.3	yes	`D/d`
0xB0–0xB7	High-resolution profile salinity	A.3	yes	`PRU/pru`
0xB8–0xBF	Drift salinity	A.3	yes	`D/d`
0xC1	Ocean surface waves: auto-spectra	C.3	yes (spectral v0–2)	`WVS`, `wvs`, `wvm`
0xC2	Auxiliary sensor data	A.4.4	no	—
0xC3	Ocean surface waves: cross-spectra	C.3	yes (spectral v0–2)	`WVS`, `wvs`, `wvm`
0xC5	WIMS configuration parameters	C.7	yes (v1, v2; inferred)	`WIP`
0xCA	Marginal-ice-zone acoustics (0xC0 before build 3249)	C.4	subtypes 3 and 5	`APS`
0xD0–0xD1	Mission configuration (binary, two parts)	C.1	yes (117 firmware versions)	`MIS`, `m`, `mis`
0xD2–0xD7	Mission configuration, further parts / SOLO-II ASCII dump	C.1	no	—
0xDC	Dynamically corrected profile salinity	A.4.2	yes	`salnCorr` column of `pro`
0xDE	Mission command echo	C.2	yes	`CMD`
0xE0	Engineering, diagnostic dive (SOLO-II)	B.1.1	validated only	—
0xE2	Engineering, normal dive (SOLO-II)	B.1.2	validated only	—
0xE3	Engineering, mission abort (SOLO-II)	B.1.3	validated only	—
0xE5	Built-in test passed	B.2	formats 6–11	`E05`
0xE6	Built-in test failed	B.2	formats 6–11	`E06`
0xE7	Engineering, descent	B.3	versions 0x10–0x11	`E07`
0xE8	Engineering, drift/park	B.4	versions 0x10–0x12	`E08`
0xE9	Engineering, ascent	B.5	versions 0x10–0x12	`E09`
0xEA	Engineering, surface	B.6	versions 0x10–0x14	`E0A`, `e0A`
0xEB	Valve operations	A.5.4	yes	`VLV/vlv`
0xEC	Engineering, ice detection	B.7	versions 0x10–0x11	`E0C`, `e0c`
0xF0	Argo data (mission summary, gains and offsets)	A.6	v2.1–v2.4	`A`
0xF1	Test pattern	—	no	—
0xFF	Extended records (event, reset, serial number, CTD exchange, fault dump)	C.6	since build 3321 (format inferred)	`XFF`

IDs not listed are unassigned. In types 0x1n–0x8n and 0x9n–0xBn, the low nibble of the ID is the part number of a multi-part series (A.3).

Part A — Transport, framing and data messages

Scope of this part: packet framing and general conventions; GPS (0x00–0x05); binned, high-resolution and drift P/T/S (0x1n/0x2n/0x3n, 0x90–0xBF); drift sample times (0x78–0x7F); dynamically corrected salinity (0xDC); auxiliary sensor data (0xC2) and the auxiliary temperature channel; fall (0x4n), rise (0x5n), pump (0x6n), surface time series (0x8n) and valve (0xEB); Argo data (0xF0).

A.0 Sources and conventions of this document

Tag	Source
[D]	XMTPS decoder source (`src/data/xmsg/*`, `src/data/XFloat.cpp`, `src/data/XFloatMailMessage.*`). Ground truth for what is implemented.
[O]	Observed in production packets (hex dumps in `lis9287-0325.txt` and `lis11186-0107.txt`, firmware 10.12.x, decoded by hand for this document).
[S]	SOLO-II X format v2.0 (`SOLO2_Xformat_v2.0_20Mar2014.pdf`).
[X]	ALAMO X-format spec v2.4 spreadsheet (`xfmt-format-spec-2.4-alamo_2018-05-15.xlsx`).
[M]	MRV documents: `nextgen_v10.5.0_blobs.txt`, `DataFormat-v7.3.5.yaml`, `xfs.txt`.
[I]	Inferred (not documented anywhere; reconstructed from data or code).

- Offsets are byte offsets counted from the blob's ID byte = 0, unless the table says otherwise.
- `u8/u16/u24/u32` are unsigned and `i8/i16/i24/i32` are two's-complement signed big-endian integers. `u4` / `u20` denote nibble-aligned fields that share a byte with a neighbor (see the Notes column).
- "counts" are raw integers converted with a gain and offset from the 0xF0 Argo message (section A.1.5).
- Listing tags (`GPS`, `PRO/pro`, `F/f`, …) refer to the XMTPS listing file (README, "Listing files").

A.1 Packet framing and general conventions

A.1.1 Transport: Iridium SBD and the mail gateway

- The float stores telemetry in an on-board "XMessage file system" (XFS) of 1500 blocks [M `xfs.txt`]. **One XFS block = one Iridium SBD mobile-originated message = one X packet.**
- An SBD message carries at most **340 bytes**. The X packet header (8 bytes: `X nn mm dd p`) and trailer (4 bytes: `$cc>`) take 12, so a block carries at most **328 bytes of blobs** [M]. (SOLO-II documentation still refers to the 189-byte 9601 limit [S]; it doesn't apply to ALAMO/ALTO.)
- A block holds one or more complete blobs, end to end with no gaps. All blobs in a block belong to the same dive. **A blob never spans two packets.** Multi-part data (profiles, time series) is split into separate blobs that carry their own part index [M][D].
- The Iridium gateway forwards each SBD message as an e-mail [D `XFloatMailMessage.h`]:
  - Sender matches `(sbdservice@sbd.iridium.com|service@sbd.pac.disa.mil)`. This is configurable.
  - Subject matches `SBD Msg From Unit: (\d+)`, which gives the IMEI.
  - The binary attachment (MIME type `application/*`) is named `<IMEI>_<MOMSN, 6 digits>.sbd` (regex `(\d+)_(\d{6})\.sbd`). Its bytes are the X packet.
  - The body text gives `MOMSN:`, `MTMSN:`, `Time of Session (UTC):`, `Session Status: NN - text`, `Message Size (bytes):`, `Unit Location: Lat = … Long = …` and `CEPradius = N` (km).
- The decoder accepts the attachment only if it has at least **8 bytes** (`XMSG_SBD_MIN_BYTES`) and the session status is **00 or 02** (transfer OK; 02 = OK but the location is of unacceptable quality) [D].
- Duplicates [D]:
  - A mail with an already-seen Message-ID is ignored.
  - A mail whose SBD bytes are identical to an already-assigned packet of the dive is a retransmission under a new MOMSN. It is kept for its Iridium position, but its blobs are not decoded again.

  - Dive data is keyed by (serial number, dive number) taken from the packet header, not from the e-mail.

- Transmission order is not packet-ID order. In the observed dives the highest `p` was sent first and `p = 0` last [O].

A.1.2 X packet: `X nn mm dd p <data> $ c c >`

Offset	Size	Type	Field	Scale/units	Notes
0	1	char	`X` (0x58)	—	Start marker. Not checked by the decoder [D].
1	2	u16	`nn`	bytes	Number of bytes after `nn` up to, but not including, `$` = 5 + len(<data>) [S]. Verified: `nn` = packet length − 7 on 27 packets [O]. Ignored by the decoder, which uses the attachment length [D].
3	2	u16	`mm`	—	Float serial number (WHOI/instrument ID) [S][D].
5	2	i16	`dd`	—	Dive (cycle) number: −1 = start-up, 0 = test/diagnostic dive, then +1 per dive [S]. Signed [D].
7	1	u8	`p`	—	Packet ID within the dive. The data of each dive starts at `p = 0` [S]. Listed as `PID` on `SBD` lines.
8	len−12	—	<data>	—	Zero or more blobs (A.1.3).
len−4	1	char	`$` (0x24)	—	Checksum delimiter.
len−3	2	2×char	`cc`	—	8-bit checksum (see below).
len−1	1	char	`>` (0x3E)	—	End marker.

Checksum. `sum = (Σ bytes from 'X' (offset 0) through the byte before '$') mod 256`. It is written as two characters, high nibble first, each nibble + 0x30. So the characters are in the range `0–?` (`0x3A–0x3F` = `:;<=>?` stand for nibble values 10–15) [S]. All 1,351,052 X packets in the production RAW files (all floats and firmware, 2014–2026) have a correct checksum [O].

> **Decoder note [D]:** since build 3310 the decoder verifies the checksum (`XFloatDiveData::DecodeXMsgData`). A packet whose checksum does not match is discarded and logged ("X-Message checksum mismatch"); a packet without the `$cc>` trailer is decoded with a warning. The blob parser reads offsets 8 … len−5.

A.1.3 Blob structure: `ID jj <data> ;`

Offset	Size	Type	Field	Notes
0	1	u8	ID	High nibble = message class; low nibble = part index or sub-type (e.g. 0x1n = binned pressure, part n). For classes with a single type (0xD0, 0xDE, 0xE0–0xEC, 0xF0, 0xF1) the whole byte is the type.
1–2	2	—	`jj`	Blob length in bytes, **including** ID, `jj` and the trailing `;`. Two layouts, see below.
3	jj−4	—	data	Type-specific.
jj−1	1	char	`;` (0x3B)	Terminator.

`jj` layouts:

- **Encoding nibble + 12-bit size**, used by the sensor/time-series families: 0x1n–0x3n, 0x4n, 0x5n, 0x6n, 0x78–0x7F, 0x8n, 0x9n, 0xAn, 0xBn and 0xDC. Bits 15–12 of `jj` (high nibble of byte 1) hold the packing/encoding format; bits 11–0 hold the size (0–4095) [S][X].
- **Full 16-bit size**, used by GPS 0x00–0x05, 0xC1, 0xD0–0xD7, 0xDE, 0xE5–0xEC (including the valve message 0xEB) and 0xF0 [X]. Here the encoding format, if any, is in byte 3 (e.g. 0xEB).
- Exception: SOLO-legacy pump format 0 (0x6n with format nibble 0) is read with the full 16-bit size by `XPump` [D], which is equivalent because the high nibble is then 0.

How the decoder splits a packet into blobs [D] (`XFloatDiveData::DecodeXMsgData`):

1. Bytes 8 … len−5 are appended one at a time to a buffer.

2. At 3 buffered bytes the size is taken as `0x0FFF & u16(buf, 1)`. **The 12-bit mask is applied to every type.** This is harmless because a blob is always smaller than 4096 bytes (SBD ≤ 340), so the high nibble of a 16-bit size is always 0.
3. When the buffer reaches the size, the blob is dispatched on its ID byte and the buffer is cleared. A trailing incomplete blob is silently dropped. A wrong size desynchronizes the rest of the packet.
4. Each class decoder checks the ID, a minimum length, and that `buf[size−1] == ';'` (most of them via `XMessage::CheckXMsgData`; the size must be at least 4). Extra bytes produce a warning; a missing terminator is an error.
5. IDs without a decoder are logged as `Unknown X message ID: 0x..` and skipped, for example 0xC2 and 0x70–0x77 (and 0xFF before build 3321, C.6). 0xF1 (test pattern) is ignored silently.

A.1.4 Byte order, bit order, epochs

- **Big-endian** everywhere: multi-byte integers are sent MSB first [S][D].
- Bit fields are numbered from the MSB of the first byte. In a byte that holds two nibbles, the **high nibble comes first**. For example, a "code + 20-bit pressure" field is `code = b[k] >> 4`, `pres = ((b[k] & 0x0F) << 16) | b[k+1] << 8 | b[k+2]`.
- **Time epochs:**
  - **2000-01-01T00:00:00Z** (`s2000`): u32 seconds since this epoch for the 0xF0 state start times and the fall/rise/surface start/reference times. **0 means "not set"**: the decoder maps it to not-a-date-time, listed as `0000-00-00T00:00:00Z` [D]. The spreadsheet's "offset 36526" on these fields is the Excel day number of 2000-01-01 [X].
  - **GPS epoch 1980-01-06**: GPS week and day in the GPS message (A.2).
  - **UNIX epoch 1970-01-01** (`s1970`): used only by the BIT v2 engineering message (`XAlamoBit2.h`, outside this part) [D].
  - Relative times: u16 seconds after a reference (fall/rise/surface), and u24/i24 seconds after the drift start (0x78).
- Leap seconds are not applied anywhere [D].

A.1.5 Counts-to-units conversion

All CTD quantities (pressure, temperature, salinity) are sent as integer counts:

```
value = counts / gain − offset
```

`gain` and `offset` are unsigned 16-bit integers from the 0xF0 Argo message of the **same dive** [D][X][M]. The SOLO-II document gives `counts · gain − offset` in one place [S]; that is a typo, and both its own 0xF0 table and the decoder use division.

Data	Gains/offsets used [D]
Binned/hi-res/drift profiles, 0xDC, aux temperature, fall, rise, pump, valve	pressure: `pGain/pOff` (0xF0 bytes 40–43); temperature: `tGain/tOff` (44–47); salinity: `sGain/sOff` (48–51).
Surface time series 0x8n	surface gains/offsets (0xF0 bytes 52–63). In v2.1 these are copies of the profile gains.

Typical values [O]:

Quantity	Gain	Offset	Resolution	Note
P	25	10	0.04 dBar	0–65535 counts covers −10 … 2611.4 dBar
T	1000	5	0.001 °C	
S	1000	1	0.001 PSU	
Surface P	25 or 100	10		

- **No 0xF0 in the dive** [D]: all gains and offsets are set to 9999 (`IRI_GAIN_INVALID` / `IRI_OFFS_INVALID`), so every converted value falls out of range and is listed as the fill value. The float-level fallback is commented out. A dive's data can only be converted once its own 0xF0 has arrived.
- **gain = 0** gives the fill value (the per-message `SampleValue` returns −100 internally).
- **Signedness [D]:** composite accessors pass gain/offset as `int16_t`, so a gain above 32767 would be misread. No observed float uses one.
- **Valid ranges [D]:** values outside these limits are replaced by the fill value (`XDefinitions.h`):

Quantity	Valid range
P	−5 … 2610 dBar
T	−5 … 40 °C
S	2 … 41 PSU

- **Pressure offset correction [D]:** for RBR floats (or WMO instrument types 866/867/875) XMTPS can subtract a surface-pressure offset `pc`, derived from the surface time series of recent dives, from profile, drift, fall, rise, pump and valve pressures. Surface (`sfc`) pressures are always listed uncorrected. See README "Surface pressure offset correction".

A.1.6 Invalid and fill values

Context	Encoding on the wire	Decoder representation / listing
Fall/rise/pump depth (SOLO)	0xFFFF = invalid pressure [S]	No special case; converted, then range-checked [D].
GPS fix	byte 3 = 0	lat 99.99999, lon 999.99999, time unset [D].
s2000 time	0	not-a-date-time → `0000-00-00T00:00:00Z`.
CTD values in listings	—	`-9.999` (pressure printed with 2 decimals) for out-of-range or missing values.
Pump/valve vacuum not sent	—	65535 cinHg → `655.35` in `vlv`/`pmp` vac0 columns [D].
Integer fields never received	—	255 / 65535 / INT_MIN (`XMSG_BAD_*`); dive −999, IDs −1.

A.2 GPS position — 0x00–0x05

Purpose. A GPS fix with its quality metrics. The low nibble of the ID is the mission phase in which the fix was taken. There is one layout for all phases.

Low nibble (ID)	Meaning [S][D]
0x00	Diagnostic dive: fix after the first surfacing following the diagnostic (test) dive.
0x01	Start of dive: just before leaving the surface.
0x02	End of dive: at the surface after the ascent/profile. The best position/time for this dive's profile.
0x03	After a mission abort.
0x05	During the built-in self-test (BIT).
0x04, 0x06–0x0F	Unused. The decoder rejects IDs > 0x05.

> **Discrepancy [X]:** the spreadsheet's overview sheet labels 0x01 "end-of-dive" and 0x02 "start-of-dive". Its GPS sheet, the SOLO-II document and the decoder all use 0x01 = start, 0x02 = end.

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID	—	0x00–0x05.
1	2	u16	size	bytes	24 (0x0018). Full 16-bit size.
3	1	i8	fix status / hemisphere	—	0 = no/invalid fix; +2 = longitude East; −2 (0xFE) = West [S][X]. The decoder only tests for 0; the sign is also carried by the i32 longitude.
4	4	i32	latitude	1e-7 deg	Positive = North.
8	4	i32	longitude	1e-7 deg	Signed, −180 … +180, positive = East.

Offset	Size	Type	Field	Scale/units	Notes
12	2	u16	GPS week	weeks	Full week number since 1980-01-06: “traditional” 10-bit week in the low 10 bits, rollover count in the high 6 bits [S], so the u16 is the continuous week number.
14	1	u8	GPS day of week	—	0 = Sunday … 6 = Saturday.
15	1	u8	hour	UTC	
16	1	u8	minute	UTC	Seconds are not sent; the decoder uses 0.
17	1	u8	time to fix	10 s	Decoded as `u8 × 10` seconds (0–2550 s).
18	1	u8	satellites used	—	The decoder keeps only the low nibble (`& 0x0F`).
19	1	u8	SNR min	—	
20	1	u8	SNR mean	—	
21	1	u8	SNR max	—	Observed as 0/0/0 on 10.12 firmware [O].
22	1	u8	HDOP	0.1	`HDOP = u8 / 10`.
23	1	char	;	—	

Example [O] (float 11186, dive 107, 0x01): `01 00 18 02 0C D2 E2 01 46 9A 69 41 09 89 05 16 16 59 05 00 00 00 0F 3B` decodes to

Field	Value
Hemisphere	East
Latitude	21.51470
Longitude	118.45246
Week / day	2441 / 5
Time	2026-10-23T22:22:00Z
Time to fix	890 s
Satellites	5
HDOP	1.5

Decoding notes [D]:

- Time = 1980-01-06T00:00 + (week·7 + day) days + hour:minute (UTC).
- A fix is set invalid when:
  - byte 3 = 0;
  - the coordinates are outside ±90/±180;
  - the time is before 2000-01-01 or in the future;
  - (when the fix is assigned) the time is later than the dive’s last Iridium session time.

  An invalid fix gets lat 99.99999, lon 999.99999 and no time.
- A valid 0x05 fix advances the float’s start-of-mission time. Packets sent before the start-of-mission or deployment time are ignored.
- **Profile position selection:**
  - A 0x02 fix of the same dive is the profile position.
  - A 0x01 fix labeled with dive N+1 is valid for dive N.
  - A 0x00 fix is used only for dive 0.
  - 0x03 and 0x05 are never used as profile positions.
- **Listing:** one `GPS` line per fix: `GPS SN dive idx date/time position TTF nSat SNRmin SNRavg SNRmax HDOP stat`. `idx` is the ID (phase) and `stat` is always 0.

A.3 Sensor series: binned profile (0x1n/0x2n/0x3n), high-resolution profile (0x90–0xB7), drift (0x98–0xBF)

Purpose. Series of integer counts from one channel, split into numbered parts.

Series	Pressure	Temperature	Salinity	Part index	Max parts [D]	Listing
Binned (bin-averaged) profile	0x10–0x1F	0x20–0x2F	0x30–0x3F	low nibble 0–F	16	`PRO/pro`
High-resolution (unbinned, ≈1 Hz) profile	0x90–0x97	0xA0–0xA7	0xB0–0xB7	low nibble 0–7	8	`PRU/pru`
Drift/park time series	0x98–0x9F	0xA8–0xAF	0xB8–0xBF	low nibble 8–F (first part = 8)	8	`D/d`

All of these use the same two encodings, selected by the high nibble of byte 1:

Format nibble	Encoding	Status
0	First-difference (“legacy”)	SOLO-II heritage; ALAMO mission parameter XAS = 1.
1	Curvature (second-difference)	Standard; XAS = 2. All observed ALAMO/ALTO data.
3	Curvature, **auxiliary temperature channel**	Only meaningful on 0x2n; see A.4.3.
2, 4–15	—	The decoder throws “Unsupported packing format”.

The same curvature body is also used by drift times (0x78, A.4.1), by corrected salinity (0xDC, after a 4-byte prefix, A.4.2) and, per the MRV YAML, by 0xC2.

A.3.1 Curvature encoding (format nibble 1 or 3)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID	—	Class nibble + part index.
1	2	u4+u12	format / size	— / bytes	High nibble = 1 (or 3); low 12 bits = blob size.
3	1	u8	B	sub-blocks	Starting sub-block number. See the notes; **ignored by the decoder**.
4	2	u16	NN	values	Number of values in this part, **counting VVV** (see reassembly).
6	3	i24	VVV	counts	First value of the part.
9	3	i24	DDD	counts	First difference = v[1] − v[0].
12	12	32 × u3	p0 … p31	—	Packing factors, one per 16-value sub-block, MSB-first: `p[8i+j] = (u24(12+3i) >> (21–3j)) & 7`, i = 0..3, j = 0..7.
24	⌈Σnibbles/2⌉	nibble stream	second differences	counts	Sub-block k holds min(16, remaining) values, each `p_k` nibbles wide.
size−1	1	char	;	—	

Packing-factor-to-width mapping: the truth is *width = p nibbles = 4·p bits*.

p	Bits per value	Range (signed)	Bytes per full 16-value sub-block
0	—	sub-block **absent**: the decoder skips it and consumes no values or nibbles	0
1	4	−8 … 7	8
2	8	−128 … 127	16
3	12	−2048 … 2047	24
4	16	−32768 … 32767	32
5	20	±2^19	40
6	24	±2^23	48
7	28	±2^27	56

Evidence: this is `AbstractXSensorData::UnpackSensorData` [D]. The documents disagree with it:

- **[S]** says “packing factor 0 = 1 nibble”, i.e. width = p+1 nibbles.
- **[X]** says width = `pN·4+4` bits (p+1 nibbles) in one sentence, then lists “p=0 → 0 nibble null, p=1 → 1 nibble, …”. Its byte counts per sub-block (16, 24, …, 64 bytes for p = 1 … 7) are 8 bytes too large.

All 24 curvature blobs in the two production dives were checked [O]. Under the decoder rule (p nibbles, p = 0 skipped) the nibble count equals the data length exactly, with at most one pad nibble. Under the p+1 rule

every blob would need 30–80 % more bytes than it has. Even all-zero second differences (constant 1800 s steps in 0x78) are sent with p = 1, never p = 0, so p = 0 is used only for unused trailing sub-blocks.

Unpacking, exactly as the decoder does it [D]:

1. Read B (ignored), NN, VVV (i24) and DDD (i24).
2. If NN = 0, throw. If NN = 1, the part's values are just [VVV]. If NN = 2, they are [VVV, VVV + DDD]. VVV is included only in the first part; see reassembly. In both cases return.
3. Otherwise, require size ≥ 24 and read the 32 packing factors.
4. Set `remaining = NN − 2` and k = 0, and set the nibble cursor to the high nibble of byte 24.
5. While `remaining > 0` and the cursor is before the terminator:
   - If k ≥ 32, error.
   - If p_k = 0, set k += 1 and continue (no data consumed).
   - Otherwise read n = min(16, remaining) values. For each value read p_k nibbles, most significant first, as an unsigned integer u, then sign-extend: `if u ≥ 2^(4p−1) then u −= 2^(4p)`. Append it to the second differences and decrement `remaining`.
   - Set k += 1.

   The nibble cursor is continuous across sub-blocks: sub-blocks are not byte-aligned. Because a full sub-block has 16·p nibbles (always even), only the last sub-block can end mid-byte. The low nibble of the last data byte is then padding.
6. First differences: `d[0] = DDD`, `d[i] = d[i−1] + s[i−1]`, i = 1 … NN−2 (cumulative sum of the second differences, seeded with DDD).
7. Values: `x = VVV` (not appended unless this is the first part), then for each first difference `x += d[i]` and append x. That is NN − 1 values, plus VVV for the first part.
8. If the stream ends before `remaining` reaches 0, the decoder logs a warning and keeps the partial part (it does not throw).

Multi-part reassembly [D] (`AbstractXSensorDataComposite`):

- **The first part includes VVV; later parts don't.** The "first part" is the one whose ID is 0x10, 0x20, 0x30, 0x90, 0xA0, 0xB0, 0x98, 0xA8, 0xB8 or 0x78. Part m > 0 repeats the last value of part m−1 as its VVV (an overlap of one value), and the decoder drops it, so part m contributes NN − 1 values. Total samples = $NN_0 + \Sigma_{m>0} (NN_m - 1)$.

  Verified [O], float 11186 dive 107, temperature 0x20/0x21/0x22 with NN = 306/290/192: 306 + 289 + 191 = 786 samples, the same count as pressure and the aux-temperature channel.
- Parts are kept per channel, sorted by part index (the low nibble of the ID), and concatenated in index order. A duplicate index is rejected. The decoder doesn't check that the overlap VVV equals the previous part's last value.
- **B** in the observed data is the number of 16-value sub-blocks in all preceding parts of the same channel: 0 for the first part, then 15, 19, 32, 37 … [O]. The SOLO-II document says B = 1 in the first part [S]. Not used by the decoder.
- At most 32 sub-blocks per part, so NN ≤ 514 per blob. Observed maximum: 514 [O].
- **Completeness** (`IsDataComplete`), required before any `pro`/`pru`/`d` rows are listed:
  - at least one sample;
  - part indices contiguous **from the lowest received index** (a missing first part isn't detected by this test alone);
  - the same number of P and T samples;
  - aux-temperature samples (if any) equal T;
  - S samples are 0 or equal P.

  Otherwise the listing prints `# (Incomplete … Data)`.

- **Missing bins:** a bin with no valid CTD data is simply not sent, so all channels skip the same bin and stay aligned [S]. The decoder has no bin numbers; `sampleIdx` is the position in the series.

A.3.2 First-difference ("legacy") encoding (format nibble 0)

Offset	Size	Type	Field	Notes
0	1	u8	ID	
1	2	u4+u12	format = 0 / size	
3	…	—	sub-block 0, 1, …	Each sub-block is `[S][v0][δ1 … δ(n−1)]`, n ≤ 25.
size−1	1	char	;	

Sub-block layout:

Sub-block byte	Type	Content
0	u8	scale S (1–255), chosen so that each scaled difference fits in an i8
1–2	u16	v0 (unsigned, MSB first)
3 … n+1	i8	δk = (v[k] − v[k−1]) / S

Decoder [D]:

- `v[k] = v[k−1] + δk·S`. Sub-blocks hold up to 25 values (1 + 24 differences, 27 bytes); only the last sub-block may be shorter.
- Sub-blocks are read back to back until the terminator. Parts concatenate by index **without any overlap**.
- Quirk: the loop runs while `start < end − 5`, so a final sub-block with only 1 or 2 values is **silently dropped**.

Discrepancies [S]:

- The SOLO-II document gives "22 bytes" for a full 25-value sub-block, which should be 27.
- It writes the last difference as `{v[n−1] − v[n]}/S`. That is a typo for `v[n−1] − v[n−2]`.

A.3.3 Listing

Series	Listing [D]
Binned	`PRD`/`prd` describes the columns, `PRO SN dive numSamples`, then `pro SN dive sampleIdx pres temp saln`. With aux temperature: `pres temp1 temp2 saln`. With complete 0xDC data: `salnCorr` is appended.
Unbinned (hi-res)	`PRU SN dive numSamples`, then `pru SN dive sampleIdx pres temp saln`.
Drift	Without sample times: `D SN dive numSamples`, then `d SN dive sampleIdx pres temp saln`. With complete 0x78 times: `D SN dive nSam driftStart` (the 0xF0 park start time), then `d SN dive i +sec pres temp saln`.

A.4 Drift sample times (0x78–0x7F), corrected salinity (0xDC), auxiliary sensor data (0xC2), auxiliary temperature channel

A.4.1 Drift sample times — 0x78–0x7F *(format inferred [I]; decoder [D] `XDriftTime.h`)*

Purpose. The time of each drift sample (0x98/0xA8/0xB8 series), in seconds after the Argo `park start time` (0xF0 bytes 20–23). Not in any MRV or spreadsheet document. The format was inferred from production packets (2026-10-24).

- **Layout:** identical to the curvature encoding of A.3.1. ID = 0x78 + part index (0x78 = first part, which includes VVV); format nibble 1.
- **Values:** u24 seconds, no gain or offset.

- **Observed [O]:**
  - VVV = 1 (the first drift sample is taken 1 s after the drift starts).
  - DDD = sample period (1800, 7200 or 14400 s; equal to the mission parameter DDT on float 11186).
  - All second differences 0 (p = 1).
  - NN equals the drift P/T/S sample count.
- **Multi-part:** 0x79 has been observed in logs, so multi-part series do occur. Reassembly is the same as A.3.1.
- **Decoder [D]:**
  - Accepts `(ID & 0xF8) == 0x78`.
  - Parts are kept with the drift composite.
  - `IsTimeDataComplete` requires contiguous indices and as many times as drift samples. Then the `D/d` lines switch to the time layout (A.3.3). Otherwise `# (Incomplete Drift Time Data)` is printed and the old layout is used.
  - `xmtps-list2json.py` uses these times instead of estimating drift times.

A.4.2 Dynamically corrected salinity — 0xDC *(format inferred [I]; decoder [D] `XProfileSalinityCorrected.h`)*

Purpose. A second binned-profile salinity channel, corrected on the float (RBR dynamic correction). It is sent only when mission parameter PPE (dynamic correction mode) = 2 [O]. Values use the salinity gain and offset, and differ from the uncorrected salinity by a few thousandths of a PSU (at most about 0.07). Not in any MRV or spreadsheet document.

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID	—	Always 0xDC. The part index is in byte 4, not in the ID.
1	2	u4+u12	format / size	— / bytes	Format 1 (curvature) observed.
3	1	u8	(unknown)	—	Always 0x00 [O]. The decoder warns if it is non-zero.
4	1	u8	part index	—	0 … 15. The decoder rejects values > 15.
5	1	u8	channel	—	Always 0x01 (salinity) [O]. The decoder rejects other values.
6	1	u8	(unknown)	—	Always 0x00 [O]. The decoder warns if it is non-zero.
7	1	u8	B	sub-blocks	As A.3.1 byte 3.
8	2	u16	NN	values	
10	3	i24	VVV	counts	
13	3	i24	DDD	counts	
16	12	32 × u3	packing factors	—	
28	…	nibbles	second differences	counts	
size−1	1	char	;	—	

Decoder [D]:

- Validates size (8 ≤ size ≤ received bytes), byte 5 = 1 and byte 4 ≤ 15.
- Then rewrites the blob as a standard salinity message: ID `0x30 | byte4`, `jj` = original format nibble + (size − 4), followed by bytes 7 … size−1. The result is decoded as A.3.1, so part 0 includes VVV.
- The total count equals the 0x3n salinity count. The parts can split differently from 0x3n [O].
- The `salnCorr` column is added to `pro` only when the corrected channel is complete (contiguous indices and the same count as salinity). Otherwise `# (Incomplete Corrected Salinity Data)` is printed.
- In the JSON output it is `saln_corr` (NaN when absent).

A.4.3 Auxiliary temperature channel (format nibble 3 on 0x2n)

Purpose. A second temperature series, aligned with the binned profile. Presumably the RBR conductivity-cell thermistor, enabled by mission parameter CTC "conductivity temperature channel mode" [M] [I]. It is sent as ordinary 0x20, 0x21, … blobs whose `jj` high nibble is **3** instead of 1, alongside the primary temperature parts, which carry the same IDs with format 1 [O].

- **Encoding:** curvature, exactly as A.3.1 (`XSENS_FMT_CURVATURE_AUX_TEMP_CHANNEL = 3` [D]).
- **Units:** the temperature gain and offset.
- **Decoder [D]:**
  - `XProfileTemperature` sets `is_aux_channel()`. `XFloatDiveData` then routes the part to a separate aux-temperature channel of the binned composite, so the shared IDs don't collide.
  - It must have the same count as the primary temperature.
  - Listed as the `temp2` column (`temp1` = CTD temperature), and as `tcnd` in the JSON.
- **Limitations [D]:**
  - The aux flag is honored only for binned 0x2n. A format-3 high-resolution (0xAn) or drift (0xA8+) part would be merged into the primary channel, and fail with a duplicate index.
  - The legacy `AXT`/`axt` listing section (`XCompositeAuxTemp`) is no longer written.
- The spreadsheet [X] and the SOLO-II document [S] don't document format 3.

A.4.4 Auxiliary sensor data — 0xC2 *(not implemented by the decoder)*

Status [D]: 0xC2 is defined in `XMessage.h` (`XMSG_TYPE_ID_AUX_SENSOR_DATA`) but has no case in the blob dispatcher, so it is logged as an unknown ID and skipped. The class `AuxiliarySensorData` (a copy of the A.3 unpacker supporting formats 0 and 1) exists but is never instantiated. No 0xC2 blob was seen in the production log sampled for this document.

Format per MRV DataFormat-v7.3.5.yaml, blob 194 = 0xC2 [M]. Byte offsets are **inferred [I]**: the YAML gives field order and types only, and assumes the usual ID + `jj` header.

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID	—	0xC2.
1	2	u16?	size	bytes	Size layout not stated. Probably a full 16-bit size like the other 0xCn types [I].
3	1	u8	packet index	—	Part number.
4	1	u8	sensor channel	—	
5	1	state code	state	—	Dive phase code (A.5.2). The YAML type `state_code` is 1 nibble in other blobs; it is assumed here to take a whole byte [I].
6	1	u8	data type	—	
7	2	u16	gain	—	Applies to `data`: value = counts/gain − offset.
9	2	u16	offset	—	
11	…	packed	data	counts	"packed" = the curvature body of A.3.1 (B, NN, VVV, DDD, packing factors, nibbles) [I].
size−1	1	char	;	—	

> **Discrepancy [X] vs [M]:** the v2.4 spreadsheet instead puts an auxiliary-sensor table into 0xF0 (sensor count N, then for each sensor: data type, message start index, gain, offset; 6 bytes each). MRV firmware 10.5/10.12 sends a fixed 66-byte 0xF0 without that table, and carries gain and offset in the 0xC2 blob itself.

A.5 Time series: fall 0x4n, rise 0x5n, pump 0x6n, surface 0x8n, valve 0xEB

A.5.1 Fall (0x40–0x4F), rise (0x50–0x5F) and surface (0x80–0x8F) time series

Purpose.

- **Fall:** pressure samples from the start of descent to the start of drift. The first sample is taken when the valve opens; others at fixed pressure steps or time intervals [S].
- **Rise:** samples from the start of the descent-to-profile, through the ascent, to the end of ascent. SOLO-II v2.0 adds a record after surface buoyancy pumping [S].
- **Surface:** CTD samples while at the surface. The surface pressure is used for the pressure-offset correction and for the `SFD` statistics.

All three share one decoder (`AbstractXFallRise::UnpackData` [D]). The low nibble of the ID is the part index (0–15). Each part carries its own reference time.

Common header:

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID	—	0x4n fall, 0x5n rise, 0x8n surface; n = part index.
1	2	u4+u12	format / size	— / bytes	Format nibble selects the record layout (below).
3	4	u32	start/reference time	s since 2000-01-01	Part 0 starts at the start of descent (fall), the start of ascent (rise) or surface arrival. The record times are u16 offsets, so the float starts a new part with a new reference time when an offset would exceed 65535 s [X].
7	…	—	records	—	Back to back.
size−1	1	char	;	—	

Record layouts by format nibble:

Format	Record size	Name	Used by
0	4	Legacy	SOLO-II fall/rise; legacy surface
1	5	Standard (phase code)	fall/rise (observed [O]); “legacy” surface
2	9	Standard with T & S	surface (observed [O]); fall/rise also accepted by the decoder
3	—	Curvature, P/T/S in separate blobs (0x80–0x83 P, 0x84–0x87 T, 0x88–0x8B S)	Spreadsheet only [X]; **not implemented** (the decoder throws “unsupported … format”)

Format 0 record (offset r = 7 + 4k):

Offset	Size	Type	Field	Scale/units	Notes
r	2	u16	time	s after reference	
r+2	2	u16	pressure	counts	No phase code (listed as 0 “unknown”). The spreadsheet says 20 bits here [X], but a 4-byte record only leaves 16 bits; the decoder reads 16.

Format 1 record (offset r = 7 + 5k):

Offset	Size	Type	Field	Scale/units	Notes
r	2	u16	time	s after reference	
r+2	½	u4	phase code	—	High nibble of byte r+2 (A.5.2).
r+2	2½	u20	pressure	counts	Low nibble of r+2, then r+3 and r+4.

Format 2 record (offset r = 7 + 9k):

Offset	Size	Type	Field	Scale/units	Notes
r	2	u16	time	s after reference	
r+2	½	u4	phase code	—	
r+2	2½	u20	pressure	counts	

Offset	Size	Type	Field	Scale/units	Notes
r+5	2	u16	temperature	counts	
r+7	2	u16	salinity	counts	

Typical sizes [X] (N = index of the last record):

Format	Size
0	4N + 12
1	5N + 13
2	9N + 17

Decoding notes [D]:

- Records are read until fewer than one record plus `;` remains. The decoder checks `size ≤ received` and the terminator at size−1.
- Formats 0 and 1 carry no T/S; those columns are listed as `-9.999`.
- Fall, rise and the profile use the profile gains/offsets and the optional pressure correction. Surface uses the surface gains/offsets and is never corrected.
- **Listing:** one group per part: `F|R|SFC SN dive idx nSam startTime`, then `sfc SN dive idx sam +sec pres temp saln phaseCode # phaseName`, or for fall and rise `f|r SN dive idx sam +sec pres phaseCode # phaseName` (the temp and saln columns were dropped from `f` and `r` in build 3321; only the 9-byte packing format carries them, last used by 9035 and 9050).
- Surface parts are also combined into the derived `SFD` line: number of valid samples, and the minimum, mean, maximum and variance of the surface pressure.

Example [O] (float 11186, 0x41, format 1): `41 10 3F 32 6E E1 E6 00 00 A0 79 DE …` decodes to

Field	Value
Part	1
Size	63
Reference time	0x326EE1E6 s after 2000-01-01
First record	t = +0 s, code 10 (descending), P counts 0x079DE

A.5.2 Phase codes (4-bit, shared by fall/rise/surface/pump/valve)

Code	ALAMO/ALTO — decoder [D] (`XPresCodeCounts`) and spreadsheet [X]	SOLO-II v2.0 [S]
0	unknown (legacy formats)	—
1	start of descent (START_OF_SINK)	START_OF_SINK
2	first descent pump (“buoyancy at 100 dbar”)	Buoyancy at 100 db
3	seek to drift pressure	SEEK
4	start of drift	BEGINNING_OF_DRIFT
5	seek during drift	BEGINNING_OF_FALL_TO_ PROFILE
6	start of descent to profile	START_OF_RISE
7	start of ascent	END_OF_RISE
8	end of ascent	ICE_TURNAROUND
9	ice turn-around	—
10	descending	SINKING
11	drift/park	DRIFTING
12	descending to profile	FALLING_TO_PROFILE
13	ascending	RISING
14	surface	SURFACE
15	“invalid phase code” (accepted by the decoder)	—

> ALAMO inserted “seek during drift” at 5, which shifts SOLO-II codes 5–8 up by one. The spreadsheet lists “9: END_OF_RISE” and “9: ICE_TURNAROUND”. The first is a typo for 8 (decoder and data: 8 = end of ascent [O]).

A.5.3 Pump — 0x60–0x6F

Purpose. One record per pump run: pressure, duration, battery voltage, current and case vacuum. The low nibble of the ID is the part index.

Header:

Offset	Size	Type	Field	Notes
0	1	u8	ID	
1	2	u4+u12	format / size	Format 0 reads the size as the full 16 bits [D].
3	…	records		
size−1	1	char	;	

Format 0 record, 10 bytes, legacy SOLO-II (offset r = 3 + 10k):

Offset	Size	Type	Field	Scale/units	Notes
r	2	u16	pressure	counts	
r+2	2	i16	pump time	s	Signed [S].
r+4	2	u16	pump battery voltage	0.01 V	
r+6	2	u16	pump current	mA	
r+8	1	u8	vac0	see note	Vacuum after the pump starts [S].
r+9	1	u8	vac1	see note	Vacuum before the pump stops [S].

Format 1 record, 11 bytes, standard with 2 vacuums (offset r = 3 + 11k):

Offset	Size	Type	Field	Scale/units	Notes
r	½	u4	phase code	—	
r	2½	u20	pressure	counts	
r+3	2	i16	pump time	s	
r+5	2	u16	voltage	0.01 V	
r+7	2	u16	current	mA	
r+9	1	u8	vac0 (before pumping)	0.1 inHg	Decoder: ×10 → cinHg.
r+10	1	u8	vac1 (after pumping)	0.1 inHg	

Format 2 record, 11 bytes, single 16-bit vacuum (offset r = 3 + 11k):

Offset	Size	Type	Field	Scale/units	Notes
r	½	u4	phase code	—	
r	2½	u20	pressure	counts	
r+3	2	i16	pump time	s	
r+5	2	u16	voltage	0.01 V	
r+7	2	u16	current	mA	
r+9	2	u16	case vacuum	0.01 inHg	The decoder stores it as vac1 (“after”); vac0 is the fill value (listed 655.35). The MRV YAML calls it `case_pressure` (u16, /100, inHg) [M].

Status of the formats:

- Format 2 is what current firmware sends (observed 0x60 `jj` = 0x205C / 0x2122 [O]).
- The spreadsheet documents only format 1 (“standard”) in full; it describes format 2 as “11-byte (Tentative) … with phase codes” without a layout [X]. The format-2 layout above is from the decoder and the YAML, and is confirmed by the data.

Example [O] (float 11186, format 2): `20 07 59 00 87 05 DA 00 C5 01 F0` decodes to

Field	Value
Phase code	2 (first descent pump)
Pressure	0x00759 counts = 65.24 dBar (gain 25, offset 10)
Pump time	135 s
Voltage	14.98 V
Current	0.197 A
Case vacuum	4.96 inHg

Decoding notes [D]:

- Format 0 vacuum bytes are taken as cinHg without scaling, so the listing shows byte/100 inHg. The SOLO-II document calls them “vacuum counts” without units, so the format-0 vacuum units are unverified.
- Phase codes are decoded but not listed.
- **Listing:** `PMP SN dive idx nSam`, then `pmp SN dive idx sam pres pumpT(s) pumpV(V) pumpI(A) vac0(inHg) vac1(inHg)`.
- **Spreadsheet typo [X]:** the terminator offset is given as 4N+14; it should be 11N+14.

A.5.4 Valve — 0xEB

Purpose. One record per valve opening: pressure, open time and case vacuum.

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID	—	0xEB.
1	2	u16	size	bytes	Full 16-bit size; 7N + 12 typical.
3	½	u4	encoding format	—	High nibble. 0 is the only format [X]; not checked [D].
3	½	u4	part index	—	Low nibble [X][D].
4 + 7k	½	u4	phase code	—	
4 + 7k	2½	u20	pressure	counts	Profile gains/offsets.
7 + 7k	2	i16	valve open time	s	
9 + 7k	2	u16	case vacuum	0.01 inHg	
size−1	1	char	;	—	

Example [O] (float 11186): `EB 00 0C 00 10 00 FE 01 5B 01 DD 3B` decodes to

Field	Value
Format / part	0 / 0
Phase code	1 (start of descent)
Pressure	254 counts = 0.16 dBar
Valve open time	347 s
Case vacuum	4.77 inHg

Decoding notes:

- **Vacuum scale [D][O]:** the spreadsheet gives gain 10 for the vacuum [X], which would put the observed values near 48 inHg, more than one atmosphere. The decoder’s /100 gives 4.77 inHg, which agrees with the pump vacuums, so /100 is correct.
- The spreadsheet says “WHEN?” for when the vacuum is measured. The decoder treats it as the vacuum after the valve operation (vac1); vac0 is listed as 655.35.
- The MRV YAML lists the record fields but omits byte 3 [M]. The observed data has byte 3.
- The spreadsheet overview marks 0xEB “decoder supported: no”; that is outdated.
- **Listing:** `VLV SN dive idx nSam`, then `vlv SN dive idx sam pres valveTime vac0 vac1`.

A.6 Argo data — 0xF0

Purpose. Per-dive mission targets, the four state start times, and the **gains and offsets that convert all CTD counts in the dive**. It is listed as `A` lines with `midx` 0–5. Without it, nothing in the dive can be converted (A.1.5).

Version byte [D][X]: byte 3. **High nibble = major, low nibble = minor** (0x24 = v2.4). The SOLO-II document says the opposite ("minor in high nibble, major in low") [S]; observed packets (`0x24` on firmware 10.12 [O]) and the decoder use high = major.

Versions the decoder supports [D]:

Version	Minimum size checked	Layout
1.x (SOLO-II)	—	Not decoded; throws "unsupported X-Message format".
2.1	≥ 40	Short layout (below). Surface gains/offsets are copies of the profile ones.
2.2, 2.3	≥ 66	Full layout (below).
2.4	== 66	Full layout (below).

For every version, fields beyond bytes 4–7 are decoded **only if the float-type byte = 3 (ALAMO)**. Otherwise the instrument type stays unknown and the listing prints `A (unknown float type)`. SOLO-II (0) and Deep Argo (1) are `@todo` in the decoder.

A.6.1 v2.2 / v2.3 / v2.4 — 66 bytes (MRV firmware 10.5+ "Argo packet version 0x24" [M][O])

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID	—	0xF0.
1	2	u16	size	bytes	66 (0x0042).
3	1	u4/u4	version	major/minor	0x22, 0x23, 0x24.
4	2	u16	target profile pressure	dBar	`profPres`. Observed 0 on 9287, 1000 on 11186 [O].
6	2	u16	target park pressure	dBar	`parkPres`.
8	2	u16	max ascent duration	min	`maxAscT`. MRV: "max pump runtime during ascent (minutes)".
10	2	u16	max descent-to-park duration	min	`maxDesc2ParkT`.
12	2	u16	max fall-to-profile duration	s	`maxDesc2ProfT`.
14	2	u16	target park duration	**5 min**	MRV "divisor 0.2"; SOLO "units = 5 minutes". The decoder lists the raw value under a `(min)` header: observed 288 = 24 h [O].
16	4	u32	descent start time	s since 2000-01-01	`start-of-descent`.
20	4	u32	park (drift) start time	s since 2000-01-01	`start-of-drift`; the reference for 0x78 drift times.
24	4	u32	ascent start time	s since 2000-01-01	`start-of-ascent`.
28	4	u32	surface start time	s since 2000-01-01	`start-of-surface`.
32	1	u8	float type	—	0 = SOLO-II, 1 = Deep Argo, 3 = ALAMO. Must be 3 for decoding.
33	1	u8	target ascent rate	cm/s	`tarAscRate`.
34	2	u16	max number of seeks	—	**The decoder reads only byte 34 (u8)**, so the listed `nSeek` is the high byte, 0 on all observed floats, where the true value is 1 [O]. *(Fixed in build 3309.)*
36	2	u16	max surface duration	s	`maxSurfT`. Observed raw value 1 [O].
38	2	u16	seek interval	min	`seekIntervalT`.
40	2	u16	pressure gain	LSB/dBar	`pGain`.
42	2	u16	pressure offset	dBar	`pOff`.
44	2	u16	temperature gain	LSB/°C	`tGain`.
46	2	u16	temperature offset	°C	`tOff`.
48	2	u16	salinity gain	LSB/PSU	`sGain`.
50	2	u16	salinity offset	PSU	`sOff`.
52	2	u16	surface pressure gain	LSB/dBar	`pGainSurf`.

Offset	Size	Type	Field	Scale/units	Notes
54	2	u16	surface pressure offset	dBar	`pOffSurf`.
56	2	u16	surface temperature gain	LSB/°C	`tGainSurf`.
58	2	u16	surface temperature offset	°C	`tOffSurf`.
60	2	u16	surface salinity gain	LSB/PSU	`sGainSurf`.
62	2	u16	surface salinity offset	PSU	`sOffSurf`.
64	1	u8	mission phase number	—	`missPhase` (MRV MPS 0–2).
65	1	char	;	—	

Example [O] (float 9287, dive 325): `F0 00 42 24 00 00 01 13 00 11 00 C4 0E 10 01 20 32 6F 1C A3 32 6F 4D 78 32 70 9E FB 32 70 AA 12 03 08 00 01 00 01 02 58 00 19 00 0A 03 E8 00 05 03 E8 00 01 00 19 00 0A 03 E8 00 05 03 E8 00 01 00 3B`

Field	Value
Park pressure	275 dBar
Max ascent	17 min
Max descent to park	196 min
Max fall to profile	3600 s
Park duration	288 × 5 min
Descent start	2026-10-24T07:37:07Z
Drift start	2026-10-24T11:05:28Z
Ascent start	2026-10-25T11:05:31Z
Surface start	2026-10-25T11:52:50Z
Float type	3 (ALAMO)
Ascent rate	8 cm/s
Seeks	1
Max surface duration	1
Seek interval	600 min
P gain/offset	25 / 10
T gain/offset	1000 / 5
S gain/offset	1000 / 1
Surface gains/offsets	same as above
Mission phase	0

A.6.2 v2.1 — short layout (decoder only [D]; no document found)

Offset	Size	Type	Field	Notes
0	1	u8	ID	0xF0
1	2	u16	size	≥ 40 required
3	1	u4/u4	version	0x21
4	2	u16	target profile pressure (dBar)	
6	2	u16	target park pressure (dBar)	
8	4	u32	descent start (s2000)	
12	4	u32	park start (s2000)	
16	4	u32	ascent start (s2000)	
20	4	u32	surface start (s2000)	
24	1	u8	float type	must be 3
25	1	u8	target ascent rate (cm/s)	
26–37	12	6 × u16	P gain, P offset, T gain, T offset, S gain, S offset	Also copied to the surface gains.
38 … size−2	—	—	not decoded [I]	
size−1	1	char	;	

v2.1 has no durations, seeks, surface gains or mission phase. Those fields keep their “bad” values (255/65535) in the listing.

A.6.3 Other documented layouts (not decoded)

SOLO-II "v1" (37 bytes) [S]. The PDF's table is misaligned; this realignment is inferred [I]:

Offset	Field
3	version
4–5	target profile depth
6–7	target park depth
8–9	max rise time (min)
10–11	max fall-to-park time (min)
12–13	max fall-from-park-to-profile time (s)
14–15	target drift time (5 min)
16	float version (0)
17	target ascent rate
18–19	number of seeks
20–21	surface time
22–23	seek interval (min)
24–35	P, T and S gain/offset pairs
36	;

Spreadsheet "v2.4" [X]. The SOLO-II layout plus surface gains/offsets (36–47), the mission phase (48), and an auxiliary-sensor table (49: N; then 6 bytes per sensor: type, start index, gain, offset), with size 51 + 6N. **It has no state start times.**

> **Discrepancy:** the version number 0x24 appears both in the spreadsheet and in real packets, but the layouts differ. Real 0x24 packets follow the MRV 10.5.0 layout (A.6.1, start times at 16–31, fixed 66 bytes), and so does the decoder. The spreadsheet's 0xF0 sheet does not describe what floats send.

Listing [D] (`A SN dive xVer typ midx …`):

midx	Fields
0	`missPhase profPres parkPres maxAscT maxDesc2ParkT maxDesc2ProfT`
1	`tarParkT nSeek seekIntervalT tarAscRate maxSurfT`
2	`start-of-descent start-of-drift`
3	`start-of-ascent start-of-surface`
4	`pGain pOff tGain tOff sGain sOff`
5	the same for the surface gains/offsets

The gains/offsets and the X version are also saved to the float record in the database.

A.7 Open issues found in part A

1. **Packing factor width.** Decoder and data: p nibbles, p = 0 = absent. The SOLO-II document and the spreadsheet's formula say p+1 nibbles, and the spreadsheet's per-sub-block byte counts are 8 bytes too large.
2. **B (starting sub-block).** Observed as 0 for the first part, then the cumulative sub-block count. The SOLO-II document says 1 for the first part. The decoder ignores it.
3. **0xF0 v2.4.** The spreadsheet layout (no start times, aux-sensor table) disagrees with the firmware, the MRV document and the decoder.
4. **0xF0 decoder bugs:**
   - `nSeek` read as a u8 at byte 34 instead of a u16 at 34–35. *(Fixed in build 3309.)*
   - `tarParkT` listed in 5-minute units under a "(min)" header.
5. **GPS 0x01/0x02 labels** are swapped in the spreadsheet overview.
6. **Phase codes:** SOLO-II numbering differs from ALAMO's from code 5 on. The spreadsheet has a duplicate "9".

7. **Pump format 2** is undocumented in the spreadsheet; its single vacuum is listed by the decoder as vac1.
8. **Pump format 0 vacuum units** are unknown (SOLO-II says “counts”).
9. **Valve vacuum gain:** 10 in the spreadsheet, 100 in the decoder, and the data supports 100.
10. **Surface format 3** (per-sensor curvature blobs 0x80–0x8B) is specified [X] but not decoded. Fall/rise format 2 (9-byte with T/S) is decoded but not documented for 0x4n/0x5n.
11. **0xC2 auxiliary sensor data** is not decoded. Its byte offsets here come from the MRV YAML field order only and are unverified.
12. **0x78 and 0xDC** are undocumented. Their formats are inferred, and the meanings of 0xDC bytes 3 and 6 (always 0) and byte 5 (always 1) are unknown.
13. **Legacy first-difference decoder:** it drops a final sub-block of 1–2 values. The SOLO-II document’s 22-byte sub-block size should be 27.
14. **Checksum:** documented and verified on production data. *(The decoder checks it since build 3310.)*

Part B — Engineering messages

This part specifies the engineering message group of the ALAMO/ALTO X-message format: the SOLO-II-heritage engineering messages (0xE0, 0xE2, 0xE3), the built-in-test (BIT) messages (0xE5, 0xE6), the ALAMO dive-phase engineering messages (0xE7 descent, 0xE8 drift/park, 0xE9 ascent, 0xEA surface), the ice-detection engineering message (0xEC) and the system error codes that the 0xEA and 0xEC error lists carry. The valve message 0xEB is outside this part.

B.0 Conventions

Offsets. Every offset in this part counts from the blob's ID byte (offset 0). Bytes 1–2 are the big-endian size `jj`, which counts the whole blob (ID, `jj`, data and the `;` terminator). Byte 3 of every message in this group is a format/version byte. The terminator `;` (0x3B) is the last byte, at offset `jj - 1`.

Byte order. All multi-byte fields are big-endian (most significant byte first). This holds for the IEEE-754 single-precision floats of BIT formats 9–11 too: the decoder assembles the four bytes big-endian into a `uint32` and bit-casts it to `float` (`f32` in `common/XDefinitions.h`).

Types. `u8`, `u16`, `u32` are unsigned integers; `i16` is a two's-complement signed integer; `f32` is an IEEE-754 binary32; `char[4]` is four ASCII bytes.

Time epochs. Two epochs are used, and each table names the one that applies:

Name	Meaning	Decoder macro
`s2000`	`u32` seconds since 2000-01-01T00:00:00Z	`s2000(d, n)`
`s1970`	`u32` seconds since 1970-01-01T00:00:00Z (Unix time)	`s1970(d, n)`

A raw value of 0 decodes to "not a date/time" and is listed as `0000-00-00T00:00:00Z`.

Sensor counts. Fields marked *CTD counts* are raw 16-bit sensor codes converted with the float's profile gains and offsets from the Argo data message (0xF0): `value = counts / gain − offset`. The decoder uses the profile (not surface) gains for every engineering message. A result outside the valid range (pressure −5…2610 dbar, temperature −5…40 °C, salinity 2…41 PSU) or a zero gain gives the fill value −9.999.

Fill values. Fields that a message version does not carry keep the decoder's fill value: `0xFF` (`u8`), `0xFFFF` (`u16`), `0xFFFFFFFF` (`u32`), `−32768` (`i16`), or not-a-date/time. Some classes leave such fields uninitialized; this is noted per message. Fill values matter only to code that reads the decoded objects: the listings print only the fields each version carries.

Version byte. For 0xE7–0xEC, byte 3 is the "engineering version", written in hex (`0x10`, `0x11`, …). `XAlamoEngineering::engineering_version_as_float()` reads the high nibble as the major and the low nibble as the minor number (0x12 → 1.2). For 0xE5/0xE6, byte 3 is the "BIT format" (decimal 6–11). Firmware documents sometimes give the engineering version in decimal (`packetVersion = 17` = 0x11, `= 20` = 0x14).

Common validation. Every class below first calls `XMessage::CheckXMsgData()`, which:

1. rejects an empty blob, or one shorter than the class minimum (given per message);
2. checks the ID byte (skipped for 0xE5/0xE6, whose ID is checked later);
3. reads the size as `u16(d, 1) & 0x0FFF` (only the low 12 bits are used), rejects a size below 4 or larger than the received blob, and only warns when the blob is longer than the size;
4. rejects the blob unless byte `size − 1` is `;`.

The per-version size checks below compare the length of the received blob (`data.size()`), not the size field. A failed check throws `util::log::MessageException`; the message is then not assigned to the dive and the failure is logged.

Listing output. Engineering lines are written in the "Engineering Data" part of the dive listing, in this order of sections: Abort, BIT failed, BIT passed, Descent, Diagnostic Dive, Park/Drift, Ascent, Ice, Standard Dive, Surface. Each data line starts with a line-type tag, the serial number (`SN`), the dive number and a minor index (`midx`); the `#` lines above it are column headings and units.

Message	Listing line type(s)	SN formatting
0xE0, 0xE2, 0xE3	none (an empty section heading only)	—
0xE5 / 0xE6	`E05` / `E06`	`%5d` (not zero-padded)
0xE7	`E07`	`%05d`
0xE8	`E08`	`%05d`
0xE9	`E09`	`%5d`
0xEA	`E0A`; error entries `e0A`	`%05d`
0xEC	`E0C`; error entries `e0c`	`%05d`

The Python tool `bin/py/xmtps-list2json.py` maps these line types to the JSON engineering sections `bit`, `descent`, `drift`, `ascent`, `surface` and `ice` (see the README).

B.1 SOLO-II-heritage engineering messages: 0xE0, 0xE2, 0xE3

Purpose. These are the SOLO-II engineering messages, one per mission phase: 0xE0 after the first (diagnostic) dive of a mission, 0xE2 at the end of a normal dive cycle and 0xE3 after a mission abort. The SOLO-II X-format v2.0 document (20 Mar 2014) defines them as version 5. ALAMO floats do not send them: the v2.4 format overview marks all three "ALAMO supported: no, decoder supported: no".

Decoder status. `XEngDiagnostic` (0xE0), `XEngStandard` (0xE2) and `XEngAbort` (0xE3) are stubs. Each runs the common validation (expected ID, no minimum length beyond the size field) and decodes nothing; the listing gets the section heading ("Engineering (Diagnostic Dive)", "Engineering (Standard Dive)", "Engineering (Abort)") with an empty body. The version byte is not checked.

The layouts below are transcribed from the SOLO-II v2.0 document, for reference only. In that document's text the byte column and the content column of some rows have drifted apart; the assignments marked *reconstructed* were made by matching the remaining contents, in order, to the remaining byte numbers so that the total equals the stated size.

B.1.1 0xE0 — diagnostic dive (SOLO-II version 5, 80 bytes)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xE0		
1	2	u16	size = 80 (0x50)	bytes	
3	1	u8	engineering message version = 5		
4	1	u8	packets in current session		
5	6	—	0 (filler)		
11	2	u16	`EP->sattime`	s	
13	2	u16	CPU battery voltage (`Vcpu`)	0.01 V	
15	2	u16	pump battery voltage at surface (`Vpmp`)	0.01 V	
17	2	u16	pump battery voltage at end of last pump (`Vple`)	0.01 V	
19	2	u16	case vacuum at beginning of BIT (`BTvac`)	0.01 inHg	
21	2	u16	vacuum before filling bladder at surface (`Air[1]`)	0.01 inHg	
23	2	u16	vacuum after filling bladder at surface (`Air[2]`)	0.01 inHg	
25	2	u16	ID of last interrupt (`ISRID`)		
27	2	u16	average pump current at bottom (`HPavgI`)	1 mA	
29	2	u16	maximum pump current at bottom (`HPmaxI`)	1 mA	
31	2	u16	total seconds pumped to surface	s	
33	2	u16	seconds pumped at surface	s	
35	2	u16	surface pressure before offset reset (`SPRX`)		previous dive
37	2	u16	pressure after offset reset (`SPRXL`)		previous dive
39	3	u24	pressure when "in water" sensed (`diagP[0]`)		
42	3	u24	temperature when "in water" sensed (`diag T[0]`)		
45	3	u24	salinity when "in water" sensed (`diagS[0]`)		

Offset	Size	Type	Field	Scale/units	Notes
48	2	u16	scans recorded by the SBE (`SBnscan`)		0xFFFF: scan count unavailable; 0xFFFE: SBE never started
50	2	u16	packed SBE status: `(SBntry&0xF)<<4 | (SBstrt&0x3)<<2 | (SBstop&0x3)`		
52	3	u24	shallowest pressure in profile (`diagP[1]`)		
55	3	u24	shallowest temperature in profile (`diagT[1]`)		
58	3	u24	shallowest salinity in profile (`diagS[1]`)		
61	2	u16	BIT vacuum (`BTvac`)	0.01 inHg	
63	2	u16	BIT motor current out (`BTPcur`)	1 mA	
65	2	u16	BIT pump seconds (`BTPsec`)	s	
67	1	u8	BIT pump vacuum before pumping (`BTPvac[0]`)		
68	1	u8	BIT pump vacuum after pumping (`BTPvac[1]`)		
69	2	u16	BIT pump battery (`BTVple`)	0.01 V	
71	2	u16	BIT CPU battery (`BTVcpu`)	0.01 V	
73	2	u16	exception flags		see 0xE2 offset 89
75	1	u8	vent data: tenths of seconds the vent motor ran	0.1 s	
76	1	u8	LLD status before/after vent ran		
77	2	u16	abort code (`AbrtCd`)		see 0xE3 offset 25
79	1	char	;		

B.1.2 0xE2 — normal dive cycle (SOLO-II version 5, 102 bytes)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xE2		
1	2	u16	size = 102 (0x66)	bytes	
3	1	u8	engineering message version = 5		
4	1	u8	packets sent in current surface session		
5	2	u16	connection tries in previous surface session		
7	2	u16	`parse_X_reply`		low byte: number of messages; high byte: error bit field
9	2	u16	ATSBD return status, previous session		
11	2	u16	seconds to send all SBD messages, previous session (`sattime`)	s	
13	2	u16	CPU battery voltage (`Vcpu`)	0.01 V	
15	2	u16	pump battery voltage at surface (`Vpmp`)	0.01 V	
17	2	u16	pump battery voltage at end of last pump (`Vple`)	0.01 V	
19	2	u16	case vacuum sinking at 50 dbar, oil inside (`Air[0]`)	0.01 inHg	
21	2	u16	case vacuum before filling oil bladder at surface (`Air[1]`)	0.01 inHg	
23	2	u16	case vacuum after filling bladder at surface (`Air[2]`)	0.01 inHg	
25	2	u16	ID of last interrupt (`ISRID`)		
27	2	u16	average pump current at bottom (`HPavgI`)	1 mA	
29	2	u16	maximum pump current at bottom (`HPmaxI`)	1 mA	
31	2	u16	total seconds pumped to surface	s	
33	2	u16	seconds pumped at surface	s	
35	2	u16	surface pressure before offset reset (`SPRX`)		previous dive
37	2	u16	pressure after offset reset (`SPRXL`)		previous dive
39	3	u24	pressure before pumping for ascent (`diagP[0]`)		
42	3	u24	temperature before pumping for ascent		
45	3	u24	salinity before pumping for ascent		
48	3	u24	last (shallowest) pressure on ascent (`diagP[1]`)		
51	3	u24	last (shallowest) temperature on ascent		
54	3	u24	last (shallowest) salinity on ascent		

Offset	Size	Type	Field	Scale/units	Notes
57	2	u16	bad bins from SBE (`SBnbad`)		
59	2	u16	scans recorded by SBE (`SBnscan`)		0xFFFF / 0xFFFE as in 0xE0
61	2	u16	packed SBE status (as in 0xE0 offset 50)		
63	3	u24	average pressure, first half of drift (`PAVG[0]`)		
66	3	u24	average temperature, first half of drift		
69	3	u24	average salinity, first half of drift		
72	3	u24	average pressure, second half of drift (`PAVG[1]`)		
75	3	u24	average temperature, second half of drift		
78	3	u24	average salinity, second half of drift		
81	2	u16	fall time: open air valve to end of settle	s	
83	2	u16	average fall rate while sinking	mm/s	
85	2	u16	tenths of seconds pumped in first settle to drift (`SeekT`)	0.1 s	
87	2	i16	depth change in first settle (`SeekP`)	0.1 dbar	signed
89	2	u16	exception flags		bit table below
91	1	u8	vent data: tenths of seconds the vent motor ran	0.1 s	*reconstructed*
92	1	u8	LLD status before and after vent ran		*reconstructed*
93	2	u16	SBE pressure offset	×800	*reconstructed*
95	2	u16	tenths of seconds pumped to target depth (`PP->SeekSc`)	0.1 s	*reconstructed*
97	2	u16	packets sent in previous cycle		*reconstructed*
99	1	u8	binning mode (high nibble; 2 = controller binning, else CTD binning), subcycle number (low nibble)		*reconstructed*
100	1	u8	ice-detect status: 0 `ICE_CHECK_OFF`, 1 `ICE_NOT_FOUND`, 2 `ICE_MIXLAYER`, 3 `ICE_BREAKUP`		*reconstructed*
101	1	char	;		

0xE2 exception flags (offset 89):

Bit	Meaning
0x0001	valve failed to open
0x0002	valve failed to close
0x0004	questionable pressure
0x0008	antenna was toggled
0x0010	antenna switch failure (no satellites even after toggling)
0x0020	GPS communication error
0x0080	float took too long to leave the surface (valve toggled)
0x1000	valve failure during sink phase
0x2000	valve failure during ascent phase

B.1.3 0xE3 — after mission abort (SOLO-II version 5, 30 bytes)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xE3		
1	2	u16	size = 30 (0x1E)	bytes	
3	1	u8	engineering message version = 5		
4	1	u8	packets sent in current surface session		
5	2	u16	connection tries in last surface session		
7	2	u16	`parse_X_reply`		as in 0xE2
9	2	u16	ATSBD return status, last session		
11	2	u16	seconds taken to send last SBD message	s	
13	2	u16	current CPU battery voltage	0.01 V	
15	2	u16	current pump battery voltage	0.01 V	
17	2	u16	case vacuum at beginning of abort (`Air[1]`)	0.01 inHg	
19	2	u16	case vacuum at end of last transmission (`Air[0]`)	0.01 inHg	previous cycle

Offset	Size	Type	Field	Scale/units	Notes
21	2	—	not described		gap in the source document
23	2	u16	ID of last interrupt (`ISRID`)		
25	2	u16	abort code (`AbrtCd`)		table below
27	2	—	empty		
29	1	char	;		

Abort codes: 0 no error; 1 current time later than `RTCabort`; 2 unable to wake OST; 3 unable to send dive number to SOLO-II; 4 Iridium ground station commanded abort; 5 final dive completed, mission done; 6 diagnostic dive failed (no GPS fix, pressure never exceeded `dBarGo`, or no Iridium message sent); 7 pressure sensor failure.

B.1.4 SOLO-II BIT messages (0xE5/0xE6 version 5)

SOLO-II uses 0xE5 (BIT passed, 58 bytes) and 0xE6 (BIT failed, 60 bytes), version 5. The decoder does **not** support them: `XEngBIT` accepts only BIT formats 6–11 and rejects version 5 with "Unsupported BIT message format: 5!". The layouts are given for reference; 0xE6 is 0xE5 with a 2-byte BIT status register inserted at offset 5.

0xE5 offset	0xE6 offset	Size	Type	Field	Scale/units
0	0	1	u8	ID (0xE5 / 0xE6)	
1	1	2	u16	size (58 / 60)	bytes
3	3	1	u8	engineering message version = 5	
4	4	1	u8	packets sent in this surface session	
—	5	2	u16	BIT status register	
5	7	2	u16	SBE pressure offset	×800
7	9	2	u16	CPU battery voltage	0.01 V
9	11	2	u16	no-load pump battery voltage	0.01 V
11	13	2	u16	pump battery voltage at end of last pump	0.01 V
13	15	2	u16	average pump current (`HPavgI`)	1 mA
15	17	2	u16	seconds pumped out during test	s
17	19	1	u8	oil vacuum before filling bladder	0.01 inHg
18	20	1	u8	oil vacuum after filling bladder	0.01 inHg
19	21	2	u16	case vacuum at beginning of BIT, oil bladder empty (`Air[0]`)	0.01 inHg
21	23	2	u16	case vacuum at end of BIT, air bladder inflated (`Air[1]`)	0.01 inHg
23	25	1	u8	tries needed to open valve	
24	26	1	u8	tries needed to close valve	
25	27	2	u16	ID of last interrupt	
27	29	30	char[30]	string returned from the SBE `pt` command	
57	59	1	char	;	

The 0xE5 rows from offset 13 on are *reconstructed* (the source text is misaligned there); they agree with the 0xE6 table, which is aligned.

B.2 Built-in-test messages: 0xE5 (passed) and 0xE6 (failed)

Purpose. An ALAMO/ALTO float runs a built-in self-test (BIT) at deployment. It exercises the air and oil pumps and valves, the CTD (resetting the pressure offset), the case vacuum sensor, the battery, GPS and Iridium, and reports the results once, as 0xE5 when every test passed or 0xE6 when at least one failed. The two IDs share one layout; only the ID differs.

Dispatch. `XEngBITPass` and `XEngBITFail` both derive from `XEngBIT`, which runs the common validation with a 13-byte minimum (ID check skipped) and dispatches on byte 3, the BIT format:

BIT format (byte 3)	Decoder class	Size (bytes)	Firmware (from documents)
6	`XAlamoBit`	13	early ALAMO (spreadsheet history)
7	`XAlamoBit`	72	X-format v2.1–2.4

BIT format (byte 3)	Decoder class	Size (bytes)	Firmware (from documents)
8	`XAlamoBit`	76	X-format v2.4; firmware 7.3.5
9	`XAlamoBit2`	256	firmware 10.5.0
10	`XAlamoBit2`	256	firmware 10.6.x (same layout as 9)
11	`XAlamoBit2`	256 or 264	firmware 10.8 and later (observed from 10.12.0.1)
other	—	—	rejected: “Unsupported BIT message format”

Both classes re-check that byte 0 is 0xE5 or 0xE6 and set the passed/failed flag from it.

B.2.1 BIT format 6 (13 bytes)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xE5 / 0xE6		
1	2	u16	size = 13	bytes	
3	1	u8	BIT format = 6		
4	4	u32	float software version		listed in hex
8	4	u32	BIT results mask		B.2.6
12	1	char	;		

The blob must be exactly 13 bytes.

B.2.2 BIT format 7 (72 bytes)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xE5 / 0xE6		
1	2	u16	size = 72	bytes	
3	1	u8	BIT format = 7		
4	4	u32	float software version		
8	4	u32	BIT results mask		B.2.6
12	2	u16	total BIT duration	s	
14	2	u16	pressure	CTD counts	
16	2	u16	temperature	CTD counts	
18	2	u16	salinity	CTD counts	
20	4	u32	CTD status		listed in hex
24	2	u16	case vacuum at start of BIT	÷100 → inHg	
26	2	u16	case vacuum at end of BIT	÷100 → inHg	
28	2	u16	battery voltage at start of BIT	÷100 → V	
30	2	u16	battery voltage at end of BIT	÷100 → V	
32	2	u16	air valve open position	÷1000 → V	
34	2	u16	air valve open calibration position	÷1000 → V	
36	2	u16	air valve closed position	÷1000 → V	
38	2	u16	air valve closed calibration position	÷1000 → V	
40	2	u16	air valve middle position	÷1000 → V	
42	2	u16	air valve middle calibration position	÷1000 → V	
44	2	u16	air pump test duration	s	
46	1	u8	air pump cycle count		
47	2	u16	maximum air pressure	÷1000 → psi	
49	2	u16	air pump maximum load current	÷1000 → A	
51	2	u16	air pump maximum load 12 V current	÷1000 → A	
53	2	u16	air pump maximum load battery voltage	÷1000 → V (decoder)	spreadsheet and 7.3.5 YAML: ÷100
55	2	u16	oil valve open position	÷1000 → V	
57	2	u16	oil valve open calibration position	÷1000 → V	
59	2	u16	oil valve closed position	÷1000 → V	
61	2	u16	oil valve closed calibration position	÷1000 → V	
63	2	u16	oil pump test duration	s	spreadsheet: gain 1000
65	2	u16	oil pump maximum load current	÷1000 → A	

Offset	Size	Type	Field	Scale/units	Notes
67	2	u16	oil pump maximum load 12 V current	÷1000 → A	
69	2	u16	oil pump maximum load battery voltage	÷1000 → V	7.3.5 YAML: ÷100
71	1	char	;		

The blob must be exactly 72 bytes.

B.2.3 BIT format 8 (76 bytes)

Format 8 replaces format 7's single pressure with the pressure before the offset reset, the offset reading and the pressure after the reset; every later field moves 4 bytes on.

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xE5 / 0xE6		
1	2	u16	size = 76	bytes	spreadsheet: "77" (its terminator offset gives 76)
3	1	u8	BIT format = 8		
4	4	u32	float software version		7.3.5 YAML: four `u8` — major, minor, patch, unused
8	4	u32	BIT results mask		B.2.6
12	2	u16	total BIT duration	s	
14	2	u16	pressure before offset reset	CTD counts	7.3.5 YAML: `i16`, surface gains
16	2	u16	pressure sensor offset reading	÷1000 → dbar	spreadsheet (0xE6) and YAML: signed `i16`; decoder reads unsigned
18	2	u16	pressure after offset reset	CTD counts	7.3.5 YAML: `i16`, surface gains
20	2	u16	temperature	CTD counts	
22	2	u16	salinity	CTD counts	
24	4	u32	CTD status		YAML: 4 ASCII characters
28	2	u16	case vacuum at start of BIT	÷100 → inHg	
30	2	u16	case vacuum at end of BIT	÷100 → inHg	
32	2	u16	battery voltage at start of BIT	÷100 → V	
34	2	u16	battery voltage at end of BIT	÷100 → V	
36	2	u16	air valve open position	÷1000 → V	
38	2	u16	air valve open calibration position	÷1000 → V	
40	2	u16	air valve closed position	÷1000 → V	
42	2	u16	air valve closed calibration position	÷1000 → V	
44	2	u16	air valve middle position	÷1000 → V	
46	2	u16	air valve middle calibration position	÷1000 → V	
48	2	u16	air pump test duration	s	
50	1	u8	air pump cycle count		
51	2	u16	maximum air pressure	÷1000 → psi	
53	2	u16	air pump maximum load current	÷1000 → A	
55	2	u16	air pump maximum load 12 V current	÷1000 → A	
57	2	u16	air pump maximum load battery voltage	÷1000 → V (decoder)	spreadsheet and YAML: ÷100
59	2	u16	oil valve open position	÷1000 → V	
61	2	u16	oil valve open calibration position	÷1000 → V	
63	2	u16	oil valve closed position	÷1000 → V	
65	2	u16	oil valve closed calibration position	÷1000 → V	
67	2	u16	oil pump test duration	s	spreadsheet: gain 1000; YAML: 1
69	2	u16	oil pump maximum load current	÷1000 → A	
71	2	u16	oil pump maximum load 12 V current	÷1000 → A	
73	2	u16	oil pump maximum load battery voltage	÷1000 → V	YAML: ÷100
75	1	char	;		

The blob must be exactly 76 bytes. The v2.4 spreadsheet's 0xE6 format 8 sheet makes the CTD status 16 bits (offsets 24–25) and shifts every later field 2 bytes earlier; this disagrees with its own 0xE5 sheet, the

7.3.5 YAML and the decoder, which all use a 4-byte status at 24–27 for both IDs. The spreadsheet also swaps the format 7/8 sizes on the 0xE6 sheet.

B.2.4 BIT formats 9 and 10 (256 bytes)

From format 9 on, measured values are sent as IEEE-754 floats in engineering units, each subsystem gets its own error bitfield, and the test thresholds are appended.

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xE5 / 0xE6		
1	2	u16	size = 256	bytes	
3	1	u8	BIT format = 9 or 10		
4	4	u32	firmware version		bytes: major, minor, patch, product (e.g. 0x0A0C0001 = 10.12.0.1)
8	1	u8	BIT type		B.2.7; decoded, not listed
9	1	u8	CTD (sensor) type		B.2.7
10	4	u32	BIT results mask		B.2.6
14	4	u32	air pump errors		B.2.6
18	4	u32	oil pump errors		B.2.6
22	4	u32	air valve errors		decoded, not listed
26	4	u32	oil valve errors		decoded, not listed
30	4	u32	GPS errors		decoded, not listed
34	4	u32	CTD errors		B.2.6
38	4	u32	satellite (Iridium) errors		decoded, not listed
42	4	u32	case pressure (vacuum) errors		
46	4	u32	battery errors		
50	4	u32	BIT start time	`s1970`	
54	4	u32	BIT end time	`s1970`	duration = end − start
58	4	f32	sea pressure before calibration	dbar	
62	4	f32	new sea pressure offset	dbar	
66	4	f32	sea pressure after calibration	dbar	
70	4	f32	temperature	°C	
74	4	f32	salinity	PSU	
78	4	char[4]	CTD status		decoded as `u32`, listed in hex (e.g. 0x64697361 = “disa”)
82	4	f32	case vacuum at start	inHg	
86	4	f32	case vacuum at end	inHg	
90	4	f32	battery voltage at start	V	
94	4	f32	battery voltage at end	V	
98	4	f32	air valve open position	V	
102	4	f32	air valve open calibration position	V	
106	4	f32	air valve closed position	V	
110	4	f32	air valve closed calibration position	V	
114	4	f32	air valve middle position	V	
118	4	f32	air valve middle calibration position	V	
122	2	u16	air pump test duration	s	
124	1	u8	air pump test cycle count		
125	4	f32	maximum air pressure	psi (decoder)	10.5.0 blob list: “inHg”
129	4	f32	maximum air pump current	A	
133	4	f32	average air pump current	A	
137	4	f32	maximum air pump 12 V current	A	
141	4	f32	maximum battery voltage during air pump test	V	
145	4	f32	oil valve open position	V	
149	4	f32	oil valve open calibration position	V	
153	4	f32	oil valve closed position	V	
157	4	f32	oil valve closed calibration position	V	
161	2	u16	oil pump test duration	s	
163	4	f32	maximum oil pump current	A	
167	4	f32	average oil pump current	A	
171	4	f32	maximum oil pump 12 V current	A	

Offset	Size	Type	Field	Scale/units	Notes
175	4	f32	maximum battery voltage during oil pump test	V	
179	76	f32 × 19	thresholds		B.2.8; not decoded
255	1	char	;		

The 10.5.0 blob list gives “divisor” values (1000, 100) for many of these floats; transmitted values are already in engineering units (for example battery 15.51 V, case vacuum 6.07 inHg, air valve 1.274 V in a 10.12.0.1 BIT), and the decoder uses them unscaled.

B.2.5 BIT format 11 (264 or 256 bytes)

Format 11 adds the traveler errors at offset 50; every later field moves 4 bytes on. The 10.12.0 definition also adds a twentieth threshold (oil pump current low warning), giving 264 bytes.

Offset	Size	Type	Field	Scale/units	Notes
0–49			as formats 9/10 (ID through battery errors)		
50	4	u32	traveler errors		added in 10.8; B.2.6; decoded, not listed
54	4	u32	BIT start time	`s1970`	
58	4	u32	BIT end time	`s1970`	
62	4	f32	sea pressure before calibration	dbar	
66	4	f32	new sea pressure offset	dbar	
70	4	f32	sea pressure after calibration	dbar	
74	4	f32	temperature	°C	
78	4	f32	salinity	PSU	
82	4	char[4]	CTD status		
86	4	f32	case vacuum at start	inHg	
90	4	f32	case vacuum at end	inHg	
94	4	f32	battery voltage at start	V	
98	4	f32	battery voltage at end	V	
102	4	f32	air valve open position	V	
106	4	f32	air valve open calibration position	V	
110	4	f32	air valve closed position	V	
114	4	f32	air valve closed calibration position	V	
118	4	f32	air valve middle position	V	
122	4	f32	air valve middle calibration position	V	
126	2	u16	air pump test duration	s	
128	1	u8	air pump test cycle count		
129	4	f32	maximum air pressure	psi	
133	4	f32	maximum air pump current	A	
137	4	f32	average air pump current	A	
141	4	f32	maximum air pump 12 V current	A	10.12.0 list names this “MaxAirPumpCurrent” (see B.10)
145	4	f32	maximum battery voltage during air pump test	V	10.12.0 list: “MaxAirV12Current”
149	4	f32	oil valve open position	V	
153	4	f32	oil valve open calibration position	V	
157	4	f32	oil valve closed position	V	
161	4	f32	oil valve closed calibration position	V	
165	2	u16	oil pump test duration	s	
167	4	f32	maximum oil pump current	A	
171	4	f32	average oil pump current	A	
175	4	f32	maximum oil pump 12 V current	A	10.12.0 list: “MaxOilPumpCurrent”
179	4	f32	maximum battery voltage during oil pump test	V	10.12.0 list: “MaxOilV12Current”
183	80	f32 × 20	thresholds (264-byte form)		B.2.8; not decoded
263	1	char	;		264-byte form

The decoder also accepts a 256-byte format 11. No document describes it. With the fields above ending at offset 182 and the terminator at 255, it would have room for 18 thresholds (183–254), one fewer than the 19 of format 9; *inferred*, unverified. The decoder reads nothing beyond offset 182 in either form.

B.2.6 BIT bitfields

BIT results mask (all formats; offset 8 in formats 6–8, offset 10 in 9–11). The spreadsheet gives bits 0x01–0x40 (`BIST_RESULT_*`); 10.5.0 and 10.12.0 add the rest.

Bit	Name (spreadsheet / 10.5.0)	10.12.0 name
0x00000000	`BIST_RESULT_OK`	
0x00000001	air pump failure	AirPumpErrors
0x00000002	oil pump failure	OilPumpErrors
0x00000004	air valve failure	AirValveErrors
0x00000008	oil valve failure	OilValveErrors
0x00000010	GPS failure	GPSErrors
0x00000020	CTD sensor failure	CTDSensorErrors
0x00000040	satellite failure	SatModemErrors
0x00000080	case pressure failure (10.5.0)	CasePressureErrors
0x00000100	battery voltage failure (10.5.0)	BatteryVoltageErrors
0x00000200	accelerometer failure (10.5.0 only)	—
0x00000400	undocumented	—

Bit 0x400 is set in a format 11 BIT failure from firmware 10.12.0.1 (results 0x00000401); it is *inferred* to flag the traveler errors added in format 11.

Per-subsystem error bitfields (formats 9–11):

Field	Bit	10.5.0 meaning	10.12.0 meaning
air pump errors	0x01	air pump sensor pressure low	lowPressure
	0x02	air pump current high	hiCurrent
	0x04	air pump cycle count	hiCycleCount
oil pump errors	0x01	oil pump current low	loCurrent
	0x02	oil pump current high	hiCurrentWarning
	0x04	oil pump current high	hiCurrentError
air valve errors	0x01	open fail	OpenFail
	0x02	close fail	CloseFail
	0x04	middle fail	MiddleFail
oil valve errors	0x01	open fail	OpenFail
	0x02	close fail	CloseFail
GPS errors	0x01	GPS communication failure	NoDatafromGPS
	0x02	GPS no fix	FixAttempt1Failed
	0x04	GPS failure	FixAttempt2Failed
	0x08	—	OtherError
CTD errors	0x01	unable to start profile	UnableToStartProfile
	0x02	unable to communicate	NoDataFromCTD
	0x04	unable to profile	UnabletoRetrieveProfile
	0x08	bad data	FieldsOutsideThresholds
satellite errors	0x01	communication failure	NoDataFromModem
	0x02	first transmit failure	TransmitFailure
case pressure errors	0x01	case pressure low	CasePressureLow
	0x02	case pressure high	CasePressureHigh
battery errors	0x01	battery voltage low	LowVoltageWarning
	0x02	battery voltage low	LowVoltageError
traveler errors (format 11)	0x01	—	MissingPumpType
	0x02	—	MissingBatteryType

B.2.7 BIT enumerations (formats 9–11)

Field (offset)	Value	Meaning	Decoder
BIT type (8)	0	no sky view	`XAlamoBIT2Type::NO_SKY_VIEW`
	1	sky view	`SKY_VIEW`
	0xFF (−1)	unknown	`UNKNOWN`
CTD type (9)	0	RBR	listed “RBR”
	1	Sea-Bird	listed “SBE”
	other	—	listed “unknown”

The CTD type uses the same code as the mission configuration message (0xD0 byte 9).

B.2.8 BIT thresholds (formats 9–11; not decoded)

Each threshold is an `f32`. The decoder skips them.

Format 9/10 offset	Format 11 (264) offset	Threshold	Units
179	183	case vacuum minimum	inHg
183	187	case vacuum maximum	inHg
187	191	battery voltage low warning	V
191	195	battery voltage low	V
195	199	oil pump current high warning	A
199	203	oil pump current high	A
—	207	oil pump current low warning	A
203	211	oil pump current low	A
207	215	air pump maximum cycle count	
211	219	air pump sensor pressure high	(10.5.0: “inHg”)
215	223	air pump sensor pressure low	(10.5.0: “inHg”)
219	227	air pump current high	A
223	231	CTD pressure maximum	dbar
227	235	CTD pressure minimum	dbar
231	239	CTD temperature maximum	°C (10.5.0 says “dBar”)
235	243	CTD temperature minimum	°C
239	247	CTD salinity maximum	PSU
243	251	CTD salinity minimum	PSU
247	255	CTD pressure offset maximum	dbar
251	259	CTD pressure offset minimum	dbar

B.2.9 Decoding notes

- **Size checks.** Formats 6, 7 and 8 must be exactly 13, 72 and 76 bytes. For formats 9–11 the build sets `XMSG_DEBUG_BIST` to true, so a wrong size (anything but 256 for 9/10, 256 or 264 for 11) only logs a warning and the fields are read anyway; the debug build also writes the full listing to the debug log. (The format 9/10 warning text says “Expected 260 bytes” although the check is for 256.) *(Fixed in build 3309.)* In build 3309, format 9/10 messages shorter than 180 bytes and format 11 messages shorter than 184 bytes are rejected, because decoding would read past their end.
- **CTD counts** (formats 7, 8) are converted with the profile gains/offsets from 0xF0 and range-checked; the 7.3.5 YAML says the format 8 pressures use the *surface* gains.
- **Pressure offset** (format 8, offset 16) is read unsigned and divided by 1000; a negative offset (signed per the spreadsheet and YAML) would list as about 65.5 dbar.
- **Fill values**: formats 6–8 start every `u16` at 0xFFFF, the status at 0xFFFFFF and the CTD status at 0xFFFFFFFF; `XAlamoBit2` does not initialize its fields (all are set by every format it accepts).

B.2.10 Listing output

Formats 6–8 (`XAlamoBit`). Line type `E05` (passed) or `E06` (failed). `midx` 0 carries `BITVer` (decimal), `SWVer` (`0x%08X`) and `BITResults` (`0x%08X`). Format 6 has only `midx` 0.

midx	Format 7 columns	Format 8 columns
0	BITVer, SWVer, BITResults	same
1	BITTime (s); pres (dbar, counts); temp (°C, counts); saln (PSU, counts)	BITTime (s), CTDStat (hex)
2	CTDStat (hex); VacStart, VacEnd (inHg); BatStart, BatEnd (V)	presPostReset (dbar, counts); presPreReset (dbar, counts); presOffset (dbar)
3	air valve: AVOpenPos, AVOpenPosCal, AVMidPos, AVMidPosCal, AVClosePos, AVClosePosCal (V)	temp (°C, counts); saln (PSU, counts)
4	air pump: APTestTime (s), APCycleCount, APMxAPres (psi), APMxLoadCur, APMxLoadCur12V (A), APMxLoadBatV (V)	VacStart, VacEnd (inHg); BatStart, BatEnd (V)
5	oil valve: OVOpenPos, OVOpenPosCal, OVClosePos, OVClosePosCal (V)	air valve (as format 7 midx 3)
6	oil pump: OPTestTime (s), OPMxLoadCur, OPMxLoadCur12V (A), OPMxLoadBatV (V)	air pump (as format 7 midx 4)
7	—	oil valve (as format 7 midx 5)
8	—	oil pump (as format 7 midx 6)

The format 8 `presOffset` column uses `%15.3d` on a floating-point value; boost::format prints it with three significant digits.

Formats 9–11 (`XAlamoBit2`). Fourteen lines, each with `BITVer` after `midx`:

midx	Columns
0	FWVer (hex), BITResults (hex), sensorType, `# sensorTypeName`
1	startTime, endTime (ISO 8601), duration (s)
2	CTD errors (hex), CTD status (hex), presPreCal, presOffset, presPostCal (dbar)
3	temp (°C), saln (PSU)
4	case pressure errors (hex), caseVacStart, caseVacEnd (inHg)
5	battery errors (hex), batVoltStart, batVoltEnd (V)
6	air valve positions: open, mid, closed (V)
7	air valve calibration positions: open, mid, closed (V)
8	air pump errors (hex), duration (s), cycles, presMax (psi)
9	air pump currAvg, currMax, 12VCurrMax (A), batMax (V), powerMax (W) = 12 × 12VCurrMax
10	oil valve positions: open, closed (V)
11	oil valve calibration positions: open, closed (V)
12	oil pump errors (hex), duration (s)
13	oil pump currAvg, currMax, 12VCurrMax (A), batMax (V), powerMax (W) = 12 × 12VCurrMax

Not listed: BIT type, air/oil valve errors, GPS, satellite and traveler errors, thresholds.

B.3 Descent engineering: 0xE7

Purpose. Sent once per dive cycle, describing the descent from the surface to the drift/park pressure: case vacuum, valve positions, and the seek (settle-to-target) results. Introduced with X-format 2.1; ALAMO/ALTO only.

Decoder: `XEngDescent`, minimum 22 bytes. Versions 0x10 and 0x11.

B.3.1 Version 0x10 (23 or 27 bytes)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xE7		
1	2	u16	size = 23 or 27	bytes	
3	1	u8	version = 0x10		
4	4	u32	descent start time	`s2000`	
8	2	u16	cumulative pump run time during descent	s	
10	2	u16	pump current	mA (÷1000 → A)	
12	2	u16	system current	mA (÷1000 → A)	
14	2	u16	system voltage	mV (÷1000 → V)	27-byte form only

Offset	Size	Type	Field	Scale/units	Notes
16	2	u16	case vacuum	0.01 inHg	27-byte form only
18	2	i16	first-pump run-time adjustment	s	27-byte form only
20	6	—	not decoded		27-byte form
26	1	char	;		27-byte form (23-byte form: bytes 14–21 not decoded, ; at 22)

Version 0x10 is not in the v2.4 spreadsheet; this table is the decoder's reading, and the undecoded bytes are undocumented.

B.3.2 Version 0x11 (22 bytes)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xE7		
1	2	u16	size = 22	bytes	
3	1	u8	version = 0x11		10.12.0: `packetVersion = 17`
4	4	u32	descent start time	`s2000`	
8	2	u16	case vacuum at start	÷100 → inHg	10.5.0 blob list: ÷1000
10	2	u16	case vacuum at end	÷100 → inHg	10.5.0 blob list: ÷1000
12	2	u16	oil valve open position at surface	÷1000 → V	
14	2	u16	oil valve closed position during fall pump	÷1000 → V	
16	2	u16	air valve middle position at surface	÷1000 → V	
18	1	u8	seek attempts		
19	2	i16	seek pump time adjustment	s	10.12.0 list: `U2`
21	1	char	;		

B.3.3 Decoding notes and listing

Exact sizes are required (23/27 for 0x10, 22 for 0x11); other versions are rejected. Unset fields keep the fill values. There is no error list.

- 0x10: one `E07` line (`midx` 0): engVer, start-of-descent, pumpT (s), pumpI, sysI (A), sysV (V), vac (inHg), pumpAdj (`%+d` s).
- 0x11: `midx` 0: engVer, start-of-descent, vac0, vac1 (inHg), seeks, seekPumpAdjT (`%+d` s); `midx` 1 ("Valve Position Measurements"): OilValveOpenPosSurf, OilValveClosePosFallPump, AirValveMidPosSurf (V).

B.4 Drift/park engineering: 0xE8

Purpose. Sent once per dive cycle for the drift (park) phase: case vacuum, the average CTD values over the drift, and the seek results. Versions before 0x12 also carried the start time and pressure of the fall to profile depth, which moved to the ascent message (0xE9 v0x12).

Decoder: `XEngDrift`, minimum 19 bytes. Versions 0x10, 0x11, 0x12.

B.4.1 Version 0x10 (19 bytes)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xE8		
1	2	u16	size = 19	bytes	
3	1	u8	version = 0x10		
4	4	u32	drift start time	`s2000`	
8	2	u16	average drift pressure	CTD counts	
10	2	u16	average drift temperature	CTD counts	
12	4	u32	fall-to-profile start time	`s2000`	

Offset	Size	Type	Field	Scale/units	Notes
16	2	u16	fall-to-profile start pressure	CTD counts	
18	1	char	;		

B.4.2 Version 0x11 (28 bytes)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xE8		
1	2	u16	size = 28	bytes	
3	1	u8	version = 0x11		
4	4	u32	drift start time	`s2000`	
8	2	u16	case vacuum at start	÷100 → inHg	
10	2	u16	case vacuum at end	÷100 → inHg	
12	2	u16	average pressure	CTD counts	
14	2	u16	average temperature	CTD counts	
16	2	u16	average salinity	CTD counts	
18	1	u8	seek attempts		
19	2	i16	seek pump time adjustment	s	decoder reads `u16` into an `int16_t` (same result)
21	4	u32	fall-to-profile start time	`s2000`	
25	2	u16	fall-to-profile start pressure	CTD counts	
27	1	char	;		

B.4.3 Version 0x12 (22 bytes)

Version 0x11 without the fall-to-profile fields.

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xE8		
1	2	u16	size = 22	bytes	
3	1	u8	version = 0x12		10.12.0 list says 17 (0x11); see B.10
4	4	u32	drift start time	`s2000`	
8	2	u16	case vacuum at start	÷100 → inHg	
10	2	u16	case vacuum at end	÷100 → inHg	
12	2	u16	average pressure	CTD counts	
14	2	u16	average temperature	CTD counts	
16	2	u16	average salinity	CTD counts	
18	1	u8	seek attempts		
19	2	i16	seek pump time adjustment	s	10.12.0 list: `U2`
21	1	char	;		

B.4.4 Decoding notes and listing

Exact sizes are required (19, 28, 22); other versions are rejected. No error list.

- 0x10: one `E08` line: engVer, start-of-drift, meanP (dbar), meanT (°C), start-of-fall, fallP (dbar).
- 0x11 and 0x12: `midx` 0: engVer, start-of-drift, vac0, vac1 (inHg), seeks, seekPumpAdjT (`%+d` s); `midx` 1 (“Drift Sensor Averages”): pres (dbar, counts), temp (°C, counts), saln (PSU, counts). The 0x11 fall-to-profile fields are decoded but not listed.

Example (README, `lis9287-0325.txt`):

```
E08 09287  325    0   0x12 2026-10-24T11:05:29Z      6.65      6.69              0             +0
E08 09287  325    1          274.72         7118        9.744        14744       34.523        35523
```

B.5 Ascent engineering: 0xE9

Purpose. Sent once per dive cycle for the ascent (profile) phase. From version 0x12 it also carries the start time and pressure of the fall from park to profile depth.

Decoder: `XEngAscent`, minimum 13 bytes. Versions 0x10, 0x11, 0x12.

B.5.1 Version 0x10 (13 bytes)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xE9		
1	2	u16	size = 13	bytes	
3	1	u8	version = 0x10		
4	4	u32	ascent start time	`s2000`	
8	2	u16	system current	mA (÷1000 → A)	
10	2	u16	cumulative pump run time during ascent	s	
12	1	char	;		

B.5.2 Version 0x11 (13 bytes)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xE9		
1	2	u16	size = 13	bytes	
3	1	u8	version = 0x11		
4	4	u32	ascent start time	`s2000`	spreadsheet text says “drift/park phase” (copy error)
8	2	u16	case vacuum at start	÷100 → inHg	
10	2	u16	case vacuum at end	÷100 → inHg	
12	1	char	;		

B.5.3 Version 0x12 (19 bytes)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xE9		
1	2	u16	size = 19	bytes	
3	1	u8	version = 0x12		10.12.0 list says 17 (0x11); see B.10
4	4	u32	ascent start time	`s2000`	
8	2	u16	case vacuum at start	÷100 → inHg	
10	2	u16	case vacuum at end	÷100 → inHg	
12	4	u32	fall-from-park start time	`s2000`	
16	2	u16	fall-from-park start pressure	CTD counts	
18	1	char	;		

B.5.4 Decoding notes and listing

Exact sizes are required (13, 13, 19); other versions are rejected. `XEngAscent` does not initialize its fields, so fields a version does not carry are undefined rather than fill values (none of them is listed). No error list.

- 0x10: `E09` `midx` 0: engVer, start-of-ascent, sysI (A), pumpT (s).
- 0x11: `midx` 0: engVer, start-of-ascent, vac0, vac1 (inHg).
- 0x12: `midx` 0 as 0x11; `midx` 1: start-of-fall, fallP (dbar).

B.6 Surface engineering: 0xEA

Purpose. Sent once per dive cycle from the surface: Iridium session statistics, case vacuum, air pump and valve measurements, the surface pressure offset reading, profile bookkeeping (first/last raw CTD sample, bin counts), the GPS–RTC clock difference, and the list of system errors recorded during the dive cycle.

Decoder: `XEngSurface`, minimum 46 bytes. Versions 0x10–0x14.

B.6.1 Error list

Every version ends with an error count `N` (`u8`) followed by `N` entries and the terminator.

Version	Entry size	Entry layout	First entry at
0x10	4	`u32` error code (no timestamp)	47
0x11–0x14	6	`u32` time (`s2000`), then `u16` error code (B.9)	45 / 49 / 67 / 75

The message size is therefore `first + entry × N + 1`. The decoder reads entries while the next entry ends before the terminator, then requires the number read to equal `N` (`XMSG_DEBUG_ENG_SURF` is false, so a mismatch rejects the message).

B.6.2 Version 0x10 (48 + 4N bytes)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xEA		
1	2	u16	size = 48 + 4N	bytes	
3	1	u8	version = 0x10		
4	4	u32	surface start time	`s2000`	
8	4	u32	CTD status		listed hex
12	2	u16	SBD packets for this dive		
14	2	u16	transmission attempts, previous dive		
16	2	u16	case vacuum	÷100 → inHg	
18	2	u16	battery voltage	mV (÷1000 → V)	
20	2	u16	system current	mA (÷1000 → A)	
22	24	—	not decoded		undocumented
46	1	u8	error count N		
47	4N	u32 × N	error codes		no timestamps
47+4N	1	char	;		

Version 0x10 is not in the v2.4 spreadsheet; the table is the decoder's reading.

B.6.3 Version 0x11 (46 + 6N bytes)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xEA		
1	2	u16	size = 46 + 6N	bytes	
3	1	u8	version = 0x11		
4	4	u32	surface start time	`s2000`	
8	2	u16	case vacuum at start	÷100 → inHg	
10	2	u16	case vacuum at end	÷100 → inHg	
12	4	u32	CTD status		
16	2	u16	packets to transmit this session		
18	2	u16	transmission attempts, previous dive		
20	2	u16	air pump current	÷1000 → A	
22	2	u16	air pump run time	s	
24	2	u16	air bladder maximum pressure	÷1000 → psi	
26	2	u16	air valve open position at start	÷1000 → V	

Offset	Size	Type	Field	Scale/units	Notes
28	2	u16	air valve middle position at end	÷1000 → V	
30	2	u16	oil valve open position during air assist	÷1000 → V	
32	2	u16	oil valve closed position at end	÷1000 → V	
34	2	u16	air valve calibration open position	÷1000 → V	
36	2	u16	air valve calibration closed position	÷1000 → V	
38	2	u16	air valve calibration middle position	÷1000 → V	
40	2	u16	oil valve calibration open position	÷1000 → V	
42	2	u16	oil valve calibration closed position	÷1000 → V	
44	1	u8	error count N		
45	6N		error entries (B.6.1)		
45+6N	1	char	;		

B.6.4 Version 0x12 (50 + 6N bytes)

Version 0x11 with an unidentified 2-byte field after the case vacuum and the pressure offset reading before the error count.

Offset	Size	Type	Field	Scale/units	Notes
0–11			as 0x11 (ID … case vacuum at end)		version = 0x12
12	2	u16	unidentified (“???” in the spreadsheet, typical value 0)		not decoded
14	4	u32	CTD status		
18	2	u16	packets to transmit this session		
20	2	u16	transmission attempts, previous dive		
22	2	u16	air pump current	÷1000 → A	
24	2	u16	air pump run time	s	
26	2	u16	air bladder maximum pressure	÷1000 → psi	
28	2	u16	air valve open position at start	÷1000 → V	
30	2	u16	air valve middle position at end	÷1000 → V	
32	2	u16	oil valve open position during air assist	÷1000 → V	
34	2	u16	oil valve closed position at end	÷1000 → V	
36	2	u16	air valve calibration open position	÷1000 → V	
38	2	u16	air valve calibration closed position	÷1000 → V	
40	2	u16	air valve calibration middle position	÷1000 → V	
42	2	u16	oil valve calibration open position	÷1000 → V	
44	2	u16	oil valve calibration closed position	÷1000 → V	
46	2	i16	pressure sensor offset reading	÷1000 → dbar	
48	1	u8	error count N		
49	6N		error entries (B.6.1)		
49+6N	1	char	;		

The spreadsheet gives the size as “6N+49”, but its own terminator offset (6N+49) makes it 6N+50, which is what the decoder requires as a minimum.

B.6.5 Version 0x13 (68 + 6N bytes)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xEA		
1	2	u16	size = 68 + 6N	bytes	
3	1	u8	version = 0x13		
4	4	u32	surface start time	`s2000`	
8	2	u16	packets to transmit this session		
10	2	u16	transmission attempts, previous dive		
12	2	u16	case vacuum at start	÷100 → inHg	
14	2	u16	case vacuum at end	÷100 → inHg	
16	2	i16	GPS − RTC time difference before synchronization	s	

Offset	Size	Type	Field	Scale/units	Notes
18	4	u32	CTD status		firmware lists: 4 characters
22	2	u16	first raw profile pressure	CTD counts	
24	2	u16	first raw profile temperature	CTD counts	
26	2	u16	first raw profile salinity	CTD counts	
28	2	u16	last raw profile pressure	CTD counts	
30	2	u16	last raw profile temperature	CTD counts	
32	2	u16	last raw profile salinity	CTD counts	
34	2	u16	number of profile bins		
36	2	u16	spreadsheet: number of raw profile samples		**decoder: number of empty bins**
38	2	u16	spreadsheet: number of empty bins		**decoder: number of raw samples**
40	2	u16	air pump current	÷1000 → A	
42	2	u16	air pump run time	s	
44	2	u16	air bladder maximum pressure	÷1000 → psi	
46	2	u16	air valve open position at start	÷1000 → V	
48	2	u16	air valve middle position at end	÷1000 → V	
50	2	u16	oil valve open position during air assist	÷1000 → V	
52	2	u16	oil valve closed position at end	÷1000 → V	
54	2	u16	air valve calibration open position	÷1000 → V	
56	2	u16	air valve calibration closed position	÷1000 → V	
58	2	u16	air valve calibration middle position	÷1000 → V	
60	2	u16	oil valve calibration open position	÷1000 → V	
62	2	u16	oil valve calibration closed position	÷1000 → V	
64	2	i16	pressure sensor offset reading	÷1000 → dbar	
66	1	u8	error count N		
67	6N		error entries (B.6.1)		
67+6N	1	char	;		

B.6.6 Version 0x14 (76 + 6N bytes)

Version 0x13 with the surface data-collection (“packetize”) start and end times inserted after the start time, and the clock/packet fields reordered. Firmware 7.3.5, 10.5.0 and 10.12.0 (`packetVersion = 20`) all use this layout.

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xEA		
1	2	u16	size = 76 + 6N	bytes	
3	1	u8	version = 0x14		
4	4	u32	surface start time	`s2000`	
8	4	u32	surface data collection (packetize) start time	`s2000`	
12	4	u32	surface data collection (packetize) end time	`s2000`	
16	2	u16	packets this dive/mission		
18	2	u16	transmission attempts, previous dive		
20	2	u16	case vacuum at start	÷100 → inHg	
22	2	u16	case vacuum at end	÷100 → inHg	
24	2	i16	GPS − RTC time difference	s	
26	4	char[4]	CTD status		decoded as `u32`
30	2	u16	first raw profile pressure	CTD counts	
32	2	u16	first raw profile temperature	CTD counts	
34	2	u16	first raw profile salinity	CTD counts	
36	2	u16	last raw profile pressure	CTD counts	
38	2	u16	last raw profile temperature	CTD counts	
40	2	u16	last raw profile salinity	CTD counts	
42	2	u16	number of CTD profile bins		
44	2	u16	firmware lists: number of CTD profile samples		**decoder: number of empty bins**
46	2	u16	firmware lists: number of empty bins		**decoder: number of raw samples**
48	2	u16	air pump current	÷1000 → A	

Offset	Size	Type	Field	Scale/units	Notes
50	2	u16	air pump run time	s	
52	2	u16	air bladder maximum pressure	÷1000 → psi	
54	2	u16	air valve open position at start	÷1000 → V	
56	2	u16	air valve middle position at end	÷1000 → V	
58	2	u16	oil valve open position during air assist	÷1000 → V	
60	2	u16	oil valve closed position at end	÷1000 → V	
62	2	u16	air valve calibration open position	÷1000 → V	
64	2	u16	air valve calibration closed position	÷1000 → V	
66	2	u16	air valve calibration middle position	÷1000 → V	
68	2	u16	oil valve calibration open position	÷1000 → V	
70	2	u16	oil valve calibration closed position	÷1000 → V	
72	2	i16	pressure sensor offset reading	÷1000 → dbar	10.5.0/10.12.0 lists: `U2`; 7.3.5 YAML: `i16`
74	1	u8	error count N		
75	6N		error entries (B.6.1)		10.5.0 lists room for 10 entries
75+6N	1	char	;		

B.6.7 Decoding notes

- Minimum sizes: 48 (0x10), 46 (0x11), 50 (0x12), 68 (0x13), 76 (0x14); longer messages carry error entries. Unknown versions are rejected.
- The pressure offset reading is listed with `XMsgFormatPressure()` (two decimals, fill −9.999 if outside −5…2610 dbar).
- First/last raw samples are converted with the profile gains/offsets and range-checked.
- Error codes are stored as 16-bit values (32-bit in 0x10) and listed as `0x%08X`, with no name lookup; B.9 gives the names. A zero timestamp lists as `0000-00-00T00:00:00Z`.

B.6.8 Listing output (`E0A`, `e0A`)

midx	0x10	0x11	0x12	0x13	0x14
0	engVer, start-of-surface, CTD-status, nSBD, nAtmpt, vac, sysV, sysI, nErr	engVer, start-of-surface, vac0, vac1, nSBD, nPrevAttempt	same as 0x11	same	same
1	—	air pump: APCur (A), APRunT (s), AirBladMxP (psi)	same	same	same
2	—	air valve: AVOpenStart, AVMidEnd, AVOpenCal, AVMidCal, AVCloseCal (V)	same	same	same
3	—	oil valve: OVOpenAirPmp, OVCloseEnd, OVOpenCal, OVClosCal (V)	same	same	same
4	—	`e0A` nErr	presOffset (dbar)	presOffset	presOffset
5	—	—	`e0A` nErr	first sample: pres (dbar, counts), temp (°C, counts), saln (PSU, counts)	same
6	—	—	—	last sample (same columns)	same
7	—	—	—	numBins, numEmptyBins, numRawSamples	same
8	—	—	—	gpsOffset (s)	gpsOffset
9	—	—	—	`E0A` nErr	surface sampling start, end
10	—	—	—	—	`E0A` nErr

When there are errors, a heading and one `e0A` line per entry follow: `e0A SN dive idx date-time 0x%08X` (`idx` counts from 0; version 0x10 prints `0000-00-00T00:00:00Z` for the missing time). The `nErr` line is tagged `e0A` in versions 0x11 and 0x12 but `E0A` in 0x13 and 0x14. The CTD status is listed only by version 0x10; the 0x11 fields `nSBD`/`nPrevAttempt` are printed for all later versions.

(Fixed in build 3309.) In version 0x10 the error heading's format string had five fields but was given four values; boost::format throws in that case, so a 0x10 message with errors is expected to fail listing generation (*inferred from the code*; not observed).

B.7 Ice-detection engineering: 0xEC

Purpose. Sent by floats with ice detection enabled, once per dive cycle. It reports the ice-avoidance mode, what happened on the way up (surfaced, ice found, temperature turnaround), how many surfacing attempts were made, how much data is waiting to be sent, and the values used by the temperature-threshold and surface-salinity tests. Version 0x11 adds an error list.

Decoder: `XEngIce`, minimum 21 bytes. Versions 0x10 and 0x11.

B.7.1 Version 0x10 (21 bytes)

Offset	Size	Type	Field	Scale/units	Notes
0	1	u8	ID = 0xEC		
1	2	u16	size = 21	bytes	
3	1	u8	version = 0x10		
4	1	u8	operation mode		B.7.3
5	1	u8	ice detection status		B.7.3
6	1	u8	urgency status		B.7.3
7	1	u8	surface attempts this dive cycle		
8	2	u16	stored SBD messages (packets)		10.5.0: "maximum stored packets"
10	4	u32	time since last successful transmission	s	a duration (the spreadsheet's "since 01 Jan 2000" is an error)
14	2	u16	minimum (shallowest) pressure during ice detection	CTD counts	
16	2	u16	threshold test temperature	see notes	spreadsheet: temperature gain/offset; 7.3.5 YAML and 10.5.0: ÷1000 → °C
18	2	u16	surface test salinity	CTD counts	valid only where the configuration measures salinity at the surface
20	1	char	;		

B.7.2 Version 0x11 (24 + 6N bytes)

Offset	Size	Type	Field	Scale/units	Notes
0–19			as 0x10		version = 0x11
20	2	u16	seek time adjustment (10.5.0 blob list)	s	decoder: undocumented, not decoded (observed 0–30)
22	1	u8	error count N		
23	6N		error entries: `u32` time (`s2000`), `u16` error code (B.9)		
23+6N	1	char	;		

The v2.4 spreadsheet and the 7.3.5 YAML have no field at 20–21: error count at 20, entries from 21, size 6N + 22 (the spreadsheet says "6N+21"). The decoder follows the 10.5.0 layout, which matches transmitted messages; a message in the spreadsheet layout would be mis-decoded.

B.7.3 Enumerations

Operation mode (offset 4): 0 summer, 1 winter.

Ice detection status (offset 5):

Value	v2.4 spreadsheet	10.5.0 blob list	SOLO-II 0xE2
0	ice detection disabled (should never be sent)	disabled	`ICE_CHECK_OFF`
1	ice not encountered during surface attempt	not found	`ICE_NOT_FOUND`
2	ice encountered during surface attempt	ice found (mixed layer)	`ICE_MIXLAYER`
3	temperature turnaround: threshold test failed, no further surfacing attempts	ice found (breakup)	`ICE_BREAKUP`
4	—	ice found (surface halt)	—
5	—	complete, not found	—

The meanings of 2 and 3 differ between the spreadsheet and the firmware list; value 5 occurs in transmitted messages (float 9213). The decoder lists the number only.

Urgency status (offset 6): 0 not triggered (disabled), 1 triggered (enabled).

B.7.4 Decoding notes and listing (`E0C`, `e0c`)

- Version 0x10 has no size check beyond the 21-byte minimum; 0x11 needs at least 24 bytes and an entry count equal to `N` (otherwise rejected). The start time is not set; the case-vacuum members exist in the class but no version fills them.
- `midx` 0: engVer, mode. `midx` 1 (“Status”): ice-status, urgency, surfAttempts, numStoredSBD, lastXmit (s). `midx` 2: minPres (dbar, from the profile pressure gain/offset; there is no zero-gain guard), testTemp, testSalinity. Version 0x11 adds `midx` 3: nErr (`E0C`), then a heading and one `e0c SN dive idx date-time 0x%08X` line per error.
- `testTemp` is printed as the raw `u16` although the heading says (degC); per 7.3.5/10.5.0 it is °C × 1000.
- `testSalinity` is converted with the salinity gain/offset and range-checked, but printed with `%15.3d`, which boost::format renders with three significant digits: the fill value −9.999 appears as `-10`.

Example (float 9213, dive 8, version 0x11):

```
E0C 09213    8    0   0x11        1
E0C 09213    8    1               5              0               3             29           96781
E0C 09213    8    2            0.16              0             -10
E0C 09213    8    3               1
e0c 09213    8    0 2026-09-29T00:56:40Z      0x00000114
```

B.8 Version summary

ID	Versions decoded	Size(s)	Documented in
0xE0, 0xE2, 0xE3	none (validated only)	—	SOLO-II v2.0 (version 5)
0xE5/0xE6	6, 7, 8, 9, 10, 11	13, 72, 76, 256, 256, 256/264	spreadsheet (7, 8); 7.3.5 (8); 10.5.0 (9); 10.12.0 (layout of 11)
0xE7	0x10, 0x11	23/27, 22	spreadsheet, 7.3.5, 10.5.0, 10.12.0 (0x11)
0xE8	0x10, 0x11, 0x12	19, 28, 22	spreadsheet (0x11); 7.3.5, 10.5.0, 10.12.0 (0x12 layout)
0xE9	0x10, 0x11, 0x12	13, 13, 19	spreadsheet (0x11); 7.3.5, 10.5.0, 10.12.0 (0x12 layout)
0xEA	0x10–0x14	48+4N, 46+6N, 50+6N, 68+6N, 76+6N	spreadsheet (0x11–0x13); 7.3.5, 10.5.0, 10.12.0 (0x14)
0xEC	0x10, 0x11	21, 24+6N	spreadsheet (0x10, 0x11); 10.5.0 (0x11)

B.9 System error codes

The 0xEA (versions 0x10–0x14) and 0xEC (version 0x11) error lists carry ALAMO firmware system error codes. A code is 16 bits on the wire (32 bits in 0xEA v0x10; the firmware enum `ASSERT_LOG_ERROR_*` is 32 bits wide). The **low byte** names the module or mission state and the **high byte** the specific error within it; a high byte of 0 is the generic error of that module. For example, 0x010C is module 0x0C (CTD), error 0x01 (pressure out of range), and 0x0114 is module 0x14 (ascend), error 0x01 (CTD).

The decoder does not translate codes: listings print them as `0x%08X` (e.g. `0x00000114`), and the Python tools pass the number through. The table merges `system-error-codes.txt` (names with the `ASSERT_LOG_ERROR_` prefix) and Appendix 2 of `nextgen_v10.5.0_blobs.txt` (`ERROR_` prefix).

Code	Name (`ERROR_…`)	Module
0x0000	NONE	—
0x0001	ACMP	analog comparator
0x0101	ACMP_XXX	
0x0201	ACMP_YYY	
0x0002	ADC	ADC
0x0003	GPIO	GPIO
0x0004	I2C	I²C
0x0104	I2C_NACK	
0x0204	I2C_HW_FAULT	
0x0005	PWR	power
0x0006	RTC	real-time clock
0x0007	TIMER	timer
0x0008	UART	UART
0x000A	ACCEL	accelerometer
0x000B	BLUETOOTH	Bluetooth
0x000C	CTD	CTD
0x010C	CTD_DATA_RANGE_PRESSURE	
0x020C	CTD_DATA_RANGE_ TEMPERATURE	
0x030C	CTD_DATA_RANGE_SALINITY	
0x040C	CTD_DATA_RANGE_PAR	
0x050C	CTD_DATA_RANGE_TEMPSAL	
0x060C	CTD_DATA_RANGE	
0x070C	CTD_DATA_CONVERSION	
0x080C	CTD_RETRY	
0x000D	EEPROM	EEPROM
0x000E	GPS	GPS
0x000F	PUMP	pump
0x010F	PUMP_MAX_RUN_TIME	
0x0010	SATELLITE	Iridium modem
0x0110	SATELLITE_CONFIG	
0x0210	SATELLITE_CMD_TIMEOUT	
0x0310	SATELLITE_PWR	
0x0410	SATELLITE_DATA	
0x0011	VALVE	valve
0x0111	VALVE_CALIBRATION	
0x0211	VALVE_CONTROL	
0x0311	VALVE_TIMEOUT	
0x0012	MISSION	mission
0x0112	MISSION_CONFIG_DATA	
0x0212	MISSION_PHASE_CONFIG	
0x0013	DESCEND	descent state
0x0113	DESCEND_CTD	
0x0213	DESCEND_AIR_OPEN_VALVE	
0x0313	DESCEND_AIR_MID_VALVE	
0x0413	DESCEND_AIR_CLOSE_VALVE	
0x0513	DESCEND_OIL_OPEN_VALVE	
0x0613	DESCEND_OIL_CLOSE_VALVE	
0x0713	DESCEND_PUMP	

Code	Name (`ERROR_…`)	Module
0x0813	DESCEND_DATA	
0x0913	DESCEND_MAX_DEPTH	
0x0014	ASCEND	ascent state
0x0114	ASCEND_CTD	
0x0214	ASCEND_CTD_PROFILE	
0x0314	ASCEND_AIR_OPEN_VALVE	
0x0414	ASCEND_AIR_MID_VALVE	
0x0514	ASCEND_AIR_CLOSE_VALVE	
0x0614	ASCEND_OIL_OPEN_VALVE	
0x0714	ASCEND_OIL_CLOSE_VALVE	
0x0814	ASCEND_PUMP	
0x0914	ASCEND_DATA	
0x0A14	ASCEND_DEPTH	
0x0015	PARK	park/drift state
0x0115	PARK_CTD	
0x0215	PARK_CTD_INVALID_ PRESSURE	
0x0515	PARK_DATA	
0x0615	PARK_DEPTH_MIN	
0x0715	PARK_DEPTH_MAX	
0x0815	PARK_DATE_TIME_PAST	
0x0016	SURFACE	surface state
0x0116	SURFACE_CTD	
0x0216	SURFACE_CTD_PROFILE	
0x0316	SURFACE_CTD_OFFSET_ RESET	
0x0416	SURFACE_AIR_OPEN_VALVE	
0x0516	SURFACE_AIR_MID_VALVE	
0x0616	SURFACE_AIR_CLOSE_VALVE	
0x0716	SURFACE_OIL_OPEN_VALVE	
0x0816	SURFACE_OIL_CLOSE_VALVE	
0x0916	SURFACE_PUMP	
0x0A16	SURFACE_DATA	
0x0B16	SURFACE_DATA_MAX_LOOP	
0x0C16	SURFACE_GPS	
0x0D16	SURFACE_SATELLITE	
0x0E16	SURFACE_CFG_STORE	
0x0F16	SURFACE_REMOTE_CMD	
0x1016	SURFACE_BEACON_DEPTH	
0x2016	SURFACE_CTD_CRC	10.5.0 appendix only
0x0020	BIT	built-in test
0x0120	BIT_SATELLITE	
0x0030	MAIN	main loop
0x00FF	EXCEPTION	`system-error-codes.txt` only

No code is defined for module 0x09 or for 0x0315/0x0415 (park); a code not in the table should be reported as its module plus the raw high byte.

B.10 Discrepancies and open questions

Between the decoder and the documents:

1. **0xEA v0x13/v0x14 bin counts.** The decoder reads offset 36 (v0x13) / 44 (v0x14) as the number of *empty bins* and 38 / 46 as the number of *raw samples*. The v2.4 spreadsheet, the 7.3.5 YAML, the 10.5.0 list and the 10.12.0 list all give the opposite order (samples, then empty bins). Listings checked so far (floats 9213, 9216, 9287, 11186) have 0 in both fields, so data cannot settle it yet; the documents suggest the decoder's `numEmptyBins` and `numRawSamples` columns are swapped. *(Fixed in build 3309: the decoder now reads samples before empty bins, as the documents give them.)*

2. **0xEC v0x11 bytes 20–21.** Decoder and 10.5.0 list: a 2-byte field (10.5.0: “seek time adjustment”) before the error count at 22. Spreadsheet v2.4 and 7.3.5 YAML: error count at

   (t) The decoder matches transmitted messages; whether early-firmware 0x11 messages used the spreadsheet layout is unknown.

3. **Ice detection status values.** Spreadsheet: 0–3 (3 = temperature turnaround). 10.5.0: 0–5, with 2 = mixed layer, 3 = breakup, 4 = surface halt, 5 = complete, not found. Value 5 is seen in data. The firmware list is presumably current.
4. **0xEC threshold test temperature.** Spreadsheet: temperature counts (gain/offset). 7.3.5 and 10.5.0: °C × 1000. The decoder prints the raw number under a (degC) heading.
5. **BIT format 7/8 scales.** Air-pump maximum-load battery voltage: decoder ÷1000, spreadsheet and YAML ÷100. Oil-pump battery voltage: decoder and spreadsheet ÷1000, YAML ÷100. Oil pump test duration: spreadsheet gain 1000, decoder and YAML seconds.
6. **BIT format 8 signedness.** The pressure offset (and, per the YAML, both pressures) are signed; the decoder reads them unsigned.
7. **BIT format 8 in the spreadsheet.** The 0xE6 sheet gives a 16-bit CTD status and shifted offsets, and the 0xE5/0xE6 sheets swap the format 7/8 sizes (72/77); the 0xE5 size “77” conflicts with its own terminator at 75.
8. **BIT format 11 sizes.** The 10.12.0 list describes the format 11 layout (traveler errors, 20 thresholds, 264 bytes) but labels it `PacketVersion = 9`; a 10.12.0.1 float sends 11. The decoder also accepts a 256-byte format 11 that no document describes.
9. **BIT format 11 field names (10.12.0 list).** Offsets 141 and 175 are named “MaxAirPumpCurrent”/”MaxOilPumpCurrent” (duplicates) and 145/179 “Max…V12Current”; by comparison with 10.5.0 they are the 12 V current maximum and the battery voltage maximum, as the decoder reads them.
10. **BIT result bit 0x400** is set in real data and undocumented; bit 0x200 (accelerometer) is in 10.5.0 only.
11. **BIT format 9 “maximum air pressure”** is “inHg” in the 10.5.0 list; the decoder lists psi.
12. **E8/E9 version in 10.12.0.** The list gives `packetVersion = 17` (0x11) for 0xE8 and 0xE9 but the 0x12 layouts (22 and 19 bytes). If a float actually sent 0x11 with those sizes the decoder would reject it (it expects 28 and 13). A 10.12.0.1 float (11186) sends 0x12 for both, so the list is wrong on this point.
13. **Seek pump time adjustment type.** `I2` in 10.5.0 and 7.3.5, `U2` in 10.12.0; the decoder treats it as signed.
14. **0xE7 case vacuum scale.** 10.5.0 says ÷1000 for 0xE7 only (÷100 for 0xE8/0xE9 and in all other documents); the decoder uses ÷100.
15. **0xEA pressure offset reading** is `U2` in the 10.x lists, signed elsewhere; the decoder reads it signed.
16. **BIT format 9–11 “divisor” values** in the 10.5.0 list do not apply to the transmitted floats, which are already in engineering units.
17. **7.3.5 YAML** omits the version byte for 0xE7–0xEC (its field lists start at offset 4) and uses the surface gains for BIT pressures; the decoder uses the profile gains.
18. **Error-code tables.** `SURFACE_CTD_CRC` (0x2016) is only in the 10.5.0 appendix and `EXCEPTION` (0x00FF) only in `system-error-codes.txt`.
19. **Unexplained bytes.** 0xE7 v0x10 bytes 14–21 (23-byte form) / 20–25 (27-byte form), 0xEA v0x10 bytes 22–45 and 0xEA v0x12 bytes 12–13 are undocumented and not decoded.

Decoder behavior worth knowing (not format discrepancies):

- 0xEA v0x10 with errors: the error heading has more format fields than values, which makes boost::format throw during listing generation (inferred).
- 0xEA `nErr` lines are tagged `e0A` in v0x11/0x12 and `E0A` in v0x13/0x14.
- BIT formats 9–11 are accepted at any size (warning only) because `XMSG_DEBUG_BIST` is on.
- `XEngAscent`, `XEngIce` and `XAlamoBit2` leave some members uninitialized; the listings do not print those

members.

Part C — Mission configuration, command echo, waves and acoustics

This part covers these blob types:

ID	Name	Decoder class	Status in the XMTPS decoder
0xD0-0xD7	Mission configuration	`XMfaMissionCfg` (`XAlamoMission Configuration` is the unused older version)	D0 and D1 are decoded for 117 firmware versions. D2-D7 are rejected. SOLO-II ASCII dumps are not decoded.
0xDE	Mission command echo	`XMissionCmdEcho`	Decoded since build 3310; one `CMD` listing line per command.
0xC1, 0xC3	Ocean surface waves	`XWave`, `XWaveSpectralSample`, `XWaveRaw Sample`	Spectral versions 0-2 are decoded. The raw IMU format is not.
0xCA	Marginal-ice-zone (MIZ) acoustics	`XAcousticPosition`	Subtypes 3 and 5 are decoded. Subtypes 1 and 2 are seen in production but ignored.
0xFF	Extended records	`XExtendedRecord`	Decoded since build 3321; one `XFF` listing line per record. The format is inferred from production messages (C.6).
0xC5	WIMS configuration parameters	`XWimsParameters`	Decoded since build 3336; one `WIP` listing line per parameter. The format and most parameter names are inferred from production messages (C.7).

C.0 Conventions

- Offsets count from the blob's ID byte, which is offset 0. The common blob header is `ID` (offset 0, U1), `jj` (offsets 1-2, U2, the blob length including ID, `jj`, data and terminator) and the `;` (0x3B) terminator at offset `jj - 1`.
- Multi-byte integers are big-endian. Types: U1, U2 and U4 are unsigned 8-, 16- and 32-bit integers; I1, I2 and I4 are two's-complement signed; R4 is an IEEE-754 single-precision float sent as a big-endian 32-bit word (the decoder reads a U4 and bit-casts it, `f32()` in `XDefinitions.h`).
- **[inferred]** marks something worked out from production messages or decoder behavior, not from a format document. **[decoder]** marks behavior of the XMTPS implementation, not of the format itself.
- Sources:
  - *Decoder*: `src/data/xmsg/` in the XMTPS repository, as of 2026-10-25.
  - *XF2.4*: the ALAMO X-format spec v2.4 spreadsheet (`xfmt-format-spec-2.4-alamo_2018-05-15.xlsx`).
  - *SOLO2*: *Format of SOLOII X messages*, v2.0, 20 Mar 2014.
  - *NG10.5*: `nextgen_v10.5.0_blobs.txt`.
  - *NG10.12*: `nextgen-10.12.0-xmessage.txt`.
  - *DF7.3.5*: `DataFormat-v7.3.5.yaml`.
  - *Production*: hex dumps (`shx` lines) in production listings under `Z:\alamo\data\list`.
- Fill values. The decoder starts every mission parameter at an "unset" value: 255 (U1), 65535 (U2), 4294967295 (U4), -32768 (I2) or `FLT_MIN` (R4). A parameter that a firmware layout doesn't contain is listed with that value.

C.1 Mission configuration (0xD0-0xD7)

C.1.1 Purpose

The float sends its mission parameters: the dive-cycle, CTD, descent, park, ascent, surface, ice-detection, IMU and calibration settings stored in its non-volatile memory. ALAMO and ALTO floats send them in a binary format at each surfacing when `APT` = 1, as two blobs, D0 and D1. SOLO-II floats send an ASCII dump of their EEPROM instead, only in answer to a ground-station `P` command. The low nibble of the ID is the part index (`D0` = part 0, `D1` = part 1, and so on).

C.1.2 The two encodings

SOLO-II ASCII EEPROM dump (SOLO2 p. 13)

Offset	Size	Type	Field	Notes
0	1	U1	ID	0xD0, 0xD1, 0xD2, 0xD3 (SOLO2: “possible values 0:d”
1	2	U2	`jj`	Variable.
3	`jj` - 4	ASCII	Parameter list	`NAMExx=vvvvv|` records, 6-character name, `=`, 5-character value and `|`, with no gaps between records. Example: `PchSec= -1|MaxHrs= 1440|dBarGo= -1|...| XP2= 0| XPdly= 0|`.
`jj` - 1	1	char	;	Terminator.

SOLO2 says the dump takes 4 SBD messages: D0, D1 and D2 of 328 bytes each, and D3 of 7 bytes. It also says each parameter takes 15 bytes, but the record template it gives has 13 characters (see C.5).

The two encodings are told apart by byte 3. XF2.4 calls it “message encoding format type identifier”: 0 = binary (ALAMO), 1-255 = ASCII (SOLO-II). In an ASCII dump, byte 3 is the first character of the first parameter name, so it is never 0. NG10.12 calls the same byte `MissionConfigVersion` and gives it the value 0.

[decoder] The only code that checks byte 3 is the unused `XMissionConfiguration` wrapper: it treats `d[3] == 0` on part 0 as binary and leaves everything else undecoded. The active decoder, `XMfaMissionCfg`, never reads byte 3. A SOLO-II ASCII D0 would go through the binary path and fail as an unsupported firmware version, because bytes 4-7 would be ASCII characters.

ALAMO/ALTO binary format

The rest of C.1 covers the binary format.

C.1.3 Decoder families

Family	Source	Firmware	Status
`XAlamoMissionConfiguration`	`XAlamoMissionConfiguration.h` (wrapped by `XMission Configuration.h`)	0x02006200 to 0x05006100. It decodes one message, using offsets equal to the MFA layouts of the same versions.	Not used: nothing creates an `XMissionConfiguration`. This is the predecessor of the MFA decoder.
`XMfaMissionCfg` (“MFA” mission parameter message)	`XMfaMissionCfg.cpp/.h`	117 firmware words (C.1.6). Each firmware word maps to one `DecodeXMsgDataFw<layout>` function and one `Generate ListingStringFw<layout>` function.	Active. `XFloat.cpp` sends every 0xD0-0xD7 blob to `XMfaMission Cfg::AssignMessage`.

C.1.4 Common header (binary, part 0)

Offset	Size	Type	Field	Notes
0	1	U1	ID	0xD0.
1	2	U2	jj	322-328 bytes depending on the firmware (C.1.6). [decoder] Not read by XMfaMissionCfg, which checks the blob's actual length instead.
3	1	U1	Message config version / encoding	0 = binary. NG10.12: MissionConfigVersion = 0. NG10.5: "Message version". XF2.4: encoding format (0 = binary). [decoder] Not read.
4	1	U1	Firmware major	For example 0x0A = 10.
5	1	U1	Firmware minor	For example 0x0C = 12.
6	1	U1	Firmware patch	
7	1	U1	Firmware build / product	NG10.5: "Float model". NG10.12: FirmwareVersion-Product. The values seen are variant codes: 0x31 = WIMS, 0x32 = MM2/Micro-ALTO (10.12.0.32), 0x33 = Micro-ALTO (10.6.1.33), 0x01 or 0x04 = RBR or SBE builds.
4	4	U4	Firmware word	[decoder] Bytes 4-7 are read as one big-endian U4, 0x MMmmppbb, which selects the layout. It is stored as the float's firmware version, and only a D0 updates it. Version strings in this document follow the decoder's convention: major and minor in decimal, patch and build written as their hex digits (0x0A0C0032 = 10.12.0.32, 0x02056200 = 2.5.62
8	1	U1	Mission parameters (packet) version	From firmware 4.2.62 on. NG10.5 and NG10.12 both document 23 (0x17). Production 10.12.0.x floats send 0x18 (24), and 8.0.1.1 floats send 0x17. Listed as CfgMsgVer. In firmware before 4.2.62 the parameters start here (MSN at offset 8
9	1	U1	CTD (sensor) type	From 4.2.62 on. 0 = RBR, 1 = SBE (Sea-Bird). XF2.4: high nibble = manufacturer (0 RBR, 1 Seabird), low nibble = sensor type (reserved). The decoder compares the whole byte, so 0x10 would be listed as unknown. The type selects how CPN/CPP are listed, and whether surface pressure offset correction applies (RBR corrected, SBE not

Production example (float 9213, 10.12.0.32, dive 3): D0 01 47 | 00 | 0A 0C 00 32 | 18 | 00 | 00 03 ..., which decodes as jj = 327, binary, firmware 10.12.0.32, parameters version 24, RBR, MSN = 3.

Part 1 (D1) has no header: its parameters start at offset 3, right after jj.

C.1.5 Splitting into D0 and D1, assembly and dispatch

- From 4.5.63 on, the parameter list is longer than one SBD-sized blob and is cut in two. D0 holds the header and the first parameters, and D1 holds the rest from offset 3. Firmware before 4.5.63 sends D0 only, except 4.6.61: its D0 ends at SAN like 4.5.63, but the decoder doesn't decode a D1 for it.
- Where the cut falls depends on the firmware:
  - 5.0.61 to 10.6.x (D0 = 328 bytes): D0 ends with PFV1 (offsets 325-326). D1 starts with PFV2. NG10.5 numbers the parameters as one contiguous blob; D1 offset = NG10.5 offset - 324 (for example PFV2: 327 → 3, PPE: 415 → 91).
  - 6.5.77 and 6.8.5: D0 ends with PFV0, and D1 starts with PFV1.
  - 5.0.61.1, and 10.7 and later (D0 = 327 bytes): D0 ends with PSM (offset 325). D1 starts with PSD. In 10.7 the PNB/PNC/PXC BIT thresholds (6 bytes) were dropped, so PFV2, PDT and PSM (5 bytes) now fit in D0.
- Assembly [decoder] (AssignMessage): the part index is ID & 0x0F. Only indexes 0 and 1 are accepted (XMSG_MAX_MFA_MISS = 2), so D2-D7 are rejected with "Message index exceeds". Each part is stored per dive, and a repeated part replaces the earlier one.
  - On D0: the firmware word is read from bytes 4-7 (the blob must be at least 8 bytes), D0 is decoded, and a stored D1 is decoded again.
  - On D1: it is decoded only if D0 is already stored. Otherwise it is kept until D0 arrives.

- [decoder] The decoder never looks at the D0 parameters version (offset 8) to choose a layout. Only the firmware word does.

C.1.6 Supported firmware versions

117 firmware words are decoded. “Decode layout” is the `DecodeXMsgDataFw…` function and “Listing function” is the `GenerateListingStringFw…` function. “D0 bytes” and “D1 bytes” are the exact blob lengths the decoder expects. A version marked “layout inferred” was mapped to an existing layout after checking production messages (README, 2026-10-25), not from an MRV document.

FW word	Version	Decode layout	D0 bytes	D1 bytes	Listing function (style)	Layout inferred	Decoder comment
`0x02006200`	2.0.62.0	`02006200`	323	-	`02006200` (mis groups)		
`0x02006300`	2.0.63.0	`02006200`	323	-	`02006200` (mis groups)		
`0x02006700`	2.0.67.0	`02006200`	323	-	`02006200` (mis groups)		
`0x02006800`	2.0.68.0	`02006200`	323	-	`02006200` (mis groups)		
`0x02006900`	2.0.69.0	`02006900`	322	-	`02006900` (mis groups)		
`0x02036200`	2.3.62.0	`02006900`	322	-	`02006900` (mis groups)		
`0x02046200`	2.4.62.0	`02006900`	322	-	`02006900` (mis groups)		
`0x02056200`	2.5.62.0	`02006900`	322	-	`02006900` (mis groups)		
`0x03006500`	3.0.65.0	`03006500`	328	-	`02006900` (mis groups)		
`0x03016700`	3.1.67.0	`03016700`	328	-	`02006900` (mis groups)		Solo-II Hybrid #10083
`0x03026100`	3.2.61.0	`03026100`	328	-	`02006900` (mis groups)		2016 NASCar floats
`0x04016100`	4.1.61.0	`04016100`	328	-	`02006900` (mis groups)		2016 NASCar floats
`0x04026200`	4.2.62.0	`04026200`	323	-	`04026200` (MIS + mis groups)		PAR floats
`0x04036100`	4.3.61.0	`04036100`	327	-	`04026200` (MIS + mis groups)		2017 S2h test float 10108
`0x04046100`	4.4.61.0	`04046100`	328	-	`04026200` (MIS + mis groups)		2017 Hurricane Floats
`0x04056300`	4.5.63.0	`04056300`	328	>= 12	`04056300` (MIS + m)		2017 S2h test float 10108
`0x04066100`	4.6.61.0	`04066100`	328	-	`04026200` (MIS + mis groups)		2017 S2h test float 10108
`0x05006100`	5.0.61.0	`05006100`	328	79	`05006100` (MIS + m)		2017 Alto test float 11011
`0x05006101`	5.0.61.1	`05006101`	328	74	`05006100` (MIS + m)		alternate (2017 Alto test float 10109)
`0x06000000`	6.0.0.0	`06000000`	328	81	`06000000` (MIS + m)		
`0x06000100`	6.0.1.0	`06000100`	328	81	`06000000` (MIS + m)		
`0x06010300`	6.1.3.0	`06010300`	328	81	`06000000` (MIS + m)		2019 ALTO RBR/O2 test float 11060
`0x06040900`	6.4.9.0	`06040900`	328	81	`06000000` (MIS + m)		v6.4.9: K. Wood’s 2019 Arctic PAR floats
`0x06050100`	6.5.1.0	`06050100`	328	77	`06050100` (MIS + m)		
`0x06057700`	6.5.77.0	`06057700`	328	79	`06057700` (MIS + m)		v6.9.w: B. Sanabia’s 2019 Hurricane floats w/ WIMS

FW word	Version	Decode layout	D0 bytes	D1 bytes	Listing function (style)	Layout inferred	Decoder comment
0x06080500	6.8.5.0	06080500	328	79	06080500 (MIS + m)		2019 ALTO #11011 test float
0x07020200	7.2.2.0	07020200	328	76	07020200 (MIS + m)		2020 ALTO #11011 test float
0x07020300	7.2.3.0	07020200	328	76	07020200 (MIS + m)		2020 ALTO #11011 test float
0x07020400	7.2.4.0	07020200	328	76	07020200 (MIS + m)		2020 ALTO #11011 test float
0x07020500	7.2.5.0	07020200	328	76	07020200 (MIS + m)		2020 ALTO #11011 test float
0x07020600	7.2.6.0	07020200	328	76	07020200 (MIS + m)		2020 ALTO #11011 test float
0x07020700	7.2.7.0	07020200	328	76	07020200 (MIS + m)		2020 ALTO #11011 test float
0x07020800	7.2.8.0	07020200	328	76	07020200 (MIS + m)		2020 ALTO #11011 test float
0x07020900	7.2.9.0	07020200	328	76	07020200 (MIS + m)		2020 ALTO #11011 test float
0x07030000	7.3.0.0	07020200	328	76	07020200 (MIS + m)		2020 ALTO RBR/ODO #11060 test float
0x07030400	7.3.4.0	07020200	328	76	07020200 (MIS + m)	yes	
0x07030500	7.3.5.0	07020200	328	76	07020200 (MIS + m)		2020 ALTO RBR/ODO #11060 test float
0x08000000	8.0.0.0	08000000	328	78	08000000 (MIS + m)		
0x08000001	8.0.0.1	08000000	328	78	08000000 (MIS + m)		
0x08000100	8.0.1.0	08000000	328	78	08000000 (MIS + m)		
0x08000101	8.0.1.1	08000000	328	78	08000000 (MIS + m)		2020 Hurricane floats
0x08000200	8.0.2.0	08000000	328	78	08000000 (MIS + m)		
0x08000300	8.0.3.0	08000000	328	78	08000000 (MIS + m)		
0x08000400	8.0.4.0	08000000	328	78	08000000 (MIS + m)		
0x08000500	8.0.5.0	08000000	328	78	08000000 (MIS + m)		
0x08010000	8.1.0.0	08010000	328	76	08010000 (MIS + m)		
0x08010001	8.1.0.1	08000000	328	78	08000000 (MIS + m)		2021 Hurricane floats
0x08020000	8.2.0.0	08010000	328	76	Unsupported (generic “unsupported” listing; see notes)		
0x09000000	9.0.0.0	08010000	328	76	08010000 (MIS + m)		
0x0A000000	10.0.0.0	08010000	328	76	08010000 (MIS + m)		
0x0A000100	10.0.1.0	08010000	328	76	08010000 (MIS + m)		
0x0A020000	10.2.0.0	08010000	328	76	08010000 (MIS + m)		
0x0A020100	10.2.1.0	08010000	328	76	08010000 (MIS + m)		
0x0A020200	10.2.2.0	08010000	328	76	08010000 (MIS + m)		
0x0A020201	10.2.2.1	08000000	328	78	Unsupported (generic “unsupported” listing; see notes)		
0x0A020301	10.2.3.1	0A050000	328	93	0A050000 (MIS + m)	yes	
0x0A030000	10.3.0.0	08010000	328	76	08010000 (MIS + m)		
0x0A030100	10.3.1.0	08010000	328	76	08010000 (MIS + m)		
0x0A040000	10.4.0.0	0A050000	328	93	0A050000 (MIS + m)	yes	
0x0A040400	10.4.4.0	0A050000	328	93	0A050000 (MIS + m)	yes	
0x0A040500	10.4.5.0	0A050000	328	93	0A050000 (MIS + m)	yes	
0x0A040600	10.4.6.0	0A050000	328	93	0A050000 (MIS + m)	yes	
0x0A050000	10.5.0.0	0A050000	328	93	0A050000 (MIS + m)		
0x0A050001	10.5.0.1	0A050000	328	93	0A050000 (MIS + m)		
0x0A050002	10.5.0.2	0A050000	328	93	0A050000 (MIS + m)		
0x0A050003	10.5.0.3	0A050000	328	93	0A050000 (MIS + m)		
0x0A050004	10.5.0.4	0A050000	328	93	0A050000 (MIS + m)		
0x0A060000	10.6.0.0	0A050000	328	93	0A050000 (MIS + m)		

FW word	Version	Decode layout	D0 bytes	D1 bytes	Listing function (style)	Layout inferred	Decoder comment
0x0A060001	10.6.0.1	0A050000	328	93	0A050000 (MIS + m)		
0x0A060002	10.6.0.2	0A050000	328	93	0A050000 (MIS + m)		
0x0A060003	10.6.0.3	0A050000	328	93	0A050000 (MIS + m)		
0x0A060004	10.6.0.4	0A050000	328	93	0A050000 (MIS + m)		
0x0A060011	10.6.0.11	0A050000	328	93	0A050000 (MIS + m)	yes	
0x0A060100	10.6.1.0	0A050000	328	93	0A050000 (MIS + m)		
0x0A060101	10.6.1.1	0A050000	328	93	0A050000 (MIS + m)		
0x0A060102	10.6.1.2	0A050000	328	93	0A050000 (MIS + m)		
0x0A060103	10.6.1.3	0A050000	328	93	0A050000 (MIS + m)		
0x0A060104	10.6.1.4	0A050000	328	93	0A050000 (MIS + m)		
0x0A060132	10.6.1.32	0A050000	328	93	0A050000 (MIS + m)	yes	
0x0A060133	10.6.1.33	0A050000	328	93	0A050000 (MIS + m)		Micro-ALTO 10.6.1
0x0A060200	10.6.2.0	0A060200	328	93	0A050000 (MIS + m)		
0x0A060201	10.6.2.1	0A060200	328	93	0A050000 (MIS + m)		
0x0A060202	10.6.2.2	0A060200	328	93	0A050000 (MIS + m)		
0x0A060203	10.6.2.3	0A060200	328	93	0A050000 (MIS + m)		
0x0A060204	10.6.2.4	0A060200	328	93	0A050000 (MIS + m)		
0x0A060231	10.6.2.31	0A060231	328	95	0A060231 (MIS + m)		WIMS
0x0A060233	10.6.2.33	0A060200	328	93	0A050000 (MIS + m)		
0x0A060300	10.6.3.0	0A060200	328	93	0A050000 (MIS + m)		
0x0A060301	10.6.3.1	0A060200	328	93	0A050000 (MIS + m)		
0x0A060302	10.6.3.2	0A060200	328	93	0A050000 (MIS + m)		
0x0A060303	10.6.3.3	0A060200	328	93	0A050000 (MIS + m)		
0x0A060304	10.6.3.4	0A060200	328	93	0A050000 (MIS + m)		
0x0A060333	10.6.3.33	0A060200	328	93	0A050000 (MIS + m)		
0x0A060401	10.6.4.1	0A050000	328	93	0A050000 (MIS + m)	yes	
0x0A070001	10.7.0.1	0A080004	327	71 or 88	0A070101 (MIS + m)	yes	
0x0A070101	10.7.1.1	0A070101	327	88	0A070101 (MIS + m)		
0x0A070104	10.7.1.4	0A080004	327	71 or 88	0A070101 (MIS + m)	yes	
0x0A070131	10.7.1.31	0A080031	327	90	0A080031 (MIS + m)	yes	WIMS
0x0A070211	10.7.2.11	0A080004	327	71 or 88	0A070101 (MIS + m)	yes	
0x0A080001	10.8.0.1	0A080004	327	71 or 88	0A070101 (MIS + m)	yes	
0x0A080004	10.8.0.4	0A080004	327	71 or 88	0A070101 (MIS + m)		SBE
0x0A080031	10.8.0.31	0A080031	327	90	0A080031 (MIS + m)		WIMS
0x0A090001	10.9.0.1	0A080004	327	71 or 88	0A070101 (MIS + m)	yes	
0x0A090101	10.9.1.1	0A080004	327	71 or 88	0A070101 (MIS + m)	yes	
0x0A090122	10.9.1.22	0A080004	327	71 or 88	0A070101 (MIS + m)	yes	
0x0A090131	10.9.1.31	0A080031	327	90	0A080031 (MIS + m)	yes	WIMS
0x0A0A0001	10.10.0.1	0A080004	327	71 or 88	0A070101 (MIS + m)	yes	
0x0A0A0004	10.10.0.4	0A080004	327	71 or 88	0A070101 (MIS + m)	yes	
0x0A0A0022	10.10.0.22	0A080004	327	71 or 88	0A070101 (MIS + m)	yes	
0x0A0A0031	10.10.0.31	0A080031	327	90	0A080031 (MIS + m)		WIMS
0x0A0C0001	10.12.0.1	0A080004	327	71 or 88	0A070101 (MIS + m)		RBR
0x0A0C0004	10.12.0.4	0A080004	327	71 or 88	0A070101 (MIS + m)	yes	
0x0A0C0032	10.12.0.32	0A080004	327	71 or 88	0A070101 (MIS + m)		ALAMO/MM2 Software version 10.12.0.32
0x0A0C0034	10.12.0.34	0A080004	327	71 or 88	0A070101 (MIS + m)	yes	
0x0A0C0131	10.12.1.31	0A080031	327	90	0A080031 (MIS + m)		WIMS
0x0A0D0013	10.13.0.13	0A080004	327	71 or 88	0A070101 (MIS + m)	yes	
0x0A0E0028	10.14.0.28	0A080004	327	71 or 88	0A070101 (MIS + m)	yes	

Notes:

- Two words decode but don’t list. 0x08020000 and 0x0A020201 are in the decode switch but missing from the listing switch, so `GenerateListingString` uses `GenerateListingStringUnsupported`. In the build configuration (`XMSG_DEBUG_MFA_MISS_GFG = true`) that only logs a warning and prints a generic `MIS/m` block (C.1.11). That block has a fixed parameter list (it includes the `PNB/PNC/PXC` BIT thresholds), so

parameters the layout doesn't contain show their unset values.

- Some words share a function whose code is identical. `0A060200` duplicates `0A050000`. `06000100` and `06010300` duplicate `06000000`.
- A firmware word that isn't in the table falls through to `DecodeXMsgDataUnsupported`. It reads only the parameters version (offset 8) and CTD type (offset 9), logs "Unsupported ALAMO Mission Parameters message version: 0x…", and gets the generic listing.

C.1.7 Current layout: 10.7 to 10.14 (`Fw0A080004`; the same D0 for `Fw0A070101` and `Fw0A080031`)

The offsets agree with NG10.12 for every parameter. The only difference is the type of `DRT`, which NG10.12 gives as I2 and the decoder reads as U2.

D0 (327 bytes; terminator at offset 326)

Offsets 0-7 are the common header (C.1.4).

Offset	Size	Type	Mnemonic	Units / scale	Meaning and notes
8	1	U1	(params ver.)	-	Mission-parameters (packet) version, MRV `MissionParams Version`. Printed as `CfgMsgVer` on the `MIS` line.
9	1	U1	(CTD type)	enum	CTD type: 0 = RBR, 1 = Sea-Bird (SBE). Printed on the `MIS` line with its name (`RBR`, `SBE`, otherwise `unknown`
10	2	U2	MSN	-	MSN: dive cycle number. Stored as signed 16-bit (10.5.0 doc: I2; 10.12.0 doc: U2
12	1	U1	MPS	-	MPS: current mission phase (0-2
13	2	U2	MPC	cycles	MPC: cycle count within the current phase (0-50,000
15	2	U2	MNM0	cycles	MNM0: number of dive cycles in phase 0.
17	2	U2	MNM1	cycles	MNM1: number of dive cycles in phase 1.
19	2	U2	MNM2	cycles	MNM2: number of dive cycles in phase 2.
21	1	U1	MNS0	-	MNS0: phase to run after phase 0.
22	1	U1	MNS1	-	MNS1: phase to run after phase 1.
23	1	U1	MNS2	-	MNS2: phase to run after phase 2.
24	2	U2	DTD0	dBar	DTD0: target drift/park pressure, phase 0.
26	2	U2	DTD1	dBar	DTD1: target drift/park pressure, phase 1.
28	2	U2	DTD2	dBar	DTD2: target drift/park pressure, phase 2.
30	2	U2	MDS	dBar	MDS: mission abort ("death") pressure, the maximum depth.
32	1	U1	APT	enum	APT: send the mission parameters at every surfacing. 0 = off, 1 = on.
33	1	U1	BTM	enum	BTM: Bluetooth mode. 0 = off, 1 = on.
34	1	U1	DDB	enum	DDB: diagnostic dive bypass. 0 = run the diagnostic dive, 1 = skip it.
35	1	U1	XAS	enum	XAS: profile compression. 1 = first difference, 2 = curvature (10.5.0 doc). Spreadsheet v2.4 says 0 = first difference, 1 = curvature.
36	1	U1	CPS	enum	CPS: CTD profiling. 0 = off, 1 = on.
37	1	U1	CST	enum	CST: CTD profile sampling. 1 = binned, 2 = raw (10.5.0 doc). Spreadsheet v2.4 says 0 = bin-averaged, 1 = raw.
38	2	U2	CTF	dBar	CTF: profile cut-off pressure.
40	2	U2	CTL	dBar	CTL: lower pressure limit for resetting the pressure-sensor offset.
42	2	U2	CTU	dBar	CTU: upper pressure limit for resetting the pressure-sensor offset.
44	1	U1	CTC	enum	CTC: conductivity-cell (auxiliary) temperature channel, RBR only. 0 = off, 1 = on.
45	4	U4	CSP	ms	CSP: CTD continuous sampling period. Listed in s.
49	2	U2	CPB1	dBar	CPB1: lower boundary of profile segment 1 (bottom
51	2	U2	CPB2	dBar	CPB2: lower boundary of profile segment 2 (middle
53	2	U2	CPB3	dBar	CPB3: lower boundary of profile segment 3 (top

Offset	Size	Type	Mnemonic	Units / scale	Meaning and notes
55	2	U2	CPN1	RBR 0.1 dBar; SBE dBar	CPN1: segment 1 bin size. Listed as value/10 dBar for RBR and as the raw value for SBE.
57	2	U2	CPN2	RBR 0.1 dBar; SBE dBar	CPN2: segment 2 bin size.
59	2	U2	CPN3	RBR 0.1 dBar; SBE dBar	CPN3: segment 3 bin size.
61	2	U2	CPP1	RBR ms; SBE dBar	CPP1: segment 1 sampling period (RBR) or sampling pressure interval (SBE). Listed as value/1000 s for RBR and the raw dBar value for SBE.
63	2	U2	CPP2	RBR ms; SBE dBar	CPP2: segment 2 sampling period or interval.
65	2	U2	CPP3	RBR ms; SBE dBar	CPP3: segment 3 sampling period or interval.
67	2	U2	CRD	dBar	CRD: pressure at which raw profile samples start.
69	2	U2	CRS	samples	CRS: maximum number of raw profile samples. 0 = off.
71	2	U2	MTD	dBar	MTD: start-of-mission trigger pressure.
73	4	U4	MTT	ms	MTT: start-of-mission pressure sample period. Listed in s.
77	4	U4	MTS	s	MTS: start-of-mission timeout.
81	1	U1	MAM	bit mask	MAM: IMU sampling phases. Spreadsheet v2.4: bit 0 surface, bit 1 descent, bit 2 drift/park, bit 3 ascent.
82	2	U2	MAT1	-	MAT1: number of IMU sampling intervals.
84	2	U2	MAT2	s	MAT2: length of one IMU sampling interval.
86	2	U2	MAP	s	MAP: IMU sample period, the time between sampling intervals.
88	2	U2	ACA	g	ACA: accelerometer full-scale range: 2, 4, 8 or 16 g.
90	2	U2	ACG	deg/s	ACG: gyroscope full-scale range: 250, 500, 1000 or 2000 deg/s.
92	2	U2	ACR	Hz	ACR: IMU sample rate.
94	2	U2	PBL	counts	PBL: calibration, battery voltage at 12 V.
96	2	U2	PBH	counts	PBH: calibration, battery voltage at 14 V.
98	2	U2	PP0	counts	PP0: calibration, oil pump current at 0 mA.
100	2	U2	PP1	counts	PP1: calibration, oil pump current at 1000 mA.
102	2	U2	PVL	counts	PVL: calibration, V12 signal at 12 V.
104	2	U2	PVH	counts	PVH: calibration, V12 signal at 14 V.
106	2	U2	PV0	counts	PV0: calibration, case vacuum at 0 inHg.
108	2	U2	PV5	counts	PV5: calibration, case vacuum at 5 inHg.
110	2	U2	PC0	counts	PC0: calibration, V12 current at 0 mA.
112	2	U2	PC1	counts	PC1: calibration, V12 current at 1000 mA.
114	1	U1	ICM	enum	ICM: ice detection. 0 = off, 1 = on.
115	2	U2	IMM	bit mask	IMM: ice-detection month mask.
117	2	U2	ICA0	attempts	ICA0: maximum surfacing attempts, summer.
119	2	U2	ICA1	attempts	ICA1: maximum surfacing attempts, winter.
121	2	U2	ICB0	cycles	ICB0: cycles between surfacing attempts, summer.
123	2	U2	ICB1	cycles	ICB1: cycles between surfacing attempts, winter.
125	2	U2	IMT0	%	IMT0: stored-message urgency threshold, summer.
127	2	U2	IMT1	%	IMT1: stored-message urgency threshold, winter.
129	2	U2	ILT0	days	ILT0: time-since-last-transmission urgency threshold, summer.
131	2	U2	ILT1	days	ILT1: time-since-last-transmission urgency threshold, winter.
133	2	U2	ICN	dBar	ICN: shallow pressure limit of the ice temperature test.
135	2	U2	ICX	dBar	ICX: deep pressure limit of the ice temperature test.
137	2	I2	ICT	0.01 degC	ICT: ice temperature threshold, signed. Listed as value/100 degC.
139	2	U2	ICS	s	ICS: ice temperature test sample interval (“ice storage timeout” in the MRV docs
141	1	U1	ISS0	enum	ISS0: surface search (multiple surfacing) mode, summer.
142	1	U1	ISS1	enum	ISS1: surface search (multiple surfacing) mode, winter.
143	1	U1	ISM	bit mask	ISM: surfacing criteria mask.
144	2	U2	IST	0.01 PSU	IST: surfacing salinity threshold. Listed as value/100 PSU. The header comment says 1/10 PSU.
146	2	U2	ISO	s	ISO: surface search timeout, the length of one surfacing attempt.
148	1	U1	ISP	enum	ISP: surface search profile mode.
149	1	U1	IUE0	enum	IUE0: urgency-mode exit criteria, summer.
150	1	U1	IUE1	enum	IUE1: urgency-mode exit criteria, winter.

Offset	Size	Type	Mnemonic	Units / scale	Meaning and notes
151	2	U2	ICD0	dBar	ICD0: turnaround hold pressure, summer.
153	2	U2	ICD1	dBar	ICD1: turnaround hold pressure, winter.
155	4	U4	ICR0	min (listing)	ICR0: turnaround hold duration, summer.
159	4	U4	ICR1	min (listing)	ICR1: turnaround hold duration, winter.
163	1	U1	ICG0	enum	ICG0: turnaround seek mode, summer.
164	1	U1	ICG1	enum	ICG1: turnaround seek mode, winter.
165	1	U1	ISA	attempts	ISA: ice descent seek attempts.
166	2	U2	ISD	dBar	ISD: ice descent seek pressure delta.
168	2	U2	IAT	s	IAT: ice descent seek timeout.
170	2	U2	ISV	s	ISV: ice descent seek velocity timeout.
172	2	U2	ISW	s	ISW: ice descent seek wait timeout.
174	2	U2	ICO0	s	ICO0: turnaround valve duration, summer.
176	2	U2	ICO1	s	ICO1: turnaround valve duration, winter.
178	4	U4	ICF	bit mask	ICF: ice-free (summer) month mask.
182	4	U4	ICW	bit mask	ICW: iced-over (winter) month mask.
186	2	U2	DNS	-	DNS: descent-phase cycle number.
188	2	U2	DRA	cm/s	DRA: target descent rate. The decoder stores it in 8 bits, so values above 255 are truncated. Listed raw, labeled `[m/s]`.
190	2	U2	DSD0	dBar	DSD0: first-pump pressure during descent, phase 0.
192	2	U2	DSD1	dBar	DSD1: first-pump pressure, phase 1.
194	2	U2	DSD2	dBar	DSD2: first-pump pressure, phase 2.
196	2	U2	DST0	s	DST0: first-pump timeout, phase 0.
198	2	U2	DST1	s	DST1: first-pump timeout, phase 1.
200	2	U2	DST2	s	DST2: first-pump timeout, phase 2.
202	2	U2	DDF	s	DDF: diagnostic dive initial pump time.
204	2	U2	DDC	s	DDC: diagnostic dive calibration timeout.
206	4	U4	DSI	ms	DSI: descent sample interval. Listed in s.
210	2	U2	DDD	dBar	DDD: fall time-series storage pressure interval.
212	2	U2	DDT	s	DDT: fall time-series storage time interval.
214	2	U2	DFT0	s	DFT0: first descent pump duration, phase 0.
216	2	U2	DFT1	s	DFT1: first descent pump duration, phase 1.
218	2	U2	DFT2	s	DFT2: first descent pump duration, phase 2.
220	1	U1	DCM	enum	DCM: descent seek/calibration mode. 0 = off, 1 = on.
221	1	U1	DAL	enum	DAL: descent algorithm. 0 = calibration, 1 = seek.
222	2	U2	DCT0	s	DCT0: wait after the first pump (calibration timeout), phase 0.
224	2	U2	DCT1	s	DCT1: same, phase 1.
226	2	U2	DCT2	s	DCT2: same, phase 2.
228	1	U1	DCA	seeks	DCA: maximum descent seeks or calibration attempts.
229	2	U2	DRC	s	DRC: calculated descent calibration rate (deprecated
231	2	U2	DRT	s	DRT: seek rate adjustment timeout. The MRV docs say I2; the decoder reads U2. The listing's DRT line printed DRA's value (fixed in build 3309; bug
233	2	U2	DCX	s	DCX: seek maximum pump time.
235	2	U2	DCP	s	DCP: seek minimum pump time.
237	2	U2	DCF	s	DCF: seek minimum valve time.
239	2	U2	DCD	dBar	DCD: calibration pressure delta (deprecated
241	2	U2	DCR	-	DCR: descent calibration rate.
243	2	U2	DSW	s	DSW: seek interval.
245	2	U2	DUC	%	DUC: seek underdamping factor.
247	2	U2	DPT	s	DPT: pump time per 100 dBar.
249	2	U2	DUT	s	DUT: valve time per 100 dBar.
251	2	U2	DPD	s	DPD: first seek pump duration.
253	2	U2	DVT	s	DVT: seek pressure sample interval (velocity measurement delay
255	2	U2	DVD	dBar	DVD: seek valve pressure delta.
257	2	U2	DVW	s	DVW: seek valve wait time.
259	2	U2	PFR	mL/min	PFR: pump flow rate.
261	2	U2	PFC	cc/m x 1000	PFC: pump flow pressure correction (listing unit: uL/dBar
263	2	U2	DDP	dBar	DDP: diagnostic dive ascent trigger pressure.

Offset	Size	Type	Mnemonic	Units / scale	Meaning and notes
265	2	U2	DDS	s	DDS: diagnostic dive ascent trigger timeout.
267	2	U2	PNS	-	PNS: park-phase cycle number.
269	4	U4	PTO0	s	PTO0: park duration, phase 0.
273	4	U4	PTO1	s	PTO1: park duration, phase 1.
277	4	U4	PTO2	s	PTO2: park duration, phase 2.
281	4	U4	PTH0	bit mask	PTH0: hour-of-day mask that ends the park phase, phase 0.
285	4	U4	PTH1	bit mask	PTH1: hour mask, phase 1.
289	4	U4	PTH2	bit mask	PTH2: hour mask, phase 2.
293	4	U4	PTD	s	PTD: park end date/time. The spreadsheet says it is set as YYYYMMDD HHMMSS and returned in seconds.
297	1	U1	PSA	enum	PSA: park sample mode.
298	4	U4	PSI	ms	PSI: park sample interval. Listed in s.
302	1	U1	PFM0	enum	PFM0: fall to profile pressure after the park phase, phase 0. 0 = off, 1 = on.
303	1	U1	PFM1	enum	PFM1: same, phase 1.
304	1	U1	PFM2	enum	PFM2: same, phase 2.
305	2	U2	PFT0	s	PFT0: fall-to-profile timeout, phase 0.
307	2	U2	PFT1	s	PFT1: same, phase 1.
309	2	U2	PFT2	s	PFT2: same, phase 2.
311	2	U2	PFD0	dBar	PFD0: target profile pressure, phase 0.
313	2	U2	PFD1	dBar	PFD1: same, phase 1.
315	2	U2	PFD2	dBar	PFD2: same, phase 2.
317	2	U2	PFV0	s	PFV0: fall valve-open duration, phase 0.
319	2	U2	PFV1	s	PFV1: same, phase 1.
321	2	U2	PFV2	s	PFV2: same, phase 2.
323	2	U2	PDT	s	PDT: park data storage interval.
325	1	U1	PSM	enum	PSM: park seek mode. 0 = off, 1 = on.
326	1	char	;		Terminator.

D1 (71 bytes without the CTD correction parameters, 88 bytes with them)

Offset	Size	Type	Mnemonic	Units / scale	Meaning and notes
0	1	U1	ID		0xD1.
1	2	U2	jj		71 or 88.
3	2	U2	PSD	dBar	PSD: park seek pressure delta.
5	2	U2	PST	s	PST: park seek interval (timeout
7	2	U2	PSV	s	PSV: park seek velocity measurement delay.
9	2	U2	PSW	s	PSW: park seek wait timeout.
11	1	U1	PSS	enum	PSS: park sampling scheme. 1 = P, 2 = T, 3 = PT, 4 = S, 7 = PTS.
12	2	U2	ASN	-	ASN: ascent cycle number.
14	2	U2	ARR	cm/s	ARR: target rise rate. Listed as value/100 m/s.
16	2	U2	APM	s	APM: maximum ascent pump run time.
18	2	U2	APS	s	APS: initial ascent pump-on duration.
20	2	U2	APN	s	APN: ascent pump-on duration.
22	2	U2	APF	s	APF: ascent pump-off duration.
24	4	U4	ASI	ms (decoder)	ASI: ascent sample interval. The 10.5.0 doc says seconds; the decoder lists value/1000 s.
28	2	U2	ASD	dBar	ASD: surface pressure threshold.
30	2	U2	ADD	dBar	ADD: rise time-series storage pressure interval.
32	2	U2	ADT	s	ADT: rise time-series storage time interval.
34	2	U2	ARC	ms (decoder)	ARC: rise-rate calculation adjustment. Listed as value/1000 s, labeled “sample delay correction”.
36	2	U2	AHR	see notes	AHR: rise rate below which the ascent halts. The 10.5.0 doc says cm/s; the decoder lists value/1000 as m/s.
38	2	U2	AHD	dBar	AHD: halt pressure delta.
40	2	U2	SFN	-	SFN: surface cycle number.
42	2	I2	SDC	dBar	SDC: surface pressure compensation, signed.

Offset	Size	Type	Mnemonic	Units / scale	Meaning and notes
44	2	U2	SGT	min	SGT: GPS timeout.
46	2	U2	SET	min	SET: Iridium timeout.
48	1	U1	SBM	enum	SBM: beacon mode. 0 = off, 1 = on.
49	2	U2	SBT	s	SBT: beacon interval.
51	2	U2	SSA	enum	SSA: surface sample averaging mode.
53	4	U4	SSI	ms	SSI: surface sample interval. Listed in s.
57	2	U2	SDS	s	SDS: surface data storage interval.
59	1	U1	SAM	enum	SAM: air-assist (air pump) mode. 0 = off, 1 = on.
60	2	U2	SAX	see notes	SAX: air-assist maximum pressure (psi in the MRV docs). Listed as value/1000 PSI.
62	2	U2	SAN	see notes	SAN: air-assist minimum pressure (psi in the MRV docs). Listed as value/1000 PSI.
64	2	U2	SAO	s	SAO: air-assist minimum (pump) duration.
66	2	U2	SPT	s	SPT: surface CTD storage timeout (sampling duration
68	2	U2	SSP	s	SSP: surface oil pump duration.
70	4	R4	PPC0	s	PPC0: RBR dynamic-correction tau (conductivity-temperature lag), IEEE-754 single. Only in the 88-byte form.
74	4	R4	PPC1	-	PPC1: RBR dynamic-correction alpha (short-term thermal mass), single. Only in the 88-byte form.
78	4	R4	PPC2	s	PPC2: RBR dynamic-correction tdelay (water-cell lag), single. Only in the 88-byte form.
82	4	R4	PPC3	-	PPC3: RBR dynamic-correction ctcoeff (long-term thermal mass), single. Only in the 88-byte form.
86	1	U1	PPE	enum	PPE: RBR post-processing (dynamic correction) enable. Floats with PPE = 2 send corrected salinity (0xDC). Only in the 88-byte form.
70 or 87	1	char	;		Terminator.

[decoder] The 88-byte form carries `PPC0`-`PPC3` and `PPE` (NG10.12; production 10.12.0.1 and 10.12.0.32 floats). The decoder accepts both lengths for the `Fw0A080004` firmware group and decodes `PPC/PPE` only when the blob is 88 bytes long.

Variants of the current D1

Layout (firmware)	D1 bytes	Difference from the table above
`Fw0A070101` (10.7.1.1)	88	Only 88 bytes is accepted. [decoder] `PPC0`-`PPE` at offsets 70-86 are not decoded (the lines are commented out), so they keep their unset values.
`Fw0A080031` (WIMS: 10.7.1.31, 10.8.0.31, 10.9.1.31, 10.10.0.31, 10.12.1.31)	90	`WMT` (U2, s, WIMS timeout) at offset 70. `PPC0` at 72, `PPC1` at 76, `PPC2` at 80, `PPC3` at 84 (R4), `PPE` at 88 (U1), terminator at 89.

C.1.8 The 10.5/10.6 layout (`Fw0A050000`, `Fw0A060200`, `Fw0A060231`)

- D0 is 328 bytes and identical to the 5.0.61 to 10.3 D0 (Appendix C.A, `Fw05006100`/`Fw06000000`/`Fw08010000`):
  - the header as in C.1.4;
  - parameters `MSN` at 10 to `ACR` at 92;
  - `PNB` 94, `PNC` 96, `PXC` 98 (U2, 0.01 V and 0.01 psi BIT thresholds);
  - calibration `PBL` to `PC1` at 100-119;
  - ice block `ICM` at 120 to `ICW` at 188;
  - descent block `DNS` at 192 to `DDS` at 271;
  - park block `PNS` at 273 to `PFV1` at 325;
  - terminator at 327.

- D1 is 93 bytes: PFV2 at 3, PDT 5, PSM 7 (U1), PSD 8, PST 10, PSV 12, PSW 14, PSS 16 (U1), ASN 17, ARR 19, APM 21, APS 23, APN 25, APF 27, ASI 29 (U4), ASD 33, ADD 35, ADT 37, ARC 39, AHR 41, AHD 43, SFN 45, SDC 47 (I2), SGT 49, SET 51, SBM 53 (U1), SBT 54, SSA 56, SSI 58 (U4), SDS 62, SAM 64 (U1), SAX 65, SAN 67, SAO 69, SPT 71, SSP 73, PPC0 75, PPC1 79, PPC2 83, PPC3 87 (R4), PPE 91 (U1), terminator at 92. Unmarked fields are U2. These offsets agree with NG10.5 for every parameter, except MSN (NG10.5: I2) and DRT (NG10.5: I2), which the decoder reads unsigned.
- Fw0A060231 (10.6.2.31, WIMS): D1 is 95 bytes, with WMT (U2) at offset 92 after PPE and the terminator at 94.

C.1.9 Layout history

Each row is the decoder's view of what changed against the layout before it. Exact offsets for every layout are in Appendix C.A.

Layout (firmware)	D0 / D1 bytes	Changes
02006200 (2.0.62-2.0.68)	323 / -	First binary layout. No parameters version or CTD-type byte: MSN is at offset 8. Single D0. Has U4 state start times (seconds since 2000-01-01): descent at 99, park at 179, ascent at 227, surface at 271. Reserved gaps at 81-98, 164-178, 216-226, 257-270 and 309-321 (XF2.4 marks them RESERVED). Single DTD, DSD, DST, DFT, DCT, PTO, PTH, PFM, PFT, PFD and PFV, all without a phase index. Has MAT (one U2), PNB/PNC/PXC, SDT, SAP and SPT.
02006900 (2.0.69, 2.3.62-2.5.62)	322 / -	Adds CTF, CTL and CTU after CST. Adds the ice block ICM to ICW (without ISA-ISW and ISM) in place of the first reserved gap. The other reserved gaps are removed and the start times kept. [decoder] ICF/ICW are read as U1 at 141 and 145, though they are 4-byte masks (XF2.4: U4), so only their most significant byte is decoded.
03006500 (3.0.65)	328 / -	A 9-byte slot at 137-145 holds ISA, ISD, IAT, ISV and ISW (XF2.4), not decoded. SPT is no longer decoded. XF2.4 gives this version 331 bytes with SPT at 326 and SSP at 328; the decoder expects 328. ICF/ICW are still read as U1.
03016700 (3.1.67, SOLO-II hybrid 10083)	328 / -	One extra byte after ISS1 (offset 116), probably ISM [inferred], not decoded.
03026100, 04016100 (3.2.61, 4.1.61)	328 / -	Adds BTM (Bluetooth mode) after APT. SAP is no longer decoded.
04026200 (4.2.62, PAR floats)	323 / -	Adds the parameters version (offset 8) and CTD type (9), so parameters start at 10. Drops the state start times. Decodes ISA-ISW. ICF/ICW become U4. Adds DVD and DVW after DVT. Decodes SAP, SPT and SSP at the end. XF2.4 differs (C.5
04036100 (4.3.61)	327 / -	A 4-byte slot after CPP3 (49-52, probably CRD/CRS [inferred]), not decoded.
04046100 (4.4.61, 2017 hurricane floats)	328 / -	A 1-byte slot after CTU (offset 27, CTC per XF2.4), not decoded.
04056300 (4.5.63)	328 / ≥ 12	Decodes CTC, CRD and CRS. The 6 bytes after MAM (61-66: MAT1, MAT2 and MAP per XF2.4) are not decoded. The slots for ISM (128) and PTD (246-249) are not decoded either. D0 ends with SAN. First D1: SAO 3, SAP 5, SPT 7, SSP 9 (U2
04066100 (4.6.61)	328 / -	Same D0 offsets as 4.5.63, but CTC, CRD and CRS are not decoded. D1 isn't decoded. XF2.4 gives a single 336-byte D0 that goes on to SSP.
05006100 (5.0.61, ALTO)	328 / 79	Restructured for mission phases. Adds MPS, MPC, MNM0-2, MNS0-2, DDB, MTS, MAT1, MAT2, MAP and ISM. Adds phase-indexed DTD0-2, DSD0-2, DST0-2, DFT0-2, DCT0-2, PTO0-2, PTH0-2, PFM0-2, PFT0-2, PFD0-2 and PFV0-2, plus PTD. D0 ends with PFV1. D1: PFV2 3, PDT 5, PSM 7, PSD to PSW 8-15, ascent block from ASN 16, surface block from SFN 42 (with SDT U4 at 53), SAO 70, SAP 72, SPT 74, SSP 76.

Layout (firmware)	D0 / D1 bytes	Changes
05006101 (5.0.61.1, ALTO 10109)	328 / 74	No DDB, MAT1 or MAP (MAT2 at 81), so later fields move back. PFV2, PDT and PSM are in D0 at 322-326. D1 starts with PSD.
06000000, 06000100, 06010300 (6.0.0, 6.0.1, 6.1.3)	328 / 81	D0 as 5.0.61. D1 adds APS (initial ascent pump-on duration) after APM.
06040900 (6.4.9, Arctic PAR floats)	328 / 81	[decoder] D1 is read with SSA as U1 at 59 and SSI as U2 at 60, SDS isn't decoded, and byte 62 is skipped. This is probably a decoder quirk rather than a real layout [inferred].
06050100 (6.5.1)	328 / 77	SDT removed from D1.
06057700 (6.5.77, "6.9.w", hurricane floats with WIMS)	328 / 79	D0 ends with PFV0. D1 starts with PFV1 at 3.
06080500 (6.8.5, ALTO 11011)	328 / 79	An unidentified 2-byte field after ACR (94-95; the decoder's comments name tsp0/tsp1/tsn0/tsn1) moves PNB and everything after it by +2. D0 ends with PFV0. D1 as 6.5.77.
07020200 (7.2.x, 7.3.x)	328 / 76	D0 as 5.0.61. D1 adds PSS (park sampling scheme) after PSW and drops SAP. DF7.3.5 has the same D0/D1 parameter order, but its field lists leave out the 5-byte D0 header (offsets 3-7
08000000 (8.0.x, 8.1.0.1, 10.2.2.1)	328 / 78	D1 adds WMT (WIMS timeout) at 75.
08010000 (8.1.0, 8.2.0, 9.0.0, 10.0.x-10.3.x)	328 / 76	D1 as 7.2 (no WMT
0A050000, 0A060200 (10.2.3.1, 10.4.x, 10.5.x, 10.6.0-10.6.3, 10.6.4.1)	328 / 93	D1 adds RBR dynamic-correction PPC0-PPC3 (R4) and PPE after SSP (C.1.8
0A060231 (10.6.2.31, WIMS)	328 / 95	Adds WMT after PPE.
0A070101, 0A080004 (10.7 and later)	327 / 71 or 88	Removes PNB, PNC and PXC. PFV2, PDT and PSM move into D0, and D1 starts with PSD (C.1.7
0A080031 (10.7.1.31 and later WIMS)	327 / 90	WMT sits between SSP and PPC0.

C.1.10 Decoding notes

- Length checks [decoder]: every layout checks the exact blob length (the size of the received blob, not jj). With XMSG_DEBUG_MFA_MISS_GFG = true, which is how XDefinitions.h is set, a mismatch only logs a warning and decoding continues. Since build 3309 a blob shorter than its layout is rejected, because decoding would read past its end. A blob shorter than its layout is then read past its end. Only the 4.5.63 D1 check (at least 12 bytes) and the 8-byte minimum for D0 always throw.
- The decoder never checks the jj field or the terminator of a mission blob (CheckXMsgData isn't called).
- [decoder] DRA is held in 8 bits, so values above 255 are truncated. DRT and MSN are read unsigned and MSN is stored in an int16. ICF/ICW are read as U1 in layouts 02006900-04016100.
- Scaling [decoder]. Only the listing scales values; stored values are raw.

Parameters	Listed as
CSP, MTT, DSI, PSI, ASI, SSI, CPP1-3 (RBR), ARC	value / 1000 s
CPN1-3 (RBR)	value / 10 dBar
ARR	value / 100 m/s
AHR	value / 1000 m/s
PNB, PNC, PXC	value / 100
ICT	value / 100 degC
IST	value / 100 PSU
SAN, SAX	value / 1000 PSI

  Where NG10.5 gives other units, the table rows in C.1.7 say so.
- PPC values are IEEE floats, printed with 3 decimals. PPE is printed as an integer.

C.1.11 Listing output

The "Mission Configuration" section is printed once per dive, from the dive's D0 and D1. Three styles exist.

1. **`mis` groups**, for layouts `02006200` and `02006900` (firmware 2.0.62 to 4.1.61). A header line `mis SN dive idx 0xFWWORD` is followed by groups of fixed-width columns, one line per group with a minor index `midx`, each group with its own column-name and unit comment lines. Example group: `mis SN dive idx midx MSN DTD MDS APT XAS`.
2. **`MIS` header and `mis` groups**, for 4.2.62, 4.3.61, 4.4.61 and 4.6.61 (listing `04026200`). The header is `MIS SN dive idx 0xVV 0xFWWORD sensor-type # name`, followed by `mis` groups as in style 1.
3. **`MIS` header and `m` lines**, for 4.5.63 and every firmware from 5.0.61 on. Format:

```
#      SN dive        CfgMsgVer       FWVer sensor-type        sensor-type-name
MIS  9213    1             0x18  0x0A0C0032           0      # RBR
# --- General ---------------------------------------------------------------------------------
#      SN dive idx param      value      units          desc
m    9213    1   0 MSN            1      [-]          # dive cycle number
m    9213    1   3 MDS         1000      [dBar]       # maximum depth
m    9213    1  42 CPP1           1.000  [sec]        # interval 1 sample period
m    9213    1 133 ICT           -1.40   [degC]       # threshold temperature
```

   - `MIS` is `MIS %5d %4d 0x%02X 0x%08X %11d # name`: serial number, dive, parameters version (offset 8), firmware word, CTD type and its name (`RBR`, `SBE` or `unknown`).
   - Each `m` line is `m SN dive idx NAME value [units] # description`. `idx` counts the `m` lines from 0, in listing order, not in message order. Integers are printed with `%10d`. Scaled values use 1-4 decimals.
   - Sections, in order: General; BIT (layouts before 10.7 only); Calibration Coefficients; Start-of-Mission Trigger; Diagnostic Dive; Mission Phase; Profile; Profile - Dynamic Salinity Correction (only in the `0A050000`, `0A060231` and `0A080031` listings); Profile - Binning Configuration (CPN/CPP depend on the CTD type: RBR scaled, SBE raw dBar, unknown raw); Profile - Raw Sampling; Profile - Surface Offset Correction; Pump Characteristics; Descent (General, Fall Time-Series, Stages 1-3); Drift/Park (General, Seek, Start of Profile Trigger, Fall to Profile Pressure); Ascent/Profile (General, Rise Time-Series, Ice-Detection, Ice-Detection Summer and Winter); Surface (General, Air Pump, Sampling, Sat./Comm., Abort/Beacon Mode); Surface - Wave-Field Measurement (`WST`, only in the WIMS listings `08000000`, `0A060231` and `0A080031`; the `06057700` listing prints `WMT` instead, which that layout never decodes, so it always shows 65535).
   - A 10.12.0.x listing has 177 `m` lines.
   - The Python tools keep each `m` line (section, index, name, value, units) in the dive JSON as `mission.entries`, and keep `cfgver`, `fwver` and `ctdtype` from the `MIS` line.

- If no D0 has arrived, the firmware word is still 0xFFFFFFFF and the section is the single line `# Incomplete mission parameters.`
- Listing bugs [decoder]:
  - The `DRT` line prints `DRA`'s value. *(Fixed in build 3309.)*
  - In the dynamic-correction section, the `PPC0` line (labeled tau) prints alpha (`PPC1`), and the `PPC1` line (labeled alpha) prints tau. The README notes this as unfixed. *(Fixed in build 3309.)*
  - The `0A070101` listing, which covers 10.7 to 10.14 RBR/SBE including 10.12.0.x, has the dynamic-correction section commented out. `PPC`/`PPE` are decoded but not listed for those floats.
  - `DRA` is labeled `[m/s]` but printed raw (cm/s per NG10.5).

C.2 Mission command echo (0xDE)

Purpose

After it receives a ground-station command upload, the float sends back the command text it received, so that the ground station can confirm the upload (SOLO2: "ASCII listing of mission command received by float").

Format

Offset	Size	Type	Field	Notes
0	1	U1	ID	0xDE.
1	2	U2	`jj`	Length including ID, `jj` and terminator. Variable. DF7.3.5 gives a text field of up to 336 characters.
3	`jj` - 4	ASCII	Command echo	The command string as received.
`jj` - 1	1	char	`;`	Terminator.

XF2.4 defines only the ID, size and terminator.

Production [inferred, from all 2026 echoes on argus]. The echo is the uploaded command text exactly as sent: the commands separated by `;`, ending with `MIC 1` (which the upload itself ends with). Besides the text, the blob carries 8-9 bytes of undocumented binary data. Its position varies:

- **After the text:** seen in most echoes. The bytes are NULs, but the last one is sometimes not (0x01, 0x08 and 0x80 have been seen).

  ```
  DE 00 37 | "dft 0 100;dft 1 100;dft 2 100;duc 100;MIC 1" | 00 00 00 00 00 00 00 00 | 3B
  ```

- **Before the text:** all NULs. Seen in the dive 0 echoes of the Lowell ALAMOs (10.12.1) and of 11190 (10.12.0.1).

  ```
  DE 00 55 | 00 00 00 00 00 00 00 00 00 | "psi 1800000;pdt 1800;...;MIC 1" | 3B
  ```

The commands use the mission-parameter mnemonics, with a phase index where the parameter has one (`dft 0 100` sets `DFT0` to 100). They are echoed in whatever case they were uploaded in. A long upload may come back as two consecutive echo blobs in the same session. When several uploads are queued, each one is echoed separately. Because `;` also appears inside the text, the end of the blob must be found from `jj`, not by looking for the first `;`.

Verification against uploads (2026-10-25). The uploads sent with `upload-alamo-cmd` since 2026-01-01 (`/data1/alamo/data/sbd/outgoing`) were compared with the echoes in the RAW files:

- 261 uploads were echoed back exactly, some in two pieces.
- 23 were not echoed, all to floats that have not called in since.
- A few were not echoed although the float called in. One, to 9213 on 2026-09-28, verifiably did not take effect: its mission parameters still show the old values.
- A handful of echoes match no recorded upload (`ppe`, `wst` commands). They were presumably sent another way.

Decoding notes [decoder]

- `XMissionCmdEcho` calls `CheckXMsgData(data, 0xDE, -1)`. That checks the ID, that `jj` (lower 12 bits) is at least 4 and no more than the data received, and that the terminator is at `jj - 1`.
- Since build 3310 the text is decoded. Leading NULs are skipped, the text ends at the next NUL, and it is split at `;` into commands. Blank entries are dropped, surrounding spaces are trimmed, and other non-printable bytes are kept as `\xNN`. The trailing `MIC 1` is listed as a command like the others. (Before build 3310 the echo was checked but not decoded, and the listing section was printed empty.)
- Listing: under the section banner “Ground Station Upload Command Echo-Back Messages” (printed whenever a dive has at least one 0xDE blob), one line per command. The parameter name (the text before the first space) is listed in uppercase, as the mission parameter listings name them; the rest of the command is listed as received:

```
#      SN dive eidx cidx  command
CMD  9213    3    0    0  DFT 0 100
CMD  9213    3    0    1  DFT 1 100
CMD  9213    3    0    2  DFT 2 100
CMD  9213    3    0    3  DUC 100
CMD  9213    3    0    4  MIC 1
```

`eidx` numbers the dive’s echo-back messages, and `cidx` the commands within one.

C.3 Ocean surface waves (0xC1, 0xC3)

C.3.1 Purpose

Wave-measuring floats (ALAMO wave floats such as 9270, 9287-9306 and 9453, and WIMS floats) sample a 3-axis IMU at the surface. The float reduces the data on board to one-sided acceleration auto- and cross-spectra and sends one spectral component per blob. Shore-side processing gives wave height and direction. The format of the raw IMU time series was defined in XF2.4 but is not decoded. The decoder treats 0xC1 and 0xC3 the same way: the blob's version and type fields decide how it is decoded, not the ID.

In production [inferred], 0xC1 blobs carry the auto-spectra (`ZZ`, `YY` and `XX` co-spectra) and 0xC3 blobs carry the cross-spectra (`XY` co-spectrum, `ZX` and `ZY` quadrature spectra). One dive sends six blobs, one per component, often one per SBD packet. Floats whose segment is too long for one blob split a component over several blobs using the first-bin index.

C.3.2 Common header and version

Offset	Size	Type	Field	Notes
0	1	U1	ID	0xC1 or 0xC3. [decoder] The ID is not checked.
1	2	U2	`jj`	[decoder] Validated with `CheckXMsgData` (minimum 17 bytes). Only the lower 12 bits are used.
3	1	U1	Message version	0 and 1 = spectral v1 layout; 2 = spectral v2 layout (with a mode byte). Any other value is ignored. The decoder has read byte 3 as the version since release 0.6.3149 (2023-08

[inferred] DF7.3.5 lists the fields of blob 193 (0xC1) as `packet_index`, `mode`, `spectral_axis`, `spectral_type`, `segment_length_power`, `base_sample_rate`, `sample_rate_divider`, `full_scale_range_accelerometer`, `sample_index` and `sample_buffer`. These match the v1 layout below, from byte 4 on, if the version byte at offset 3 is assumed. DF7.3.5 leaves out the D0 header in the same way (C.1.9).

C.3.3 Spectral format, version 1 (and 0)

Used by the 8.x firmware (for example float 9270, 8.0.1.1, 273-byte blobs).

Offset	Size	Type	Field	Scale / units	Notes
0	1	U1	ID		0xC1 or 0xC3.
1	2	U2	`jj`	bytes	16 + 2·n + 1 for n bins.
3	1	U1	Version		0 or 1.
4	1	U1	Packet index		[inferred] DF7.3.5 `packet_index`. 0 in every production blob seen. Not decoded.
5	1	U1	Mode	enum	1 = spectral (0 = raw). [decoder] Not read for v0/v1: the mode is taken to be spectral.
6	1	U1	Spectral axis	enum, C.3.5	
7	1	U1	Spectral type	enum, C.3.5	
8	1	U1	Segment length power p		Segment (FFT) length N = 2^p. The blob carries one-sided FFT bins 0 to N/2 - 1: bin 0 is the DC bin, and there is no Nyquist bin (WIMS firmware v1.x and v2.x). Production: 8 (N = 256) or 9 (N = 512
9	2	U2	Base sample rate f_base	Hz	40 (v1 floats) or 45 (v2 floats
11	2	U2	Sample rate divider d		Effective sample rate f_s = f_base / (d + 1). For example 40/(9+1) = 4 Hz, 45/(17+1) = 2.5 Hz, 45/(8+1) = 5 Hz.
13	1	U1	Accelerometer full-scale range	g	2, 4, 8 or 16. Selects ASSF (C.3.6

Offset	Size	Type	Field	Scale / units	Notes
14	2	U2	First bin index k0		Index of the first bin in this blob (DF7.3.5 `sample_index`
16 + 2i	2	U2 / I2	Spectral value, bin k0 + i	counts, ÷ ESSF → g²/Hz	Co-spectra `ZZ`, `XX` and `YY` are read unsigned. `XY` (co) and `ZX`/`ZY` (quadrature) are read signed. i = 0 … n - 1, where n = (`jj` - 17)/2 and n ≤ N/2.
`jj` - 1	1	char	;		Terminator.

C.3.4 Spectral format, version 2

Used by the 10.x wave and WIMS firmware (for example 9287 and 9291 with 121-byte blobs of 51 bins, and 9293-9305 and 9453 with 275-byte blobs of 128 bins). It is version 1 with an accelerometer gain inserted at offset 14 and the mode byte made explicit.

Offset	Size	Type	Field	Scale / units	Notes
0	1	U1	ID		0xC1 or 0xC3.
1	2	U2	`jj`	bytes	18 + 2·n + 1. [decoder] Must be at least 19.
3	1	U1	Version		2.
4	1	U1	Packet index		[inferred] 0 in every production blob seen. Not decoded.
5	1	U1	Mode	enum	0 = raw, 1 = spectral. Above 1 throws “Invalid sample mode!”. For mode 0 nothing is decoded.
6	1	U1	Spectral axis	enum, C.3.5	
7	1	U1	Spectral type	enum, C.3.5	
8	1	U1	Segment length power p		N = 2^p.
9	2	U2	Base sample rate f_base	Hz	
11	2	U2	Sample rate divider d		f_s = f_base / (d + 1
13	1	U1	Accelerometer full-scale range	g	2, 4, 8 or 16.
14	2	U2	Accelerometer software gain (Q3)	÷ 8	Gain G = value / 8. Production: 0x0020, so G = 4.0.
16	2	U2	First bin index k0		
18 + 2i	2	U2 / I2	Spectral value, bin k0 + i	counts, ÷ ESSF → g²/Hz	Signedness as in v1. n = (`jj` - 19)/2.
`jj` - 1	1	char	;		Terminator.

Production v2 header (float 9293, dive 306, 275 bytes): `C1 01 13 | 02 | 00 | 01 | 05 | 01 | 08 | 00 2D | 00 11 | 04 | 00 20 | 00 00 | 00 03 00 02` … That is version 2, packet 0, spectral, axis ZZ, type CO, N = 256, f_s = 45/18 = 2.5 Hz, ±4 g, G = 4.0, k0 = 0, then 128 bins.

C.3.5 Enumerations

Spectral axis (byte 6):

Value	Name	Decoded	Value	Name	Decoded
0	undefined	no	8	X	no
1	Z′	no	9	XZ	no
2	Y′	no	10	XX	co (U2
3	X′	no	11	XY	co (I2
4	Z	no	12	Y	no
5	ZZ	co (U2)	13	YZ	no
6	ZX	quad (I2)	14	YX	no
7	ZY	quad (I2)	15	YY	co (U2

Spectral type (byte 7): 0 undefined (throws), 1 CO (co-spectrum), 2 QU (quadrature spectrum), 3 RE, 4 IM,

5 MAG and 6 PHASE (3-6 throw “Spectral type not supported!”). [decoder] An axis/type pair other than CO with ZZ, XX, YY or XY, or QU with ZX or ZY, throws, and the blob is discarded.

Mode (v2 byte 5): 0 = raw, 1 = spectral.

C.3.6 Scale factors

Table	Index or key	Values	Units	Used
`ASSF_SPECTRAL_V1`	full-scale range 2, 4, 8, 16 g	8192, 2048, 512, 128	counts per g²/Hz	v0, v1
`ASSF_SPECTRAL_V2`	2, 4, 8, 16 g	8192, 2048, 512, 128	counts per g²/Hz	v2. Before build 3238 (2025-10) it was {4096, 1024, 256, 64}, which made `wvs`/`wvm` values twice as large.
`ASSF_RAW`	AFS_SEL 0-3 (±2, 4, 8, 16 g)	16384, 8192, 4096, 2048	LSB/g	raw format (not decoded
`GSSF`	FS_SEL 0-3 (±250, 500, 1000, 2000 °/s)	131.072, 65.536, 32.768, 16.384	LSB/(°/s)	raw format (not decoded
`MSSF`	0	30/9 ≈ 3.333	LSB/µT	raw format (not decoded
`FSAM`	0	1	Hz	raw format (not decoded

- An invalid full-scale-range byte logs an error and falls back to ASSF = 2048 (the 4 g entry).
- Effective spectral scale (WIMS user manual rev. 3.15, “Scale Sensitivity”): ESSF = 32768 · ASSF · G² · U · f_e² / (2 N²), in counts per g²/Hz, where:
  - G = Q / 8 is the acceleration gain (Q the raw Q3 gain; G = 1 when Q is 0, and for v0/v1);
  - U = 1 - (5L/4 + 1) / N is the mean square of the firmware’s Tukey window, with taper length L = round((N - 1) · 64 / 256) (`wav.twalp` = 64, which the header does not carry; U = 0.68359 for N = 256);
  - f_e = f_base / (d + 1) is the true effective sample rate, not the doubled rate of C.3.8.

  Example: v2 at ±4 g, G = 4, N = 256, f_e = 2.5 Hz gives ESSF = 35000; at f_e = 5 Hz, 140000. Before 2026-10-28 (build 3332) the decoder used ESSF = ⌊ASSF · Q² / 64⌋ = ASSF · G², the formula of manual revisions up to 3.14, which ignores f_e, N and the window (32768 in the example).
- Acceleration spectrum: S_aa [g²/Hz] = counts / ESSF.
- Displacement spectrum [decoder]: S_ηη [m²/Hz] = g² · S_aa / (2πf)⁴, with g = 9.80665 m/s² and π = 3.14159265359. It is clamped to at most +999.9999. The cross terms `Cxy`, `Qzx` and `Qzy` are also clamped to at least -999.9999.
- [decoder] ASSF, gain, f_base, d and N are kept per dive and overwritten by each blob, so the last blob decoded sets the scale for all components.

C.3.7 Frequency bins

For bin k (0 … N/2 - 1), f_k = f_s · k / N, with f_s = f_base / (d + 1). Bin 0 is the DC bin; the last bin is f_s/2 - f_s/N. WIMS firmware up to 2.01.00614b labels its own output one bin high, (k + 1) · f_s / N, and the decoder did the same until 2026-10-26 (see below).

- Build 3238 (2025-10) went back to `fs·(i+1)/seglen`.
- 2026-10-26: `fs·(k0+i)/seglen`, with bin 0 the DC bin. The WIMS firmware copies FFT bins 0 to N/2 - 1, so `fs·(i+1)/seglen` was one bin too high; an intercomparison with NDBC buoy 41047 confirmed the offset.
- The 2026-10-24 fix added k0 to k, so bins in a blob with k0 > 0 get the right frequency. Before that, the frequency ignored k0.
- [decoder] The spectral array holds N/2 bins per dive. A bin that no blob filled keeps frequency -99 and fill values. For example 9287 and 9291 send only bins 0-50 of N/2 = 256.

- Newer sensor firmware truncates the spectrum to a range of frequencies, so a blob may carry fewer than N/2 bins (n from `jj`, as in C.3.3 and C.3.4). [decoder] Only the n bins in the blob are decoded; before build 3315 the decoder tried all N/2 and logged "wave sample index exceeds message size" for every truncated blob.
- [decoder] Bins that would reach past N/2 (k0 + i ≥ N/2) are not decoded, and a warning is logged; before build 3315 `vector::at` threw and the whole blob was discarded.

C.3.8 Every-other-value correction (WIMS firmware)

[inferred from production; README 0.6.3235] Firmware 0x0A060231, 0x0A080031, 0x0A090131 and 0x0A0A0031 send only every other spectral value: the first half of the N/2 bins holds the spectrum at twice the bin spacing, and the second half is zeros. [decoder] `XFloatDiveData::ResolveWaveBinSpacing()` runs after each 0xC1/0xC3 blob and after each mission configuration message, so the result does not depend on message order. It takes the firmware word from the dive's own D0 or, without one, the float's latest; if that word is one of these, `XWave::set_every_other_bin(true)`:

- sets f_s' = 2 · f_base / (d + 1), which also becomes the `WVS fsam`; the ESSF (C.3.6) still uses the true f_e;
- sets the frequency of every received bin, i = 0 … N/2 - 1, to f_i = f_s' · i / N (f_s' · (i + 1) / N before 2026-10-26); bins not received keep -99. [inferred from the NDBC 41047 intercomparison, whose offset matches one decoded bin] The floats send the even WIMS bins (0, 2, 4, …), so bin i holds WIMS FFT bin 2i.

When the firmware word is unknown (it is not kept across decoder restarts), `XWave::LooksLikeEveryOtherBin()` decides from the data: every bin of the upper half was received and is zero in all six components, and some bin of the lower half has a nonzero Szz. Truncated spectra leave the upper bins unreceived, not zero, so they do not match. Before 2026-10-28 (build 3332) the correction was keyed only on the float's latest known firmware word, so blobs decoded before any D0 had been received were not corrected.

The spectral values are left as they are, so the zero second half is still listed, at frequencies above f_s/2. 10.12.1.31 (0x0A0C0131) and 10.7.1.31 (0x0A070131) are WIMS firmware too, but they are not in the correction list (C.5).

C.3.9 Raw IMU format (XF2.4 "Raw Wave Field"; not decoded)

Offset	Size	Type	Field	Scale	Notes
0	1	U1	ID		0xC1.
1	2	U2	`jj`		
3	1	U1	Packet index		Packet order for rebuilding the time series.
4	1	U1	Sample frequency index	`FSAM`	0 = 1 Hz.
5	1	U1	ASSF index	`ASSF_RAW`	
6	1	U1	GSSF index	`GSSF`	
7	1	U1	MSSF index	`MSSF`	
8	1	U1	Mode (high nibble) / phase code (low nibble)		Mode 0 = raw, 1 = spectral. Phase code 1-14 (14 = surface). The nibble order is [inferred] from the commented-out decoder, which takes the phase code from `d[8] & 0x0F`.
9 + 18n	2	I2	Accel x, sample n	÷ ASSF → g	Starboard.
11 + 18n	2	I2	Accel y	÷ ASSF → g	Bow.
13 + 18n	2	I2	Accel z	÷ ASSF → g	Up.
15 + 18n	2	I2	Gyro x	÷ GSSF → °/s	XF2.4: bow-up (pitch). The decoder comment calls it gamma, starboard-up.
17 + 18n	2	I2	Gyro y	÷ GSSF → °/s	XF2.4: starboard-up (roll). The decoder comment calls it beta, bow-up.
19 + 18n	2	I2	Gyro z	÷ GSSF → °/s	Increasing azimuth.
21 + 18n	2	I2	Mag x	÷ MSSF → µT	
23 + 18n	2	I2	Mag y	÷ MSSF → µT	
25 + 18n	2	I2	Mag z	÷ MSSF → µT	

Offset	Size	Type	Field	Scale	Notes
18N + 27	1	char	;		N = index of the last sample.

This layout has no version byte. Byte 3 is the packet index and byte 5 the ASSF index. The current decoder would therefore read a raw packet as a spectral version (packet index 0 or 1) or ignore it. In the v2 layout, byte 5 is the mode, and mode 0 (raw) is accepted but nothing is decoded. The old raw decoder and its `WAV`/`wav` listing lines are commented out in `XWave.cpp`.

C.3.10 Validation summary [decoder]

1. `CheckXMsgData(data, -1, 17)`: the blob is at least 17 bytes, `jj` (lower 12 bits) is at least 4 and no more than the data received, and the byte at `jj - 1` is 0x3B. Extra bytes after `jj` only give a warning.
2. Version 0, 1 or 2; any other version is ignored without an error.
3. v2: mode 0 or 1 (otherwise it throws), and `jj` at least 19.
4. Each bin is read only while its two bytes lie before the terminator. For v1 the loop stops when 2i + 17 ≥ `jj`; for v2, when 2i + 20 ≥ `jj`. At most N/2 bins are read. The 2026-10-24 fix corrected the v2 bound, which used to read one bin past the terminator and broke the 51-bin floats' products.
5. Unsupported axis/type combinations throw, and the blob is dropped (`CATCH_BLOCK`).

C.3.11 Listing output (`WVS`, `wvs`, `wvm`)

The section "Ocean Surface Wave-Field Data" is printed when the dive has wave data with N > 0. Only versions 1 and 2 print anything: a dive whose wave blobs are all version 0 gets an empty section. In the example, the `wvs` and `wvm` values were listed before build 3332 with ESSF 32768; the current decoder lists them about 7% lower (ESSF 35100).

```
#      SN dive vers slen     fsam fbase fdiv     assf  again       essf
#                           (Hz)  (Hz)      (LSB/g^2)   (LSB/(g^2/Hz))
WVS  9287  325 0x02  512   5.0000   45     9     2048  4.000    35100.0
#      SN dive  idx     freq         Szz         Syy         Sxx         Cxy         Qzx         Qzy
#                      (Hz)    (g^2/Hz)    (g^2/Hz)    (g^2/Hz)    (g^2/Hz)    (g^2/Hz)    (g^2/Hz)
wvs  9287  325    0   0.0098    +0.00018    +0.00098    +0.00156    +0.00046    +0.00000    +0.00000
wvs  9287  325   51 -99.0000  -999.99999  -999.99999  -999.99999  -999.99999  -999.99999  -999.99999
#      SN dive  idx     freq         Szz         Syy         Sxx         Cxy         Qzx         Qzy
#                      (Hz)    (m^2/Hz)    (m^2/Hz)    (m^2/Hz)    (m^2/Hz)    (m^2/Hz)    (m^2/Hz)
wvm  9287  325   50   0.4980     +0.0056     +0.0159     +0.0109     +0.0003     -0.0038     -0.0024
wvm  9287  325   51 -99.0000  -1000.0000  -1000.0000  -1000.0000  -1000.0000  -1000.0000  -1000.0000
```

- `WVS` (`WVS %5d %4d 0x%02X %4d %8.4f %4d %5d %8d %6.3f %10.1f`; `essf` was `%8d` before build 3332) has these columns:
  - `vers`: the version;
  - `slen`: N;
  - `fsam`: f_s, doubled by the C.3.8 correction;
  - `fbase`: f_base;
  - `fdiv`: d + 1, not the raw divider;
  - `assf`;
  - `again`: G;
  - `essf`.
- `wvs` (`%8.4f` frequency, `%+11.5f` values) gives one line per bin, idx = 0 … N/2 - 1, with the acceleration spectra in g²/Hz. The columns are always in the order **zz, yy, xx, xy, zx, zy**: Szz, Syy, Sxx, Cxy, Qzx, Qzy.

- `wvm` (`%+11.4f`) gives the same bins as displacement spectra in m²/Hz. It **must use the same column order as `wvs`** (zz, yy, xx, xy, zx, zy) under the same header. Builds 3234-3289 (2025-09 to 2026-10-24) printed `wvm` as xx, xy, yy, zz, zx, zy under the zz-first header, so `wvm` columns in listings from those builds are mislabeled. The 2026-10-24 fix restored the `wvs` order. Nothing downstream parses `wvm`.
- Fill: a component never received is -999.99999 in `wvs`. In `wvm` the same fill prints as `-1000.0000`, because `%+11.4f` rounds -999.99999. Unfilled bins have frequency -99.0000.

C.4 Marginal-ice-zone acoustic messages (0xCA)

C.4.1 Purpose

MIZ floats (for example 9213 and 9216 with firmware 10.12.0.32) listen for acoustic navigation and communication signals (CACST, "communication and navigation" pings) in scheduled listening windows. Each listening window and each received signal is reported in its own 0xCA blob, with a subtype byte at offset 3. Shore-side processing uses the times of arrival to position the float under ice. The type ID was 0xC0 in decoder builds before 3249 (2026-05/06); the decoder now uses 0xCA (`XMSG_TYPE_ID_MIZ_ACOUSTIC`). XF2.4 lists 0xC0 "acoustic position" as unsupported, with an empty sheet. DF7.3.5's blob 192 (0xC0) is a different text payload (`sensor_type`, `payload_index`, 333-character text).

C.4.2 Common header

Offset	Size	Type	Field	Notes
0	1	U1	ID	0xCA. [decoder] Checked.
1	2	U2	`jj`	[decoder] `CheckXMsgData(data, 0xCA, 4)`: `jj` (lower 12 bits) at least 4 and no more than the data received, terminator at `jj - 1`.
3	1	U1	Subtype	3 = listening window, 5 = CACST receive record. Other subtypes are ignored without an error (1 and 2 seen in production, C.4.5

C.4.3 Subtype 3: listening window (37 bytes)

Offset	Size	Type	Field	Units	Notes
0	1	U1	ID		0xCA.
1	2	U2	`jj`		37 (0x0025
3	1	U1	Subtype		3.
4	4	U4	Window start time	UNIX s (UTC)	
8	4	U4	"Pressure" at start	unknown	[decoder] Named `p_start`, not listed. Production values 0x00E5-0x02DF (229-735) on floats parked at 250-300 dBar; the scale is unknown.
12	4	U4	"Temperature" at start	unknown	[decoder] Named `t_start`, not listed. Always 0 in production.
16	4	U4	"Salinity" at start	unknown	[decoder] Named `s_start`, not listed. In production the high 16 bits are about 0x0021-0x0022 (33-34) and the low 16 bits 0x002A-0x002D (42-45). [inferred] It may be two 16-bit quantities, not one U4.
20	4	U4	Window end time	UNIX s (UTC)	
24	4	U4	"Pressure" at end	unknown	As offset 8.
28	4	U4	"Temperature" at end	unknown	As offset 12.
32	4	U4	"Salinity" at end	unknown	In production the low 16 bits are 0.
36	1	char	;		

Example (9213): `CA 00 25 03 6AB85C02 00000137 00000000 0022002D 6AB863B9 0000007F 00000000 00200000 3B` is a window from 2026-09-26T23:57:54Z to 2026-09-27T00:30:49Z.

C.4.4 Subtype 5: CACST receive record (33 bytes)

Offset	Size	Type	Field	Units / scale	Notes
0	1	U1	ID		0xCA.
1	2	U2	`jj`		33 (0x0021
3	1	U1	Subtype		5.
4	4	U4	Counter		[decoder] Named `counter`, not listed. In production it goes up by 1 per record, and its first value in each session looks like a UNIX time shortly before the first window (for example 0x6AB826EB = 2026-09-26T19:58:03Z) [inferred].
8	4	U4	Time of arrival, seconds	UNIX s (UTC)	
12	4	U4	Time of arrival, microseconds	µs	Values above 999999 make the listed time invalid.
16	4	U4	Matched-filter peak power	dB	Production values 41-45.
20	2	I2	Source ID		-1 (0xFFFF) in every production record. [decoder] Read as U2 and stored as int16.
22	2	U2	PSK error code	enum	4 in every production record. The codes are not documented.
24	4	I4	Input SNR	0.1 dB	Production: 0xFFFFFC19 = -999, so -99.9 dB, apparently a “no value” marker [inferred].
28	4	I4	Output SNR (after the equalizer)	0.1 dB	Production: 0xFFFFFF9D = -99, so -9.9 dB.
32	1	char	;		

Example (9213): `CA 00 21 05 6AB826ED 6AB85CEF 00033D5F 0000002D FFFF 0004 FFFFFC19 FFFFFF9D 3B` is a reception at 2026-09-27T00:01:51.212319Z with MFD power 45 dB, source -1, PSK error 4, SNR in -99.9 dB and SNR out -9.9 dB.

C.4.5 Subtypes seen in production but not decoded [inferred]

Subtype	`jj`	Layout as seen	Notes
1	10	`CA 00 0A 01` · U1 (always 0x23) · U4 UNIX time · `;`	The times are on a 4-hour grid, for example 0x6AB82404 and 0x6AB85C44. Possibly a schedule or event marker.
2	14	`CA 00 0E 02` · U1 (always 0x21) · U4 UNIX time · I4 · `;`	The times fall 120 s before the subtype 1 times. The I4 values are small and signed (-120 to +49), so possibly a clock offset or a measurement.

C.4.6 Decoding notes [decoder]

- Every 0xCA blob of a dive goes into one `XAcousticPosition` object. A window or CACST blob must be exactly 37 or 33 bytes (the received blob length); otherwise it is rejected with an error.
- Before listing, `AssignWindowIndices()` numbers the windows 0, 1, ... in the order they were received, not in time order. Each CACST record gets the index of the first window with start ≤ ToA seconds ≤ end, and that window’s receive count goes up. A record outside every window keeps index -1.
- Window P/T/S, the counter and the per-window receive count are decoded but not listed.

C.4.7 Listing output (`APS`)

`APS` lines appear under “Position Data / Acoustic Position”. That section is printed only when the dive also has GPS fixes or mail messages; acoustic data alone no longer counts as position data.

```
# --- Acoustic Listening Windows -------------------------------------------------------------
#      SN dive midx widx      start-date/time        end-date/time
#                                       (UTC)                (UTC)
APS  9213    1    3    0 2026-09-26T23:57:54Z 2026-09-27T00:30:49Z
# --- Acoustic Communication Cycle Receive Statistics ----------------------------------------
#      SN dive midx widx                  date/time    src MFDPwr PSKErr  SNRIn SNROut
#                                             (UTC)          (dB)          (dB)   (dB)
APS  9213    1    5    0 2026-09-27T00:01:51.212319Z     -1     45      4  -99.9  -99.9
```

- Window lines (`APS %5d %4d %4d %4d %20s %20s`): `midx` is the subtype (3), `widx` the window index, then the start and end times. An unset time prints as `0000-00-00T00:00:00Z`.
- Receive lines (`APS %5d %4d %4d %4d %27s %6d %6d %6d %6.1f %6.1f`): `midx` = 5, `widx` (-1 if the record is in no window), ToA with microseconds (an invalid time prints as `0000-00-00T00:00:00.000000Z`), source ID, MFD power, PSK error code, SNRIn and SNROut in dB.
- **Bug:** the SNROut column prints SNRIn again. The example above should show -9.9 for SNROut. The README notes this as still present. *(Fixed in build 3309.)*

C.5 Discrepancies and open questions

Mission configuration:

1. **Parameters version.** NG10.5 and NG10.12 document `MissionParamsVersion` = 23 (0x17). Production 10.12.0.x floats send 0x18. The decoder doesn't use the field.
2. **Byte 3 of D0.** XF2.4 calls it the encoding type (0 = binary, otherwise ASCII). NG10.5 calls it "Message version" and NG10.12 `MissionConfigVersion` = 0. The active decoder ignores it, so a SOLO-II ASCII dump would be misread (C.1.2).
3. **Byte 7 of the firmware word.** The task calls it "build". NG10.5 calls it "Float model" and NG10.12 "Product". Values such as 0x31 (WIMS) and 0x32/0x33 (MM2/Micro-ALTO) suggest a product or variant code. The decoder uses all 32 bits as the key.
4. **`DRT` type.** NG10.5 and NG10.12 give I2; the decoder reads U2. **`MSN`**: NG10.5 says I2, NG10.12 U2.
5. **XF2.4 against the decoder for old firmware:**
   - 2.0.69 (XF2.4: 323 bytes with `CTF`/`CTL`/`CTU` but no ice block; decoder: 322 bytes with ice, the same as XF2.4's 2.5.62 layout).
   - 3.0.65 (XF2.4: 331 bytes ending `SPT`, `SSP`; decoder: 328 bytes, no `SPT`/`SSP`).
   - 4.2.62 (XF2.4: 327 bytes with no gap at 119 and a 5-byte "empty?" field at 161; decoder: 323 bytes, `IST` at 120, `DNS` at 162).
   - 4.6.61 (XF2.4: one 336-byte D0 ending `SSP`; decoder: 328 bytes ending `SAN`, and D1 not decoded).

   The decoder's sizes presumably come from real messages; the spreadsheet looks like intent.
6. **`ICF`/`ICW` as U1** in the 2.0.69 to 4.1.61 decoders: only the most significant byte of each 4-byte mask is decoded.
7. **10.7.1.1 D1** is 88 bytes and probably carries `PPC`/`PPE`, but the decoder doesn't decode them.
8. **0x08020000 and 0x0A020201** decode but have no listing function, so they get the generic "unsupported" listing.
9. **Listing bugs:**
   - `DRT` prints `DRA`. *(Fixed in build 3309.)*
   - The `PPC0`/`PPC1` values are swapped against their labels. *(Fixed in build 3309.)*
   - `PPC`/`PPE` are not listed for 10.7-10.14 non-WIMS floats.
   - The 6.5.77 listing shows `WMT` = 65535 because that layout doesn't decode it.
10. **Units:**
    - `AHR` (NG10.5: cm/s; listed value/1000 m/s, so 1000 lists as 1.000 m/s).
    - `ASI` (NG10.5: s; listed value/1000 s).
    - `IST` (header comment: 1/10 PSU; listed value/100).
    - `SAN`/`SAX` (NG10.5: psi; listed value/1000).
    - `DRA` labeled m/s, printed raw.
    - `XAS` and `CST` codes differ between NG10.5 and XF2.4.

These need confirming with MRV.

11. **SOLO-II ASCII record length.** SOLO2 says 15 bytes per parameter, but its template `NAMExx=vvvvv|` is 13 characters. And 324 bytes of text per 328-byte blob is not a whole number of either.
12. **D2-D7.** SOLO-II used D0-D3 for the ASCII dump. The binary decoder rejects anything above D1.

Command echo:

13. **0xDE.** The meaning of the trailing `MIC 1` and the 8 NUL bytes is unknown. Neither the decoder nor the listing shows the echo text.

Waves:

14. **0xC1 against 0xC3.** No document gives the difference. In production, C1 carries the auto-spectra and C3 the cross-spectra; the decoder treats them the same.
15. **Byte 4 of the wave header** (packet index per DF7.3.5, after alignment) is always 0 in production and not decoded. DF7.3.5 leaves out the version byte.
16. **Raw IMU format (XF2.4)** has no version byte and conflicts with the version/mode bytes of the spectral formats. It is not decoded.
17. **The every-other-value correction** covers 0x0A060231, 0x0A080031, 0x0A090131 and 0x0A0A0031. The other WIMS words, 0x0A070131 and 0x0A0C0131, are not covered; whether they need it is unverified.
18. **ASSF_SPECTRAL** values are an empirical table (counts per g^2/Hz). The v2 table was changed in build 3238. The derivation from the IMU LSB/g is not documented.
19. **Raw gyro axis names:** XF2.4 and the decoder comments disagree on pitch and roll.

Acoustics:

20. **0xCA P/T/S fields** don't look like plain pressure, temperature and salinity in production (temperature always 0, salinity seemingly two 16-bit halves). Their scaling is unknown.
21. **The CACST "counter"** looks like a UNIX-time seed plus a running count.
22. **PSK error codes and source ID** semantics are undocumented.
23. **Subtypes 1 and 2** are not decoded.
24. **The SNROut listing column** repeats SNRIn. *(Fixed in build 3309.)*

C.6 Extended records (0xFF)

No format document describes ID 0xFF; the decoder's ID list calls it `UNKNOWN`. Everything below is **[inferred]** from the 6,262 0xFF blobs in the production RAW files (survey of 2026-10-27, 160 floats, all firmware from 10.2 to 10.14, WIMS and not). Before build 3321 the decoder logged each one as `Unknown X message ID: 0xff` and dropped it (several hundred warnings a day).

Layout. The usual ID and 12-bit size (bytes 0-2, A.1.3) and `;` terminator, with a record type at byte 3 and, except for 0xDB, its version at byte 4. Several record types are also X-message IDs (0xCA, 0xD1, 0xDB, 0xEC), but the records do not match those messages: the type is a separate code space.

Type	Version	Size	Blobs	Floats	When	Record (byte offsets in the blob)
0xCA	0x20	17	2,821	160	95% pre-mission	event: u24 code @5 (0x004002 in 83%; also 0x000100, 0x004000-0x004003, 0x002000, 0x000600, 0x001000), u32 time @8, f32 value @12
0xD1	0x10	22	2,809	155	95% pre-mission	u16 counter @5, u32 time @7, u16 count @11, u32 time @13, u32 value @17 (0 in nearly all)
0xAE	0x11 (0x10)	13 (12)	483	117	pre-mission only	u16 0 @5, u16 float serial number @7, then 3 (v0x10: 2) configuration bytes

Type	Version	Size	Blobs	Floats	When	Record (byte offsets in the blob)
0xEC	0x05, 0x08	15, 10	89	4	in mission (2 pre-mission)	not decoded, e.g. `05 01 0e 00 01 07 01 0b 00 01` (11151)
0xFA	0x0A, 0x02	281	24	10	23 pre-mission	text: a processor fault dump, or all NUL
0xDB	—	12-328	36	4	in mission	CTD exchange: u16 command length @4, command, u16 reply length, reply

- **Times** are UNIX seconds (e.g. 0x62BF7945 = 2022-07-01T22:46:29Z, 23 minutes before that packet's Iridium session), or seconds since the controller started when they are below 2000-01-01 (37% of the 0xCA times, e.g. 402). The decoder lists the latter as `+402s`.
- **0xCA** records come in pairs with a 0xD1 record 8-17 s earlier, several pairs per startup packet. The value is a few tenths of a unit around zero (-0.32 to +0.10 in the examples), most likely the surface pressure in dBar measured at the event; code 0x004002 is the most common event. The 0xD1 counter increases by one per pair across a float's life (4 on 9166 in 2022; 71-73 on 9304 in 2026), so it is probably a controller start (reset) count, with the start time and, in the second time field, an uptime in seconds (`+207s`). What `count` counts is unknown.
- **0xAE** carries the float's serial number (0x23FB = 9211, 0x245B = 9307, 0x2D6D = 11629) and 2-3 small configuration values, sent during the startup tests. They may be the pump and battery types that the BIT format 11 "traveler" errors (B.2.6, MissingPumpType, MissingBatteryType) refer to.
- **0xFA** is an ARM Cortex-M fault report: `r7`, `sp`, `excrt` (EXC_RETURN), `psr`, the fault status registers `hfsr`, `cfsr`, `mmfar`, `bfar`, `shcsr`, and the stacked `s.r0`-`s.r3`, `s.r12`, `s.lr`, `s.pc`, `s.psr`, one per line (CR LF), in hex. It records a controller crash, for example 9304 dive 0 (firmware 10.12.1.31) with `cfsr 00000082` (a data access violation at a precise address). Empty reports are all NUL.
- **0xDB** records the RBR CTD exchanges of 9280 (10.6.2.1) and three other floats (11148, 11151, 11330): the command text, e.g. `ctdrw fetch\r|5` (a literal backslash-r; the `|` digit is perhaps a timeout), and the CTD's reply, e.g. `2024-03-15 22:57:12.000, 25.1567, 10.1491, 117.6462, 25.6469\r\n` or a calibration (`calibration 3 label = pressure_00, ...`) or `id model = RBRargo3, ...` line. The lengths account for every byte of the blob.

Listing. Under the section banner "Extended Record Messages (0xFF)", one line per record: `XFF SN dive ridx 0xTT 0xVV record`, where `record` is the decoded fields of the record types above, the text of 0xDB and 0xFA records (CR, LF and other non-printable bytes escaped as `\r`, `\n` and `\xNN`), or `hex` and the record bytes for other types:

```
#      SN dive ridx type  ver  record
XFF  9166   -1    0 0xCA 0x20  event 0x004002  time 2022-07-01T22:46:29Z  value -0.0100
XFF  9166   -1    1 0xD1 0x10  counter 4  time 2022-07-01T22:46:12Z  count 1  time +62s  value 0
XFF  9211  -15    0 0xAE 0x11  serial 9211  config 0x01 0x02 0x01
XFF  9280    2    5 0xDB 0x00  command "ctdrw fetch\r|5"  reply "2024-03-15 22:57:12.000, 25.1567, 10.1491, 117.6462,
↪  25.6469\r\n"
XFF  9304    0   35 0xFA 0x0A  text "\nr7    20016E60\r\nsp    20016E30\r\nexcrt FFFFFFFD\r\n..."
```

Open questions: the meaning of the 0xCA event codes and value, the 0xD1 `count` and last field, the 0xAE configuration bytes, the 0xEC records, and the `|` digit of the CTD commands.

C.7 WIMS configuration parameters (0xC5)

[inferred from production] No format document describes this message. A WIMS float sends the configuration of its WIMS wave sensor in it:

- once in its diagnostic dive (dive 0), and
- in the next dive after a `WCG -1` upload (WIMS Command Get, all parameters).

The production RAW files (2014–2026) hold 54, from 21 floats:

- dive 0 of 9246, 9248, 9249, 9265–9271, 9273, 9291–9293, 9296, 9297, 9299, 9303 and 9304;

- after `WCG -1` uploads to 9239 and 9245 (2020), and 9292, 9296 and 9297 (2024).

9310 echoed a `WCG -1` upload in dive 201 (2026-09-24) but sent no 0xC5. Before build 3336 the decoder logged 0xC5 as an unknown X message.

Offset	Size	Type	Field	Notes
0	1	U1	ID	0xC5.
1	2	U2	`jj`	12-bit size.
3	1	U1	Version	1: 42 parameters (2020 floats, 90 bytes). 2: 43 parameters (2024 floats, 92 bytes).
4	1	U1	Part index	0. One dive of 9239 had three copies with parts 0, 1 and 2, otherwise identical except parameters 36 and 37.
5 + 2i	2	U2 / I2	Parameter i	One value per WIMS parameter, by the float's parameter index (the index of the `WCG` and `WCS` commands).
jj − 1	1	char	;	

The parameter index is the float's, not the WIMS user manual's: `WCS` uploads to 9296 and 9297 (`WCS 24 45; 28 10; 35 5; 39 17; 42 32`) appear at exactly those positions in the next 0xC5.

Uploads. `WCG <index>` (WIMS Command Get; 0-41, or -1 for all) and `WCV` (WIMS Command saVe) appear to relay the WIMS firmware's `get` and `save` commands; `WCS <index> <value>` relays `set`, which changes a parameter in RAM until `save` writes it to the sensor's EEPROM. The Deployment and Piloting Guide v1.0 lists all three as Reserved (Auxiliary Sensor Systems). Every upload in `/data1/alamo/data/sbd/outgoing` that uses them, but the first one to 9239, is zero or more `WCS` commands followed by `WCG -1; WCV;`:

- 2020-07 to 09, 9239 and 9245 (about 20 uploads): `wcs 22` 2 or 4, `wcs 30` 7 or 14, `wcs 34 180`, `wcs 36` 0, 65, 98, 131 or 164, `wcs 39` 9, 19 or 39, `wcs 40` 64, 128 or 256.
- 2020-09-16, 9105, 9246-9249 and 9265-9273: `wcs 22 2; 30 14; 34 180; 36 131; 40 256` (9105 also `wcs 20 0`, then `wcs 31 6`).
- 2024-09-26, 9292, 9296 and 9297: `WCS 24 45; 28 10; 35 5; 39 17; 42 32`, then `WCS 31 2`.
- 2026-09-24, 9310: `WCS 39 17; 42 32`.

Index	Name	Confirmed by
0	`brd.leds?`	always 0
1	unknown	always 0
2–4	`cal.acofx`, `cal.acofy`, `cal.acofz`	accelerometer offsets, e.g. -213, -352, -1193 (signed)
5–7	`cal.acscx` … `cal.acscz`	accelerometer scales, 0 in every message (signed)
8–10	`cal.gcofx` … `cal.gcofz`	gyroscope offsets (signed)
11–13	`cal.gcscx` … `cal.gcscz`	gyroscope scales, 0 in every message (signed)
14–16	`cal.mcofx` … `cal.mcofz`	magnetometer offsets (signed)
17–19	`cal.mcscx?` … `cal.mcscz?`	per-float values such as 1300, 0, 181 (signed)
20	unknown	1 (v1), 0 (v2); `wcs 20 0`
21	`imu.dlpf?`	20 or 24
22	`imu.fsra`	2 or 4 g; matches the `WVS` ASSF (8192 at 2 g, 2048 at 4 g); `wcs 22` 2 and 4
23	`imu.fsrg`	500 °/s
24	`imu.srate`	40, 45 or 125 Hz; matches the `WVS fbase`, and `WCS 24 45`
25–26	unknown	0x0A62, 0xEE57 in every message
27	unknown	per float and dive, e.g. 5, 8, 25, 57, 66
28	`wav.afilt?`	2 or 10; `WCS 28 10`
29	`wav.mxsam?`	32768
30	`wav.mxseg?`	7, 12 or 14; `wcs 30` 7 and 14
31	unknown	0, 2 or 6; `wcs 31` 2 and 6
32	`wav.nfwax?`	0
33	`wav.obeta?`	1343, the manual's default
34	`wav.ostsc?`	90 or 180 s; `wcs 34 180`
35	`wav.omode?`	5; `WCS 35 5`
36	`wav.ozeta?`	65, 98, 131 or 164; changes between messages of the same float because it was tuned with `wcs 36` 0, 65, 98, 131 and 164
37	unknown	changes between messages of the same float, 1–16
38	`wav.smode?`	2

Index	Name	Confirmed by
39	wav.srdiv	4, 9 or 17; WVS fdiv = value + 1 (doubled for every-other-bin firmware, C.3.8), and wcs 39 9, 17, 19 and 39
40	wav.slen	64 or 256; matches the WVS slen; wcs 40 64, 128 and 256
41	wav.twalp?	64, the value the ESSF (C.3.6) assumes
42	wav.gaina	v2 only; 32 = 4.0, matching the WVS again, and WCS 42 32

Names ending in "?" are inferred from their values (defaults in the WIMS user manual, rev. 3.15) and position; the order is close to, but not the same as, the manual's alphabetical order. Values are listed signed for the calibration coefficients and unsigned otherwise.

Listing. Under the section banner "WIMS Configuration Parameters (0xC5)", one line per parameter: `WIP SN dive midx ver part pidx name value hex`, with `midx` the message's index in the dive and - for an unknown name:

```
#      SN dive midx  ver part pidx  name         value     hex
WIP  9297   15    0    2    0    2  cal.acofx     -213  0xFF2B
WIP  9297   15    0    2    0   24  imu.srate       45  0x002D
WIP  9297   15    0    2    0   25  -             2658  0x0A62
WIP  9297   15    0    2    0   39  wav.srdiv       17  0x0011
WIP  9297   15    0    2    0   42  wav.gaina       32  0x0020
```

Open questions: parameters 1, 20, 25–27, 31 and 37, the inferred names, and why 9310 sent no reply to `WCG -1`.

Appendix C.A: Offset maps of every MFA layout (decoder)

`NAME@offset` gives the decoder's read offset, counted from the ID byte. Fields are U2 unless a suffix says otherwise (`:U1`, `:U4`, `:I2`, `:R4`). `T_DESCENT`, `T_PARK`, `T_ASCENT` and `T_SURFACE` are U4 state start times in seconds since 2000-01-01. Gaps between fields are bytes that exist but aren't decoded (see C.1.9). In layouts from 4.2.62 on, offset 8 is the parameters version and offset 9 the CTD type (shown as `(params ver.)` and `(CTD type)`).

- **Fw02006200 D0**: MSN@8, DTD@10, MDS@12, APT@14:U1, XAS@15:U1, CPS@16:U1, CST@17:U1, CSP@18:U4, CPB1@22, CPB2@24, CPB3@26, CPN1@28, CPN2@30, CPN3@32, CPP1@34, CPP2@36, CPP3@38, MTD@40, MTT@42:U4, MAM@46:U1, MAT@47, ACA@49, ACG@51, ACR@53, PNB@55, PNC@57, PXC@59, PBL@61, PBH@63, PP0@65, PP1@67, PVL@69, PVH@71, PV0@73, PV5@75, PC0@77, PC1@79, T_DESCENT@99:U4, DNS@103, DRA@105, DSD@107, DST@109, DDF@111, DDC@113, DSI@115:U4, DDD@119, DDT@121, DFT@123, DCM@125:U1, DAL@126:U1, DCT@127, DCA@129:U1, DRC@130, DRT@132, DCX@134, DCP@136, DCF@138, DCD@140, DCR@142, DSW@144, DUC@146, DPT@148, DUT@150, DPD@152, DVT@154, PFR@156, PFC@158, DDP@160, DDS@162, T_PARK@179:U4, PNS@183, PTO@185:U4, PTH@189:U4, PSA@193:U1, PSI@194:U4, PFM@198:U1, PFT@199, PFD@201, PFV@203, PDT@205, PSM@207:U1, PSD@208, PST@210, PSV@212, PSW@214, T_ASCENT@227:U4, ASN@231, ARR@233, APM@235, APN@237, APF@239, ASI@241:U4, ASD@245, ADD@247, ADT@249, ARC@251, AHR@253, AHD@255, T_SURFACE@271:U4, SFN@275, SDC@277:I2, SGT@279, SET@281, SBM@283:U1, SBT@284, SDT@286:U4, SSA@290, SSI@292:U4, SDS@296, SAM@298:U1, SAX@299, SAN@301, SAO@303, SAP@305, SPT@307
- **Fw02006900 D0**: MSN@8, DTD@10, MDS@12, APT@14:U1, XAS@15:U1, CPS@16:U1, CST@17:U1, CTF@18, CTL@20, CTU@22, CSP@24:U4, CPB1@28, CPB2@30, CPB3@32, CPN1@34, CPN2@36, CPN3@38, CPP1@40, CPP2@42, CPP3@44, MTD@46, MTT@48:U4, MAM@52:U1, MAT@53, ACA@55, ACG@57, ACR@59, PNB@61, PNC@63, PXC@65, PBL@67, PBH@69, PP0@71, PP1@73, PVL@75, PVH@77, PV0@79, PV5@81, PC0@83, PC1@85, ICM@87:U1, IMM@88, ICA0@90, ICA1@92, ICB0@94, ICB1@96, IMT0@98, IMT1@100, ILT0@102, ILT1@104, ICN@106, ICX@108, ICT@110:I2, ICS@112, ISS0@114:U1, ISS1@115:U1, IST@116, ISO@118, ISP@120:U1, IUE0@121:U1, IUE1@122:U1, ICD0@123, ICD1@125, ICR0@127:U4, ICR1@131:U4, ICG0@135:U1, ICG1@136:U1, ICO0@137, ICO1@139, ICF@141:U1, ICW@145:U1, T_DESCENT@149:U4,

DNS@153, DRA@155, DSD@157, DST@159, DDF@161, DDC@163, DSI@165:U4, DDD@169, DDT@171, DFT@173, DCM@175:U1, DAL@176:U1, DCT@177, DCA@179:U1, DRC@180, DRT@182, DCX@184, DCP@186, DCF@188, DCD@190, DCR@192, DSW@194, DUC@196, DPT@198, DUT@200, DPD@202, DVT@204, PFR@206, PFC@208, DDP@210, DDS@212, T_PARK@214:U4, PNS@218, PTO@220:U4, PTH@224:U4, PSA@228:U1, PSI@229:U4, PFM@233:U1, PFT@234, PFD@236, PFV@238, PDT@240, PSM@242:U1, PSD@243, PST@245, PSV@247, PSW@249, T_ASCENT@251:U4, ASN@255, ARR@257, APM@259, APN@261, APF@263, ASI@265:U4, ASD@269, ADD@271, ADT@273, ARC@275, AHR@277, AHD@279, T_SURFACE@281:U4, SFN@285, SDC@287:I2, SGT@289, SET@291, SBM@293:U1, SBT@294, SDT@296:U4, SSA@300, SSI@302:U4, SDS@306, SAM@308:U1, SAX@309, SAN@311, SAO@313, SAP@315, SPT@317

- **Fw03006500 D0**: MSN@8, DTD@10, MDS@12, APT@14:U1, XAS@15:U1, CPS@16:U1, CST@17:U1, CTF@18, CTL@20, CTU@22, CSP@24:U4, CPB1@28, CPB2@30, CPB3@32, CPN1@34, CPN2@36, CPN3@38, CPP1@40, CPP2@42, CPP3@44, MTD@46, MTT@48:U4, MAM@52:U1, MAT@53, ACA@55, ACG@57, ACR@59, PNB@61, PNC@63, PXC@65, PBL@67, PBH@69, PP0@71, PP1@73, PVL@75, PVH@77, PV0@79, PV5@81, PC0@83, PC1@85, ICM@87:U1, IMM@88, ICA0@90, ICA1@92, ICB0@94, ICB1@96, IMT0@98, IMT1@100, ILT0@102, ILT1@104, ICN@106, ICX@108, ICT@110:I2, ICS@112, ISS0@114:U1, ISS1@115:U1, IST@116, ISO@118, ISP@120:U1, IUE0@121:U1, IUE1@122:U1, ICD0@123, ICD1@125, ICR0@127:U4, ICR1@131:U4, ICG0@135:U1, ICG1@136:U1, ICO0@146, ICO1@148, ICF@150:U1, ICW@154:U1, T_DESCENT@158:U4, DNS@162, DRA@164, DSD@166, DST@168, DDF@170, DDC@172, DSI@174:U4, DDD@178, DDT@180, DFT@182, DCM@184:U1, DAL@185:U1, DCT@186, DCA@188:U1, DRC@189, DRT@191, DCX@193, DCP@195, DCF@197, DCD@199, DCR@201, DSW@203, DUC@205, DPT@207, DUT@209, DPD@211, DVT@213, PFR@215, PFC@217, DDP@219, DDS@221, T_PARK@223:U4, PNS@227, PTO@229:U4, PTH@233:U4, PSA@237:U1, PSI@238:U4, PFM@242:U1, PFT@243, PFD@245, PFV@247, PDT@249, PSM@251:U1, PSD@252, PST@254, PSV@256, PSW@258, T_ASCENT@260:U4, ASN@264, ARR@266, APM@268, APN@270, APF@272, ASI@274:U4, ASD@278, ADD@280, ADT@282, ARC@284, AHR@286, AHD@288, T_SURFACE@290:U4, SFN@294, SDC@296:I2, SGT@298, SET@300, SBM@302:U1, SBT@303, SDT@305:U4, SSA@309, SSI@311:U4, SDS@315, SAM@317:U1, SAX@318, SAN@320, SAO@322, SAP@324
- **Fw03016700 D0**: MSN@8, DTD@10, MDS@12, APT@14:U1, XAS@15:U1, CPS@16:U1, CST@17:U1, CTF@18, CTL@20, CTU@22, CSP@24:U4, CPB1@28, CPB2@30, CPB3@32, CPN1@34, CPN2@36, CPN3@38, CPP1@40, CPP2@42, CPP3@44, MTD@46, MTT@48:U4, MAM@52:U1, MAT@53, ACA@55, ACG@57, ACR@59, PNB@61, PNC@63, PXC@65, PBL@67, PBH@69, PP0@71, PP1@73, PVL@75, PVH@77, PV0@79, PV5@81, PC0@83, PC1@85, ICM@87:U1, IMM@88, ICA0@90, ICA1@92, ICB0@94, ICB1@96, IMT0@98, IMT1@100, ILT0@102, ILT1@104, ICN@106, ICX@108, ICT@110:I2, ICS@112, ISS0@114:U1, ISS1@115:U1, IST@117, ISO@119, ISP@121:U1, IUE0@122:U1, IUE1@123:U1, ICD0@124, ICD1@126, ICR0@128:U4, ICR1@132:U4, ICG0@136:U1, ICG1@137:U1, ICO0@147, ICO1@149, ICF@151:U1, ICW@155:U1, T_DESCENT@159:U4, DNS@163, DRA@165, DSD@167, DST@169, DDF@171, DDC@173, DSI@175:U4, DDD@179, DDT@181, DFT@183, DCM@185:U1, DAL@186:U1, DCT@187, DCA@189:U1, DRC@190, DRT@192, DCX@194, DCP@196, DCF@198, DCD@200, DCR@202, DSW@204, DUC@206, DPT@208, DUT@210, DPD@212, DVT@214, PFR@216, PFC@218, DDP@220, DDS@222, T_PARK@224:U4, PNS@228, PTO@230:U4, PTH@234:U4, PSA@238:U1, PSI@239:U4, PFM@243:U1, PFT@244, PFD@246, PFV@248, PDT@250, PSM@252:U1, PSD@253, PST@255, PSV@257, PSW@259, T_ASCENT@261:U4, ASN@265, ARR@267, APM@269, APN@271, APF@273, ASI@275:U4, ASD@279, ADD@281, ADT@283, ARC@285, AHR@287, AHD@289, T_SURFACE@291:U4, SFN@295, SDC@297:I2, SGT@299, SET@301, SBM@303:U1, SBT@304, SDT@306:U4, SSA@310, SSI@312:U4, SDS@316, SAM@318:U1, SAX@319, SAN@321, SAO@323, SAP@325
- **Fw03026100 D0**: MSN@8, DTD@10, MDS@12, APT@14:U1, BTM@15:U1, XAS@16:U1, CPS@17:U1, CST@18:U1, CTF@19, CTL@21, CTU@23, CSP@25:U4, CPB1@29, CPB2@31, CPB3@33, CPN1@35, CPN2@37, CPN3@39, CPP1@41, CPP2@43, CPP3@45, MTD@47, MTT@49:U4, MAM@53:U1, MAT@54, ACA@56, ACG@58, ACR@60, PNB@62, PNC@64, PXC@66, PBL@68, PBH@70, PP0@72, PP1@74, PVL@76, PVH@78, PV0@80, PV5@82, PC0@84, PC1@86, ICM@88:U1, IMM@89, ICA0@91, ICA1@93, ICB0@95, ICB1@97, IMT0@99, IMT1@101, ILT0@103, ILT1@105, ICN@107, ICX@109, ICT@111:I2, ICS@113, ISS0@115:U1, ISS1@116:U1, IST@118,

ISO@120, ISP@122:U1, IUE0@123:U1, IUE1@124:U1, ICD0@125, ICD1@127, ICR0@129:U4, ICR1@133:U4, ICG0@137:U1, ICG1@138:U1, ICO0@148, ICO1@150, ICF@152:U1, ICW@156:U1, T_DESCENT@160:U4, DNS@164, DRA@166, DSD@168, DST@170, DDF@172, DDC@174, DSI@176:U4, DDD@180, DDT@182, DFT@184, DCM@186:U1, DAL@187:U1, DCT@188, DCA@190:U1, DRC@191, DRT@193, DCX@195, DCP@197, DCF@199, DCD@201, DCR@203, DSW@205, DUC@207, DPT@209, DUT@211, DPD@213, DVT@215, PFR@217, PFC@219, DDP@221, DDS@223, T_PARK@225:U4, PNS@229, PTO@231:U4, PTH@235:U4, PSA@239:U1, PSI@240:U4, PFM@244:U1, PFT@245, PFD@247, PFV@249, PDT@251, PSM@253:U1, PSD@254, PST@256, PSV@258, PSW@260, T_ASCENT@262:U4, ASN@266, ARR@268, APM@270, APN@272, APF@274, ASI@276:U4, ASD@280, ADD@282, ADT@284, ARC@286, AHR@288, AHD@290, T_SURFACE@292:U4, SFN@296, SDC@298:I2, SGT@300, SET@302, SBM@304:U1, SBT@305, SDT@307:U4, SSA@311, SSI@313:U4, SDS@317, SAM@319:U1, SAX@320, SAN@322, SAO@324

- **Fw04016100 D0**: MSN@8, DTD@10, MDS@12, APT@14:U1, BTM@15:U1, XAS@16:U1, CPS@17:U1, CST@18:U1, CTF@19, CTL@21, CTU@23, CSP@25:U4, CPB1@29, CPB2@31, CPB3@33, CPN1@35, CPN2@37, CPN3@39, CPP1@41, CPP2@43, CPP3@45, MTD@47, MTT@49:U4, MAM@53:U1, MAT@54, ACA@56, ACG@58, ACR@60, PNB@62, PNC@64, PXC@66, PBL@68, PBH@70, PP0@72, PP1@74, PVL@76, PVH@78, PV0@80, PV5@82, PC0@84, PC1@86, ICM@88:U1, IMM@89, ICA0@91, ICA1@93, ICB0@95, ICB1@97, IMT0@99, IMT1@101, ILT0@103, ILT1@105, ICN@107, ICX@109, ICT@111:I2, ICS@113, ISS0@115:U1, ISS1@116:U1, IST@118, ISO@120, ISP@122:U1, IUE0@123:U1, IUE1@124:U1, ICD0@125, ICD1@127, ICR0@129:U4, ICR1@133:U4, ICG0@137:U1, ICG1@138:U1, ICO0@148, ICO1@150, ICF@152:U1, ICW@156:U1, T_DESCENT@160:U4, DNS@164, DRA@166, DSD@168, DST@170, DDF@172, DDC@174, DSI@176:U4, DDD@180, DDT@182, DFT@184, DCM@186:U1, DAL@187:U1, DCT@188, DCA@190:U1, DRC@191, DRT@193, DCX@195, DCP@197, DCF@199, DCD@201, DCR@203, DSW@205, DUC@207, DPT@209, DUT@211, DPD@213, DVT@215, PFR@217, PFC@219, DDP@221, DDS@223, T_PARK@225:U4, PNS@229, PTO@231:U4, PTH@235:U4, PSA@239:U1, PSI@240:U4, PFM@244:U1, PFT@245, PFD@247, PFV@249, PDT@251, PSM@253:U1, PSD@254, PST@256, PSV@258, PSW@260, T_ASCENT@262:U4, ASN@266, ARR@268, APM@270, APN@272, APF@274, ASI@276:U4, ASD@280, ADD@282, ADT@284, ARC@286, AHR@288, AHD@290, T_SURFACE@292:U4, SFN@296, SDC@298:I2, SGT@300, SET@302, SBM@304:U1, SBT@305, SDT@307:U4, SSA@311, SSI@313:U4, SDS@317, SAM@319:U1, SAX@320, SAN@322, SAO@324
- **Fw04026200 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, DTD@12, MDS@14, APT@16:U1, BTM@17:U1, XAS@18:U1, CPS@19:U1, CST@20:U1, CTF@21, CTL@23, CTU@25, CSP@27:U4, CPB1@31, CPB2@33, CPB3@35, CPN1@37, CPN2@39, CPN3@41, CPP1@43, CPP2@45, CPP3@47, MTD@49, MTT@51:U4, MAM@55:U1, MAT@56, ACA@58, ACG@60, ACR@62, PNB@64, PNC@66, PXC@68, PBL@70, PBH@72, PP0@74, PP1@76, PVL@78, PVH@80, PV0@82, PV5@84, PC0@86, PC1@88, ICM@90:U1, IMM@91, ICA0@93, ICA1@95, ICB0@97, ICB1@99, IMT0@101, IMT1@103, ILT0@105, ILT1@107, ICN@109, ICX@111, ICT@113:I2, ICS@115, ISS0@117:U1, ISS1@118:U1, IST@120, ISO@122, ISP@124:U1, IUE0@125:U1, IUE1@126:U1, ICD0@127, ICD1@129, ICR0@131:U4, ICR1@135:U4, ICG0@139:U1, ICG1@140:U1, ISA@141:U1, ISD@142, IAT@144, ISV@146, ISW@148, ICO0@150, ICO1@152, ICF@154:U4, ICW@158:U4, DNS@162, DRA@164, DSD@166, DST@168, DDF@170, DDC@172, DSI@174:U4, DDD@178, DDT@180, DFT@182, DCM@184:U1, DAL@185:U1, DCT@186, DCA@188:U1, DRC@189, DRT@191, DCX@193, DCP@195, DCF@197, DCD@199, DCR@201, DSW@203, DUC@205, DPT@207, DUT@209, DPD@211, DVT@213, DVD@215, DVW@217, PFR@219, PFC@221, DDP@223, DDS@225, PNS@227, PTO@229:U4, PTH@233:U4, PSA@237:U1, PSI@238:U4, PFM@242:U1, PFT@243, PFD@245, PFV@247, PDT@249, PSM@251:U1, PSD@252, PST@254, PSV@256, PSW@258, ASN@260, ARR@262, APM@264, APN@266, APF@268, ASI@270:U4, ASD@274, ADD@276, ADT@278, ARC@280, AHR@282, AHD@284, SFN@286, SDC@288:I2, SGT@290, SET@292, SBM@294:U1, SBT@295, SDT@297:U4, SSA@301, SSI@303:U4, SDS@307, SAM@309:U1, SAX@310, SAN@312, SAO@314, SAP@316, SPT@318, SSP@320
- **Fw04036100 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, DTD@12, MDS@14, APT@16:U1, BTM@17:U1, XAS@18:U1, CPS@19:U1, CST@20:U1, CTF@21, CTL@23, CTU@25, CSP@27:U4, CPB1@31, CPB2@33, CPB3@35, CPN1@37, CPN2@39, CPN3@41, CPP1@43, CPP2@45, CPP3@47, MTD@53, MTT@55:U4, MAM@59:U1, MAT@60, ACA@62, ACG@64, ACR@66,

PNB@68, PNC@70, PXC@72, PBL@74, PBH@76, PP0@78, PP1@80, PVL@82, PVH@84, PV0@86, PV5@88, PC0@90, PC1@92, ICM@94:U1, IMM@95, ICA0@97, ICA1@99, ICB0@101, ICB1@103, IMT0@105, IMT1@107, ILT0@109, ILT1@111, ICN@113, ICX@115, ICT@117:I2, ICS@119, ISS0@121:U1, ISS1@122:U1, IST@124, ISO@126, ISP@128:U1, IUE0@129:U1, IUE1@130:U1, ICD0@131, ICD1@133, ICR0@135:U4, ICR1@139:U4, ICG0@143:U1, ICG1@144:U1, ISA@145:U1, ISD@146, IAT@148, ISV@150, ISW@152, ICO0@154, ICO1@156, ICF@158:U4, ICW@162:U4, DNS@166, DRA@168, DSD@170, DST@172, DDF@174, DDC@176, DSI@178:U4, DDD@182, DDT@184, DFT@186, DCM@188:U1, DAL@189:U1, DCT@190, DCA@192:U1, DRC@193, DRT@195, DCX@197, DCP@199, DCF@201, DCD@203, DCR@205, DSW@207, DUC@209, DPT@211, DUT@213, DPD@215, DVT@217, DVD@219, DVW@221, PFR@223, PFC@225, DDP@227, DDS@229, PNS@231, PTO@233:U4, PTH@237:U4, PSA@241:U1, PSI@242:U4, PFM@246:U1, PFT@247, PFD@249, PFV@251, PDT@253, PSM@255:U1, PSD@256, PST@258, PSV@260, PSW@262, ASN@264, ARR@266, APM@268, APN@270, APF@272, ASI@274:U4, ASD@278, ADD@280, ADT@282, ARC@284, AHR@286, AHD@288, SFN@290, SDC@292:I2, SGT@294, SET@296, SBM@298:U1, SBT@299, SDT@301:U4, SSA@305, SSI@307:U4, SDS@311, SAM@313:U1, SAX@314, SAN@316, SAO@318, SAP@320, SPT@322, SSP@324

- **Fw04046100 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, DTD@12, MDS@14, APT@16:U1, BTM@17:U1, XAS@18:U1, CPS@19:U1, CST@20:U1, CTF@21, CTL@23, CTU@25, CSP@28:U4, CPB1@32, CPB2@34, CPB3@36, CPN1@38, CPN2@40, CPN3@42, CPP1@44, CPP2@46, CPP3@48, MTD@54, MTT@56:U4, MAM@60:U1, MAT@61, ACA@63, ACG@65, ACR@67, PNB@69, PNC@71, PXC@73, PBL@75, PBH@77, PP0@79, PP1@81, PVL@83, PVH@85, PV0@87, PV5@89, PC0@91, PC1@93, ICM@95:U1, IMM@96, ICA0@98, ICA1@100, ICB0@102, ICB1@104, IMT0@106, IMT1@108, ILT0@110, ILT1@112, ICN@114, ICX@116, ICT@118:I2, ICS@120, ISS0@122:U1, ISS1@123:U1, IST@125, ISO@127, ISP@129:U1, IUE0@130:U1, IUE1@131:U1, ICD0@132, ICD1@134, ICR0@136:U4, ICR1@140:U4, ICG0@144:U1, ICG1@145:U1, ISA@146:U1, ISD@147, IAT@149, ISV@151, ISW@153, ICO0@155, ICO1@157, ICF@159:U4, ICW@163:U4, DNS@167, DRA@169, DSD@171, DST@173, DDF@175, DDC@177, DSI@179:U4, DDD@183, DDT@185, DFT@187, DCM@189:U1, DAL@190:U1, DCT@191, DCA@193:U1, DRC@194, DRT@196, DCX@198, DCP@200, DCF@202, DCD@204, DCR@206, DSW@208, DUC@210, DPT@212, DUT@214, DPD@216, DVT@218, DVD@220, DVW@222, PFR@224, PFC@226, DDP@228, DDS@230, PNS@232, PTO@234:U4, PTH@238:U4, PSA@242:U1, PSI@243:U4, PFM@247:U1, PFT@248, PFD@250, PFV@252, PDT@254, PSM@256:U1, PSD@257, PST@259, PSV@261, PSW@263, ASN@265, ARR@267, APM@269, APN@271, APF@273, ASI@275:U4, ASD@279, ADD@281, ADT@283, ARC@285, AHR@287, AHD@289, SFN@291, SDC@293:I2, SGT@295, SET@297, SBM@299:U1, SBT@300, SDT@302:U4, SSA@306, SSI@308:U4, SDS@312, SAM@314:U1, SAX@315, SAN@317, SAO@319, SAP@321, SPT@323, SSP@325
- **Fw04056300 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, DTD@12, MDS@14, APT@16:U1, BTM@17:U1, XAS@18:U1, CPS@19:U1, CST@20:U1, CTF@21, CTL@23, CTU@25, CTC@27:U1, CSP@28:U4, CPB1@32, CPB2@34, CPB3@36, CPN1@38, CPN2@40, CPN3@42, CPP1@44, CPP2@46, CPP3@48, CRD@50, CRS@52, MTD@54, MTT@56:U4, MAM@60:U1, ACA@67, ACG@69, ACR@71, PNB@73, PNC@75, PXC@77, PBL@79, PBH@81, PP0@83, PP1@85, PVL@87, PVH@89, PV0@91, PV5@93, PC0@95, PC1@97, ICM@99:U1, IMM@100, ICA0@102, ICA1@104, ICB0@106, ICB1@108, IMT0@110, IMT1@112, ILT0@114, ILT1@116, ICN@118, ICX@120, ICT@122:I2, ICS@124, ISS0@126:U1, ISS1@127:U1, IST@129, ISO@131, ISP@133:U1, IUE0@134:U1, IUE1@135:U1, ICD0@136, ICD1@138, ICR0@140:U4, ICR1@144:U4, ICG0@148:U1, ICG1@149:U1, ISA@150:U1, ISD@151, IAT@153, ISV@155, ISW@157, ICO0@159, ICO1@161, ICF@163:U4, ICW@167:U4, DNS@171, DRA@173, DSD@175, DST@177, DDF@179, DDC@181, DSI@183:U4, DDD@187, DDT@189, DFT@191, DCM@193:U1, DAL@194:U1, DCT@195, DCA@197:U1, DRC@198, DRT@200, DCX@202, DCP@204, DCF@206, DCD@208, DCR@210, DSW@212, DUC@214, DPT@216, DUT@218, DPD@220, DVT@222, DVD@224, DVW@226, PFR@228, PFC@230, DDP@232, DDS@234, PNS@236, PTO@238:U4, PTH@242:U4, PSA@250:U1, PSI@251:U4, PFM@255:U1, PFT@256, PFD@258, PFV@260, PDT@262, PSM@264:U1, PSD@265, PST@267, PSV@269, PSW@271, ASN@273, ARR@275, APM@277, APN@279, APF@281, ASI@283:U4, ASD@287, ADD@289, ADT@291, ARC@293, AHR@295, AHD@297, SFN@299, SDC@301:I2, SGT@303, SET@305, SBM@307:U1, SBT@308, SDT@310:U4, SSA@314, SSI@316:U4, SDS@320, SAM@322:U1, SAX@323, SAN@325

- **Fw04056300 D1**: SAO@3, SAP@5, SPT@7, SSP@9
- **Fw04066100 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, DTD@12, MDS@14, APT@16:U1, BTM@17:U1, XAS@18:U1, CPS@19:U1, CST@20:U1, CTF@21, CTL@23, CTU@25, CSP@28:U4, CPB1@32, CPB2@34, CPB3@36, CPN1@38, CPN2@40, CPN3@42, CPP1@44, CPP2@46, CPP3@48, MTD@54, MTT@56:U4, MAM@60:U1, ACA@67, ACG@69, ACR@71, PNB@73, PNC@75, PXC@77, PBL@79, PBH@81, PP0@83, PP1@85, PVL@87, PVH@89, PV0@91, PV5@93, PC0@95, PC1@97, ICM@99:U1, IMM@100, ICA0@102, ICA1@104, ICB0@106, ICB1@108, IMT0@110, IMT1@112, ILT0@114, ILT1@116, ICN@118, ICX@120, ICT@122:I2, ICS@124, ISS0@126:U1, ISS1@127:U1, IST@129, ISO@131, ISP@133:U1, IUE0@134:U1, IUE1@135:U1, ICD0@136, ICD1@138, ICR0@140:U4, ICR1@144:U4, ICG0@148:U1, ICG1@149:U1, ISA@150:U1, ISD@151, IAT@153, ISV@155, ISW@157, ICO0@159, ICO1@161, ICF@163:U4, ICW@167:U4, DNS@171, DRA@173, DSD@175, DST@177, DDF@179, DDC@181, DSI@183:U4, DDD@187, DDT@189, DFT@191, DCM@193:U1, DAL@194:U1, DCT@195, DCA@197:U1, DRC@198, DRT@200, DCX@202, DCP@204, DCF@206, DCD@208, DCR@210, DSW@212, DUC@214, DPT@216, DUT@218, DPD@220, DVT@222, DVD@224, DVW@226, PFR@228, PFC@230, DDP@232, DDS@234, PNS@236, PTO@238:U4, PTH@242:U4, PSA@250:U1, PSI@251:U4, PFM@255:U1, PFT@256, PFD@258, PFV@260, PDT@262, PSM@264:U1, PSD@265, PST@267, PSV@269, PSW@271, ASN@273, ARR@275, APM@277, APN@279, APF@281, ASI@283:U4, ASD@287, ADD@289, ADT@291, ARC@293, AHR@295, AHD@297, SFN@299, SDC@301:I2, SGT@303, SET@305, SBM@307:U1, SBT@308, SDT@310:U4, SSA@314, SSI@316:U4, SDS@320, SAM@322:U1, SAX@323, SAN@325
- **Fw05006100 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69, MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PNB@94, PNC@96, PXC@98, PBL@100, PBH@102, PP0@104, PP1@106, PVL@108, PVH@110, PV0@112, PV5@114, PC0@116, PC1@118, ICM@120:U1, IMM@121, ICA0@123, ICA1@125, ICB0@127, ICB1@129, IMT0@131, IMT1@133, ILT0@135, ILT1@137, ICN@139, ICX@141, ICT@143:I2, ICS@145, ISS0@147:U1, ISS1@148:U1, ISM@149:U1, IST@150, ISO@152, ISP@154:U1, IUE0@155:U1, IUE1@156:U1, ICD0@157, ICD1@159, ICR0@161:U4, ICR1@165:U4, ICG0@169:U1, ICG1@170:U1, ISA@171:U1, ISD@172, IAT@174, ISV@176, ISW@178, ICO0@180, ICO1@182, ICF@184:U4, ICW@188:U4, DNS@192, DRA@194, DSD0@196, DSD1@198, DSD2@200, DST0@202, DST1@204, DST2@206, DDF@208, DDC@210, DSI@212:U4, DDD@216, DDT@218, DFT0@220, DFT1@222, DFT2@224, DCM@226:U1, DAL@227:U1, DCT0@228, DCT1@230, DCT2@232, DCA@234:U1, DRC@235, DRT@237, DCX@239, DCP@241, DCF@243, DCD@245, DCR@247, DSW@249, DUC@251, DPT@253, DUT@255, DPD@257, DVT@259, DVD@261, DVW@263, PFR@265, PFC@267, DDP@269, DDS@271, PNS@273, PTO0@275:U4, PTO1@279:U4, PTO2@283:U4, PTH0@287:U4, PTH1@291:U4, PTH2@295:U4, PTD@299:U4, PSA@303:U1, PSI@304:U4, PFM0@308:U1, PFM1@309:U1, PFM2@310:U1, PFT0@311, PFT1@313, PFT2@315, PFD0@317, PFD1@319, PFD2@321, PFV0@323, PFV1@325
- **Fw05006100 D1**: PFV2@3, PDT@5, PSM@7:U1, PSD@8, PST@10, PSV@12, PSW@14, ASN@16, ARR@18, APM@20, APN@22, APF@24, ASI@26:U4, ASD@30, ADD@32, ADT@34, ARC@36, AHR@38, AHD@40, SFN@42, SDC@44:I2, SGT@46, SET@48, SBM@50:U1, SBT@51, SDT@53:U4, SSA@57, SSI@59:U4, SDS@63, SAM@65:U1, SAX@66, SAN@68, SAO@70, SAP@72, SPT@74, SSP@76
- **Fw05006101 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, XAS@34:U1, CPS@35:U1, CST@36:U1, CTF@37, CTL@39, CTU@41, CTC@43:U1, CSP@44:U4, CPB1@48, CPB2@50, CPB3@52, CPN1@54, CPN2@56, CPN3@58, CPP1@60, CPP2@62, CPP3@64, CRD@66, CRS@68, MTD@70, MTT@72:U4, MTS@76:U4, MAM@80:U1, MAT2@81, ACA@83, ACG@85, ACR@87, PNB@89, PNC@91, PXC@93, PBL@95, PBH@97, PP0@99, PP1@101, PVL@103, PVH@105, PV0@107, PV5@109, PC0@111, PC1@113, ICM@115:U1, IMM@116, ICA0@118, ICA1@120, ICB0@122, ICB1@124, IMT0@126,

IMT1@128, ILT0@130, ILT1@132, ICN@134, ICX@136, ICT@138:I2, ICS@140, ISS0@142:U1, ISS1@143:U1, ISM@144:U1, IST@145, ISO@147, ISP@149:U1, IUE0@150:U1, IUE1@151:U1, ICD0@152, ICD1@154, ICR0@156:U4, ICR1@160:U4, ICG0@164:U1, ICG1@165:U1, ISA@166:U1, ISD@167, IAT@169, ISV@171, ISW@173, ICO0@175, ICO1@177, ICF@179:U4, ICW@183:U4, DNS@187, DRA@189, DSD0@191, DSD1@193, DSD2@195, DST0@197, DST1@199, DST2@201, DDF@203, DDC@205, DSI@207:U4, DDD@211, DDT@213, DFT0@215, DFT1@217, DFT2@219, DCM@221:U1, DAL@222:U1, DCT0@223, DCT1@225, DCT2@227, DCA@229:U1, DRC@230, DRT@232, DCX@234, DCP@236, DCF@238, DCD@240, DCR@242, DSW@244, DUC@246, DPT@248, DUT@250, DPD@252, DVT@254, DVD@256, DVW@258, PFR@260, PFC@262, DDP@264, DDS@266, PNS@268, PTO0@270:U4, PTO1@274:U4, PTO2@278:U4, PTH0@282:U4, PTH1@286:U4, PTH2@290:U4, PTD@294:U4, PSA@298:U1, PSI@299:U4, PFM0@303:U1, PFM1@304:U1, PFM2@305:U1, PFT0@306, PFT1@308, PFT2@310, PFD0@312, PFD1@314, PFD2@316, PFV0@318, PFV1@320, PFV2@322, PDT@324, PSM@326:U1

- **Fw05006101 D1**: PSD@3, PST@5, PSV@7, PSW@9, ASN@11, ARR@13, APM@15, APN@17, APF@19, ASI@21:U4, ASD@25, ADD@27, ADT@29, ARC@31, AHR@33, AHD@35, SFN@37, SDC@39:I2, SGT@41, SET@43, SBM@45:U1, SBT@46, SDT@48:U4, SSA@52, SSI@54:U4, SDS@58, SAM@60:U1, SAX@61, SAN@63, SAO@65, SAP@67, SPT@69, SSP@71
- **Fw06000000 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69, MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PNB@94, PNC@96, PXC@98, PBL@100, PBH@102, PP0@104, PP1@106, PVL@108, PVH@110, PV0@112, PV5@114, PC0@116, PC1@118, ICM@120:U1, IMM@121, ICA0@123, ICA1@125, ICB0@127, ICB1@129, IMT0@131, IMT1@133, ILT0@135, ILT1@137, ICN@139, ICX@141, ICT@143:I2, ICS@145, ISS0@147:U1, ISS1@148:U1, ISM@149:U1, IST@150, ISO@152, ISP@154:U1, IUE0@155:U1, IUE1@156:U1, ICD0@157, ICD1@159, ICR0@161:U4, ICR1@165:U4, ICG0@169:U1, ICG1@170:U1, ISA@171:U1, ISD@172, IAT@174, ISV@176, ISW@178, ICO0@180, ICO1@182, ICF@184:U4, ICW@188:U4, DNS@192, DRA@194, DSD0@196, DSD1@198, DSD2@200, DST0@202, DST1@204, DST2@206, DDF@208, DDC@210, DSI@212:U4, DDD@216, DDT@218, DFT0@220, DFT1@222, DFT2@224, DCM@226:U1, DAL@227:U1, DCT0@228, DCT1@230, DCT2@232, DCA@234:U1, DRC@235, DRT@237, DCX@239, DCP@241, DCF@243, DCD@245, DCR@247, DSW@249, DUC@251, DPT@253, DUT@255, DPD@257, DVT@259, DVD@261, DVW@263, PFR@265, PFC@267, DDP@269, DDS@271, PNS@273, PTO0@275:U4, PTO1@279:U4, PTO2@283:U4, PTH0@287:U4, PTH1@291:U4, PTH2@295:U4, PTD@299:U4, PSA@303:U1, PSI@304:U4, PFM0@308:U1, PFM1@309:U1, PFM2@310:U1, PFT0@311, PFT1@313, PFT2@315, PFD0@317, PFD1@319, PFD2@321, PFV0@323, PFV1@325
- **Fw06000000 D1**: PFV2@3, PDT@5, PSM@7:U1, PSD@8, PST@10, PSV@12, PSW@14, ASN@16, ARR@18, APM@20, APS@22, APN@24, APF@26, ASI@28:U4, ASD@32, ADD@34, ADT@36, ARC@38, AHR@40, AHD@42, SFN@44, SDC@46:I2, SGT@48, SET@50, SBM@52:U1, SBT@53, SDT@55:U4, SSA@59, SSI@61:U4, SDS@65, SAM@67:U1, SAX@68, SAN@70, SAO@72, SAP@74, SPT@76, SSP@78
- **Fw06010300 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69, MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PNB@94, PNC@96, PXC@98, PBL@100, PBH@102, PP0@104, PP1@106, PVL@108, PVH@110, PV0@112, PV5@114, PC0@116, PC1@118, ICM@120:U1, IMM@121, ICA0@123, ICA1@125, ICB0@127, ICB1@129, IMT0@131, IMT1@133, ILT0@135, ILT1@137, ICN@139, ICX@141, ICT@143:I2, ICS@145, ISS0@147:U1, ISS1@148:U1, ISM@149:U1, IST@150, ISO@152, ISP@154:U1, IUE0@155:U1, IUE1@156:U1, ICD0@157, ICD1@159, ICR0@161:U4, ICR1@165:U4, ICG0@169:U1, ICG1@170:U1, ISA@171:U1, ISD@172, IAT@174, ISV@176,

ISW@178, ICO0@180, ICO1@182, ICF@184:U4, ICW@188:U4, DNS@192, DRA@194, DSD0@196, DSD1@198, DSD2@200, DST0@202, DST1@204, DST2@206, DDF@208, DDC@210, DSI@212:U4, DDD@216, DDT@218, DFT0@220, DFT1@222, DFT2@224, DCM@226:U1, DAL@227:U1, DCT0@228, DCT1@230, DCT2@232, DCA@234:U1, DRC@235, DRT@237, DCX@239, DCP@241, DCF@243, DCD@245, DCR@247, DSW@249, DUC@251, DPT@253, DUT@255, DPD@257, DVT@259, DVD@261, DVW@263, PFR@265, PFC@267, DDP@269, DDS@271, PNS@273, PTO0@275:U4, PTO1@279:U4, PTO2@283:U4, PTH0@287:U4, PTH1@291:U4, PTH2@295:U4, PTD@299:U4, PSA@303:U1, PSI@304:U4, PFM0@308:U1, PFM1@309:U1, PFM2@310:U1, PFT0@311, PFT1@313, PFT2@315, PFD0@317, PFD1@319, PFD2@321, PFV0@323, PFV1@325

- **Fw06010300 D1**: PFV2@3, PDT@5, PSM@7:U1, PSD@8, PST@10, PSV@12, PSW@14, ASN@16, ARR@18, APM@20, APS@22, APN@24, APF@26, ASI@28:U4, ASD@32, ADD@34, ADT@36, ARC@38, AHR@40, AHD@42, SFN@44, SDC@46:I2, SGT@48, SET@50, SBM@52:U1, SBT@53, SDT@55:U4, SSA@59, SSI@61:U4, SDS@65, SAM@67:U1, SAX@68, SAN@70, SAO@72, SAP@74, SPT@76, SSP@78
- **Fw06000100 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69, MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PNB@94, PNC@96, PXC@98, PBL@100, PBH@102, PP0@104, PP1@106, PVL@108, PVH@110, PV0@112, PV5@114, PC0@116, PC1@118, ICM@120:U1, IMM@121, ICA0@123, ICA1@125, ICB0@127, ICB1@129, IMT0@131, IMT1@133, ILT0@135, ILT1@137, ICN@139, ICX@141, ICT@143:I2, ICS@145, ISS0@147:U1, ISS1@148:U1, ISM@149:U1, IST@150, ISO@152, ISP@154:U1, IUE0@155:U1, IUE1@156:U1, ICD0@157, ICD1@159, ICR0@161:U4, ICR1@165:U4, ICG0@169:U1, ICG1@170:U1, ISA@171:U1, ISD@172, IAT@174, ISV@176, ISW@178, ICO0@180, ICO1@182, ICF@184:U4, ICW@188:U4, DNS@192, DRA@194, DSD0@196, DSD1@198, DSD2@200, DST0@202, DST1@204, DST2@206, DDF@208, DDC@210, DSI@212:U4, DDD@216, DDT@218, DFT0@220, DFT1@222, DFT2@224, DCM@226:U1, DAL@227:U1, DCT0@228, DCT1@230, DCT2@232, DCA@234:U1, DRC@235, DRT@237, DCX@239, DCP@241, DCF@243, DCD@245, DCR@247, DSW@249, DUC@251, DPT@253, DUT@255, DPD@257, DVT@259, DVD@261, DVW@263, PFR@265, PFC@267, DDP@269, DDS@271, PNS@273, PTO0@275:U4, PTO1@279:U4, PTO2@283:U4, PTH0@287:U4, PTH1@291:U4, PTH2@295:U4, PTD@299:U4, PSA@303:U1, PSI@304:U4, PFM0@308:U1, PFM1@309:U1, PFM2@310:U1, PFT0@311, PFT1@313, PFT2@315, PFD0@317, PFD1@319, PFD2@321, PFV0@323, PFV1@325
- **Fw06000100 D1**: PFV2@3, PDT@5, PSM@7:U1, PSD@8, PST@10, PSV@12, PSW@14, ASN@16, ARR@18, APM@20, APS@22, APN@24, APF@26, ASI@28:U4, ASD@32, ADD@34, ADT@36, ARC@38, AHR@40, AHD@42, SFN@44, SDC@46:I2, SGT@48, SET@50, SBM@52:U1, SBT@53, SDT@55:U4, SSA@59, SSI@61:U4, SDS@65, SAM@67:U1, SAX@68, SAN@70, SAO@72, SAP@74, SPT@76, SSP@78
- **Fw06040900 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69, MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PNB@94, PNC@96, PXC@98, PBL@100, PBH@102, PP0@104, PP1@106, PVL@108, PVH@110, PV0@112, PV5@114, PC0@116, PC1@118, ICM@120:U1, IMM@121, ICA0@123, ICA1@125, ICB0@127, ICB1@129, IMT0@131, IMT1@133, ILT0@135, ILT1@137, ICN@139, ICX@141, ICT@143:I2, ICS@145, ISS0@147:U1, ISS1@148:U1, ISM@149:U1, IST@150, ISO@152, ISP@154:U1, IUE0@155:U1, IUE1@156:U1, ICD0@157, ICD1@159, ICR0@161:U4, ICR1@165:U4, ICG0@169:U1, ICG1@170:U1, ISA@171:U1, ISD@172, IAT@174, ISV@176, ISW@178, ICO0@180, ICO1@182, ICF@184:U4, ICW@188:U4, DNS@192, DRA@194, DSD0@196, DSD1@198, DSD2@200, DST0@202, DST1@204, DST2@206, DDF@208, DDC@210, DSI@212:U4, DDD@216, DDT@218, DFT0@220, DFT1@222, DFT2@224, DCM@226:U1, DAL@227:U1,

DCT0@228, DCT1@230, DCT2@232, DCA@234:U1, DRC@235, DRT@237, DCX@239, DCP@241, DCF@243, DCD@245, DCR@247, DSW@249, DUC@251, DPT@253, DUT@255, DPD@257, DVT@259, DVD@261, DVW@263, PFR@265, PFC@267, DDP@269, DDS@271, PNS@273, PTO0@275:U4, PTO1@279:U4, PTO2@283:U4, PTH0@287:U4, PTH1@291:U4, PTH2@295:U4, PTD@299:U4, PSA@303:U1, PSI@304:U4, PFM0@308:U1, PFM1@309:U1, PFM2@310:U1, PFT0@311, PFT1@313, PFT2@315, PFD0@317, PFD1@319, PFD2@321, PFV0@323, PFV1@325

- **Fw06040900 D1**: PFV2@3, PDT@5, PSM@7:U1, PSD@8, PST@10, PSV@12, PSW@14, ASN@16, ARR@18, APM@20, APS@22, APN@24, APF@26, ASI@28:U4, ASD@32, ADD@34, ADT@36, ARC@38, AHR@40, AHD@42, SFN@44, SDC@46:I2, SGT@48, SET@50, SBM@52:U1, SBT@53, SDT@55:U4, SSA@59:U1, SSI@60, SAM@63:U1, SAX@64, SAN@66, SAO@68, SAP@70, SPT@72, SSP@74
- **Fw06050100 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69, MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PNB@94, PNC@96, PXC@98, PBL@100, PBH@102, PP0@104, PP1@106, PVL@108, PVH@110, PV0@112, PV5@114, PC0@116, PC1@118, ICM@120:U1, IMM@121, ICA0@123, ICA1@125, ICB0@127, ICB1@129, IMT0@131, IMT1@133, ILT0@135, ILT1@137, ICN@139, ICX@141, ICT@143:I2, ICS@145, ISS0@147:U1, ISS1@148:U1, ISM@149:U1, IST@150, ISO@152, ISP@154:U1, IUE0@155:U1, IUE1@156:U1, ICD0@157, ICD1@159, ICR0@161:U4, ICR1@165:U4, ICG0@169:U1, ICG1@170:U1, ISA@171:U1, ISD@172, IAT@174, ISV@176, ISW@178, ICO0@180, ICO1@182, ICF@184:U4, ICW@188:U4, DNS@192, DRA@194, DSD0@196, DSD1@198, DSD2@200, DST0@202, DST1@204, DST2@206, DDF@208, DDC@210, DSI@212:U4, DDD@216, DDT@218, DFT0@220, DFT1@222, DFT2@224, DCM@226:U1, DAL@227:U1, DCT0@228, DCT1@230, DCT2@232, DCA@234:U1, DRC@235, DRT@237, DCX@239, DCP@241, DCF@243, DCD@245, DCR@247, DSW@249, DUC@251, DPT@253, DUT@255, DPD@257, DVT@259, DVD@261, DVW@263, PFR@265, PFC@267, DDP@269, DDS@271, PNS@273, PTO0@275:U4, PTO1@279:U4, PTO2@283:U4, PTH0@287:U4, PTH1@291:U4, PTH2@295:U4, PTD@299:U4, PSA@303:U1, PSI@304:U4, PFM0@308:U1, PFM1@309:U1, PFM2@310:U1, PFT0@311, PFT1@313, PFT2@315, PFD0@317, PFD1@319, PFD2@321, PFV0@323, PFV1@325
- **Fw06050100 D1**: PFV2@3, PDT@5, PSM@7:U1, PSD@8, PST@10, PSV@12, PSW@14, ASN@16, ARR@18, APM@20, APS@22, APN@24, APF@26, ASI@28:U4, ASD@32, ADD@34, ADT@36, ARC@38, AHR@40, AHD@42, SFN@44, SDC@46:I2, SGT@48, SET@50, SBM@52:U1, SBT@53, SSA@55, SSI@57:U4, SDS@61, SAM@63:U1, SAX@64, SAN@66, SAO@68, SAP@70, SPT@72, SSP@74
- **Fw06057700 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69, MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PNB@94, PNC@96, PXC@98, PBL@100, PBH@102, PP0@104, PP1@106, PVL@108, PVH@110, PV0@112, PV5@114, PC0@116, PC1@118, ICM@120:U1, IMM@121, ICA0@123, ICA1@125, ICB0@127, ICB1@129, IMT0@131, IMT1@133, ILT0@135, ILT1@137, ICN@139, ICX@141, ICT@143:I2, ICS@145, ISS0@147:U1, ISS1@148:U1, ISM@149:U1, IST@150, ISO@152, ISP@154:U1, IUE0@155:U1, IUE1@156:U1, ICD0@157, ICD1@159, ICR0@161:U4, ICR1@165:U4, ICG0@169:U1, ICG1@170:U1, ISA@171:U1, ISD@172, IAT@174, ISV@176, ISW@178, ICO0@180, ICO1@182, ICF@184:U4, ICW@188:U4, DNS@192, DRA@194, DSD0@196, DSD1@198, DSD2@200, DST0@202, DST1@204, DST2@206, DDF@208, DDC@210, DSI@212:U4, DDD@216, DDT@218, DFT0@220, DFT1@222, DFT2@224, DCM@226:U1, DAL@227:U1, DCT0@228, DCT1@230, DCT2@232, DCA@234:U1, DRC@235, DRT@237, DCX@239, DCP@241, DCF@243, DCD@245, DCR@247, DSW@249, DUC@251, DPT@253, DUT@255, DPD@257, DVT@259, DVD@261, DVW@263, PFR@265, PFC@267, DDP@269, DDS@271, PNS@273,

PTO0@275:U4, PTO1@279:U4, PTO2@283:U4, PTH0@287:U4, PTH1@291:U4, PTH2@295:U4, PTD@299:U4, PSA@303:U1, PSI@304:U4, PFM0@308:U1, PFM1@309:U1, PFM2@310:U1, PFT0@311, PFT1@313, PFT2@315, PFD0@317, PFD1@319, PFD2@321, PFV0@323

- **Fw06057700 D1**: PFV1@3, PFV2@5, PDT@7, PSM@9:U1, PSD@10, PST@12, PSV@14, PSW@16, ASN@18, ARR@20, APM@22, APS@24, APN@26, APF@28, ASI@30:U4, ASD@34, ADD@36, ADT@38, ARC@40, AHR@42, AHD@44, SFN@46, SDC@48:I2, SGT@50, SET@52, SBM@54:U1, SBT@55, SSA@57, SSI@59:U4, SDS@63, SAM@65:U1, SAX@66, SAN@68, SAO@70, SAP@72, SPT@74, SSP@76
- **Fw06080500 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69, MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PNB@96, PNC@98, PXC@100, PBL@102, PBH@104, PP0@106, PP1@108, PVL@110, PVH@112, PV0@114, PV5@116, PC0@118, PC1@120, ICM@122:U1, IMM@123, ICA0@125, ICA1@127, ICB0@129, ICB1@131, IMT0@133, IMT1@135, ILT0@137, ILT1@139, ICN@141, ICX@143, ICT@145:I2, ICS@147, ISS0@149:U1, ISS1@150:U1, ISM@151:U1, IST@152, ISO@154, ISP@156:U1, IUE0@157:U1, IUE1@158:U1, ICD0@159, ICD1@161, ICR0@163:U4, ICR1@167:U4, ICG0@171:U1, ICG1@172:U1, ISA@173:U1, ISD@174, IAT@176, ISV@178, ISW@180, ICO0@182, ICO1@184, ICF@186:U4, ICW@190:U4, DNS@194, DRA@196, DSD0@198, DSD1@200, DSD2@202, DST0@204, DST1@206, DST2@208, DDF@210, DDC@212, DSI@214:U4, DDD@218, DDT@220, DFT0@222, DFT1@224, DFT2@226, DCM@228:U1, DAL@229:U1, DCT0@230, DCT1@232, DCT2@234, DCA@236:U1, DRC@237, DRT@239, DCX@241, DCP@243, DCF@245, DCD@247, DCR@249, DSW@251, DUC@253, DPT@255, DUT@257, DPD@259, DVT@261, DVD@263, DVW@265, PFR@267, PFC@269, DDP@271, DDS@273, PNS@275, PTO0@277:U4, PTO1@281:U4, PTO2@285:U4, PTH0@289:U4, PTH1@293:U4, PTH2@297:U4, PTD@301:U4, PSA@305:U1, PSI@306:U4, PFM0@310:U1, PFM1@311:U1, PFM2@312:U1, PFT0@313, PFT1@315, PFT2@317, PFD0@319, PFD1@321, PFD2@323, PFV0@325
- **Fw06080500 D1**: PFV1@3, PFV2@5, PDT@7, PSM@9:U1, PSD@10, PST@12, PSV@14, PSW@16, ASN@18, ARR@20, APM@22, APS@24, APN@26, APF@28, ASI@30:U4, ASD@34, ADD@36, ADT@38, ARC@40, AHR@42, AHD@44, SFN@46, SDC@48:I2, SGT@50, SET@52, SBM@54:U1, SBT@55, SSA@57, SSI@59:U4, SDS@63, SAM@65:U1, SAX@66, SAN@68, SAO@70, SAP@72, SPT@74, SSP@76
- **Fw07020200 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69, MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PNB@94, PNC@96, PXC@98, PBL@100, PBH@102, PP0@104, PP1@106, PVL@108, PVH@110, PV0@112, PV5@114, PC0@116, PC1@118, ICM@120:U1, IMM@121, ICA0@123, ICA1@125, ICB0@127, ICB1@129, IMT0@131, IMT1@133, ILT0@135, ILT1@137, ICN@139, ICX@141, ICT@143:I2, ICS@145, ISS0@147:U1, ISS1@148:U1, ISM@149:U1, IST@150, ISO@152, ISP@154:U1, IUE0@155:U1, IUE1@156:U1, ICD0@157, ICD1@159, ICR0@161:U4, ICR1@165:U4, ICG0@169:U1, ICG1@170:U1, ISA@171:U1, ISD@172, IAT@174, ISV@176, ISW@178, ICO0@180, ICO1@182, ICF@184:U4, ICW@188:U4, DNS@192, DRA@194, DSD0@196, DSD1@198, DSD2@200, DST0@202, DST1@204, DST2@206, DDF@208, DDC@210, DSI@212:U4, DDD@216, DDT@218, DFT0@220, DFT1@222, DFT2@224, DCM@226:U1, DAL@227:U1, DCT0@228, DCT1@230, DCT2@232, DCA@234:U1, DRC@235, DRT@237, DCX@239, DCP@241, DCF@243, DCD@245, DCR@247, DSW@249, DUC@251, DPT@253, DUT@255, DPD@257, DVT@259, DVD@261, DVW@263, PFR@265, PFC@267, DDP@269, DDS@271, PNS@273, PTO0@275:U4, PTO1@279:U4, PTO2@283:U4, PTH0@287:U4, PTH1@291:U4, PTH2@295:U4, PTD@299:U4, PSA@303:U1, PSI@304:U4, PFM0@308:U1, PFM1@309:U1, PFM2@310:U1, PFT0@311, PFT1@313, PFT2@315, PFD0@317, PFD1@319, PFD2@321, PFV0@323, PFV1@325

- **Fw07020200 D1**: PFV2@3, PDT@5, PSM@7:U1, PSD@8, PST@10, PSV@12, PSW@14, PSS@16:U1, ASN@17, ARR@19, APM@21, APS@23, APN@25, APF@27, ASI@29:U4, ASD@33, ADD@35, ADT@37, ARC@39, AHR@41, AHD@43, SFN@45, SDC@47:I2, SGT@49, SET@51, SBM@53:U1, SBT@54, SSA@56, SSI@58:U4, SDS@62, SAM@64:U1, SAX@65, SAN@67, SAO@69, SPT@71, SSP@73
- **Fw08000000 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69, MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PNB@94, PNC@96, PXC@98, PBL@100, PBH@102, PP0@104, PP1@106, PVL@108, PVH@110, PV0@112, PV5@114, PC0@116, PC1@118, ICM@120:U1, IMM@121, ICA0@123, ICA1@125, ICB0@127, ICB1@129, IMT0@131, IMT1@133, ILT0@135, ILT1@137, ICN@139, ICX@141, ICT@143:I2, ICS@145, ISS0@147:U1, ISS1@148:U1, ISM@149:U1, IST@150, ISO@152, ISP@154:U1, IUE0@155:U1, IUE1@156:U1, ICD0@157, ICD1@159, ICR0@161:U4, ICR1@165:U4, ICG0@169:U1, ICG1@170:U1, ISA@171:U1, ISD@172, IAT@174, ISV@176, ISW@178, ICO0@180, ICO1@182, ICF@184:U4, ICW@188:U4, DNS@192, DRA@194, DSD0@196, DSD1@198, DSD2@200, DST0@202, DST1@204, DST2@206, DDF@208, DDC@210, DSI@212:U4, DDD@216, DDT@218, DFT0@220, DFT1@222, DFT2@224, DCM@226:U1, DAL@227:U1, DCT0@228, DCT1@230, DCT2@232, DCA@234:U1, DRC@235, DRT@237, DCX@239, DCP@241, DCF@243, DCD@245, DCR@247, DSW@249, DUC@251, DPT@253, DUT@255, DPD@257, DVT@259, DVD@261, DVW@263, PFR@265, PFC@267, DDP@269, DDS@271, PNS@273, PTO0@275:U4, PTO1@279:U4, PTO2@283:U4, PTH0@287:U4, PTH1@291:U4, PTH2@295:U4, PTD@299:U4, PSA@303:U1, PSI@304:U4, PFM0@308:U1, PFM1@309:U1, PFM2@310:U1, PFT0@311, PFT1@313, PFT2@315, PFD0@317, PFD1@319, PFD2@321, PFV0@323, PFV1@325
- **Fw08000000 D1**: PFV2@3, PDT@5, PSM@7:U1, PSD@8, PST@10, PSV@12, PSW@14, PSS@16:U1, ASN@17, ARR@19, APM@21, APS@23, APN@25, APF@27, ASI@29:U4, ASD@33, ADD@35, ADT@37, ARC@39, AHR@41, AHD@43, SFN@45, SDC@47:I2, SGT@49, SET@51, SBM@53:U1, SBT@54, SSA@56, SSI@58:U4, SDS@62, SAM@64:U1, SAX@65, SAN@67, SAO@69, SPT@71, SSP@73, WMT@75
- **Fw08010000 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69, MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PNB@94, PNC@96, PXC@98, PBL@100, PBH@102, PP0@104, PP1@106, PVL@108, PVH@110, PV0@112, PV5@114, PC0@116, PC1@118, ICM@120:U1, IMM@121, ICA0@123, ICA1@125, ICB0@127, ICB1@129, IMT0@131, IMT1@133, ILT0@135, ILT1@137, ICN@139, ICX@141, ICT@143:I2, ICS@145, ISS0@147:U1, ISS1@148:U1, ISM@149:U1, IST@150, ISO@152, ISP@154:U1, IUE0@155:U1, IUE1@156:U1, ICD0@157, ICD1@159, ICR0@161:U4, ICR1@165:U4, ICG0@169:U1, ICG1@170:U1, ISA@171:U1, ISD@172, IAT@174, ISV@176, ISW@178, ICO0@180, ICO1@182, ICF@184:U4, ICW@188:U4, DNS@192, DRA@194, DSD0@196, DSD1@198, DSD2@200, DST0@202, DST1@204, DST2@206, DDF@208, DDC@210, DSI@212:U4, DDD@216, DDT@218, DFT0@220, DFT1@222, DFT2@224, DCM@226:U1, DAL@227:U1, DCT0@228, DCT1@230, DCT2@232, DCA@234:U1, DRC@235, DRT@237, DCX@239, DCP@241, DCF@243, DCD@245, DCR@247, DSW@249, DUC@251, DPT@253, DUT@255, DPD@257, DVT@259, DVD@261, DVW@263, PFR@265, PFC@267, DDP@269, DDS@271, PNS@273, PTO0@275:U4, PTO1@279:U4, PTO2@283:U4, PTH0@287:U4, PTH1@291:U4, PTH2@295:U4, PTD@299:U4, PSA@303:U1, PSI@304:U4, PFM0@308:U1, PFM1@309:U1, PFM2@310:U1, PFT0@311, PFT1@313, PFT2@315, PFD0@317, PFD1@319, PFD2@321, PFV0@323, PFV1@325
- **Fw08010000 D1**: PFV2@3, PDT@5, PSM@7:U1, PSD@8, PST@10, PSV@12, PSW@14, PSS@16:U1, ASN@17, ARR@19, APM@21, APS@23, APN@25, APF@27, ASI@29:U4, ASD@33, ADD@35, ADT@37, ARC@39, AHR@41, AHD@43, SFN@45, SDC@47:I2, SGT@49, SET@51, SBM@53:U1,

SBT@54, SSA@56, SSI@58:U4, SDS@62, SAM@64:U1, SAX@65, SAN@67, SAO@69, SPT@71, SSP@73

- **Fw0A050000 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69, MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PNB@94, PNC@96, PXC@98, PBL@100, PBH@102, PP0@104, PP1@106, PVL@108, PVH@110, PV0@112, PV5@114, PC0@116, PC1@118, ICM@120:U1, IMM@121, ICA0@123, ICA1@125, ICB0@127, ICB1@129, IMT0@131, IMT1@133, ILT0@135, ILT1@137, ICN@139, ICX@141, ICT@143:I2, ICS@145, ISS0@147:U1, ISS1@148:U1, ISM@149:U1, IST@150, ISO@152, ISP@154:U1, IUE0@155:U1, IUE1@156:U1, ICD0@157, ICD1@159, ICR0@161:U4, ICR1@165:U4, ICG0@169:U1, ICG1@170:U1, ISA@171:U1, ISD@172, IAT@174, ISV@176, ISW@178, ICO0@180, ICO1@182, ICF@184:U4, ICW@188:U4, DNS@192, DRA@194, DSD0@196, DSD1@198, DSD2@200, DST0@202, DST1@204, DST2@206, DDF@208, DDC@210, DSI@212:U4, DDD@216, DDT@218, DFT0@220, DFT1@222, DFT2@224, DCM@226:U1, DAL@227:U1, DCT0@228, DCT1@230, DCT2@232, DCA@234:U1, DRC@235, DRT@237, DCX@239, DCP@241, DCF@243, DCD@245, DCR@247, DSW@249, DUC@251, DPT@253, DUT@255, DPD@257, DVT@259, DVD@261, DVW@263, PFR@265, PFC@267, DDP@269, DDS@271, PNS@273, PTO0@275:U4, PTO1@279:U4, PTO2@283:U4, PTH0@287:U4, PTH1@291:U4, PTH2@295:U4, PTD@299:U4, PSA@303:U1, PSI@304:U4, PFM0@308:U1, PFM1@309:U1, PFM2@310:U1, PFT0@311, PFT1@313, PFT2@315, PFD0@317, PFD1@319, PFD2@321, PFV0@323, PFV1@325
- **Fw0A050000 D1**: PFV2@3, PDT@5, PSM@7:U1, PSD@8, PST@10, PSV@12, PSW@14, PSS@16:U1, ASN@17, ARR@19, APM@21, APS@23, APN@25, APF@27, ASI@29:U4, ASD@33, ADD@35, ADT@37, ARC@39, AHR@41, AHD@43, SFN@45, SDC@47:I2, SGT@49, SET@51, SBM@53:U1, SBT@54, SSA@56, SSI@58:U4, SDS@62, SAM@64:U1, SAX@65, SAN@67, SAO@69, SPT@71, SSP@73, PPC0@75:R4, PPC1@79:R4, PPC2@83:R4, PPC3@87:R4, PPE@91:U1
- **Fw0A060231 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69, MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PNB@94, PNC@96, PXC@98, PBL@100, PBH@102, PP0@104, PP1@106, PVL@108, PVH@110, PV0@112, PV5@114, PC0@116, PC1@118, ICM@120:U1, IMM@121, ICA0@123, ICA1@125, ICB0@127, ICB1@129, IMT0@131, IMT1@133, ILT0@135, ILT1@137, ICN@139, ICX@141, ICT@143:I2, ICS@145, ISS0@147:U1, ISS1@148:U1, ISM@149:U1, IST@150, ISO@152, ISP@154:U1, IUE0@155:U1, IUE1@156:U1, ICD0@157, ICD1@159, ICR0@161:U4, ICR1@165:U4, ICG0@169:U1, ICG1@170:U1, ISA@171:U1, ISD@172, IAT@174, ISV@176, ISW@178, ICO0@180, ICO1@182, ICF@184:U4, ICW@188:U4, DNS@192, DRA@194, DSD0@196, DSD1@198, DSD2@200, DST0@202, DST1@204, DST2@206, DDF@208, DDC@210, DSI@212:U4, DDD@216, DDT@218, DFT0@220, DFT1@222, DFT2@224, DCM@226:U1, DAL@227:U1, DCT0@228, DCT1@230, DCT2@232, DCA@234:U1, DRC@235, DRT@237, DCX@239, DCP@241, DCF@243, DCD@245, DCR@247, DSW@249, DUC@251, DPT@253, DUT@255, DPD@257, DVT@259, DVD@261, DVW@263, PFR@265, PFC@267, DDP@269, DDS@271, PNS@273, PTO0@275:U4, PTO1@279:U4, PTO2@283:U4, PTH0@287:U4, PTH1@291:U4, PTH2@295:U4, PTD@299:U4, PSA@303:U1, PSI@304:U4, PFM0@308:U1, PFM1@309:U1, PFM2@310:U1, PFT0@311, PFT1@313, PFT2@315, PFD0@317, PFD1@319, PFD2@321, PFV0@323, PFV1@325
- **Fw0A060231 D1**: PFV2@3, PDT@5, PSM@7:U1, PSD@8, PST@10, PSV@12, PSW@14, PSS@16:U1, ASN@17, ARR@19, APM@21, APS@23, APN@25, APF@27, ASI@29:U4, ASD@33, ADD@35, ADT@37, ARC@39, AHR@41, AHD@43, SFN@45, SDC@47:I2, SGT@49, SET@51, SBM@53:U1, SBT@54, SSA@56, SSI@58:U4, SDS@62, SAM@64:U1, SAX@65, SAN@67, SAO@69, SPT@71, SSP@73, PPC0@75:R4, PPC1@79:R4, PPC2@83:R4, PPC3@87:R4, PPE@91:U1, WMT@92
- **Fw0A060200 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13,

MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69, MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PNB@94, PNC@96, PXC@98, PBL@100, PBH@102, PP0@104, PP1@106, PVL@108, PVH@110, PV0@112, PV5@114, PC0@116, PC1@118, ICM@120:U1, IMM@121, ICA0@123, ICA1@125, ICB0@127, ICB1@129, IMT0@131, IMT1@133, ILT0@135, ILT1@137, ICN@139, ICX@141, ICT@143:I2, ICS@145, ISS0@147:U1, ISS1@148:U1, ISM@149:U1, IST@150, ISO@152, ISP@154:U1, IUE0@155:U1, IUE1@156:U1, ICD0@157, ICD1@159, ICR0@161:U4, ICR1@165:U4, ICG0@169:U1, ICG1@170:U1, ISA@171:U1, ISD@172, IAT@174, ISV@176, ISW@178, ICO0@180, ICO1@182, ICF@184:U4, ICW@188:U4, DNS@192, DRA@194, DSD0@196, DSD1@198, DSD2@200, DST0@202, DST1@204, DST2@206, DDF@208, DDC@210, DSI@212:U4, DDD@216, DDT@218, DFT0@220, DFT1@222, DFT2@224, DCM@226:U1, DAL@227:U1, DCT0@228, DCT1@230, DCT2@232, DCA@234:U1, DRC@235, DRT@237, DCX@239, DCP@241, DCF@243, DCD@245, DCR@247, DSW@249, DUC@251, DPT@253, DUT@255, DPD@257, DVT@259, DVD@261, DVW@263, PFR@265, PFC@267, DDP@269, DDS@271, PNS@273, PTO0@275:U4, PTO1@279:U4, PTO2@283:U4, PTH0@287:U4, PTH1@291:U4, PTH2@295:U4, PTD@299:U4, PSA@303:U1, PSI@304:U4, PFM0@308:U1, PFM1@309:U1, PFM2@310:U1, PFT0@311, PFT1@313, PFT2@315, PFD0@317, PFD1@319, PFD2@321, PFV0@323, PFV1@325

- **Fw0A060200 D1**: PFV2@3, PDT@5, PSM@7:U1, PSD@8, PST@10, PSV@12, PSW@14, PSS@16:U1, ASN@17, ARR@19, APM@21, APS@23, APN@25, APF@27, ASI@29:U4, ASD@33, ADD@35, ADT@37, ARC@39, AHR@41, AHD@43, SFN@45, SDC@47:I2, SGT@49, SET@51, SBM@53:U1, SBT@54, SSA@56, SSI@58:U4, SDS@62, SAM@64:U1, SAX@65, SAN@67, SAO@69, SPT@71, SSP@73, PPC0@75:R4, PPC1@79:R4, PPC2@83:R4, PPC3@87:R4, PPE@91:U1
- **Fw0A070101 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69, MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PBL@94, PBH@96, PP0@98, PP1@100, PVL@102, PVH@104, PV0@106, PV5@108, PC0@110, PC1@112, ICM@114:U1, IMM@115, ICA0@117, ICA1@119, ICB0@121, ICB1@123, IMT0@125, IMT1@127, ILT0@129, ILT1@131, ICN@133, ICX@135, ICT@137:I2, ICS@139, ISS0@141:U1, ISS1@142:U1, ISM@143:U1, IST@144, ISO@146, ISP@148:U1, IUE0@149:U1, IUE1@150:U1, ICD0@151, ICD1@153, ICR0@155:U4, ICR1@159:U4, ICG0@163:U1, ICG1@164:U1, ISA@165:U1, ISD@166, IAT@168, ISV@170, ISW@172, ICO0@174, ICO1@176, ICF@178:U4, ICW@182:U4, DNS@186, DRA@188, DSD0@190, DSD1@192, DSD2@194, DST0@196, DST1@198, DST2@200, DDF@202, DDC@204, DSI@206:U4, DDD@210, DDT@212, DFT0@214, DFT1@216, DFT2@218, DCM@220:U1, DAL@221:U1, DCT0@222, DCT1@224, DCT2@226, DCA@228:U1, DRC@229, DRT@231, DCX@233, DCP@235, DCF@237, DCD@239, DCR@241, DSW@243, DUC@245, DPT@247, DUT@249, DPD@251, DVT@253, DVD@255, DVW@257, PFR@259, PFC@261, DDP@263, DDS@265, PNS@267, PTO0@269:U4, PTO1@273:U4, PTO2@277:U4, PTH0@281:U4, PTH1@285:U4, PTH2@289:U4, PTD@293:U4, PSA@297:U1, PSI@298:U4, PFM0@302:U1, PFM1@303:U1, PFM2@304:U1, PFT0@305, PFT1@307, PFT2@309, PFD0@311, PFD1@313, PFD2@315, PFV0@317, PFV1@319, PFV2@321, PDT@323, PSM@325:U1
- **Fw0A070101 D1**: PSD@3, PST@5, PSV@7, PSW@9, PSS@11:U1, ASN@12, ARR@14, APM@16, APS@18, APN@20, APF@22, ASI@24:U4, ASD@28, ADD@30, ADT@32, ARC@34, AHR@36, AHD@38, SFN@40, SDC@42:I2, SGT@44, SET@46, SBM@48:U1, SBT@49, SSA@51, SSI@53:U4, SDS@57, SAM@59:U1, SAX@60, SAN@62, SAO@64, SPT@66, SSP@68
- **Fw0A080004 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69,

MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PBL@94, PBH@96, PP0@98, PP1@100, PVL@102, PVH@104, PV0@106, PV5@108, PC0@110, PC1@112, ICM@114:U1, IMM@115, ICA0@117, ICA1@119, ICB0@121, ICB1@123, IMT0@125, IMT1@127, ILT0@129, ILT1@131, ICN@133, ICX@135, ICT@137:I2, ICS@139, ISS0@141:U1, ISS1@142:U1, ISM@143:U1, IST@144, ISO@146, ISP@148:U1, IUE0@149:U1, IUE1@150:U1, ICD0@151, ICD1@153, ICR0@155:U4, ICR1@159:U4, ICG0@163:U1, ICG1@164:U1, ISA@165:U1, ISD@166, IAT@168, ISV@170, ISW@172, ICO0@174, ICO1@176, ICF@178:U4, ICW@182:U4, DNS@186, DRA@188, DSD0@190, DSD1@192, DSD2@194, DST0@196, DST1@198, DST2@200, DDF@202, DDC@204, DSI@206:U4, DDD@210, DDT@212, DFT0@214, DFT1@216, DFT2@218, DCM@220:U1, DAL@221:U1, DCT0@222, DCT1@224, DCT2@226, DCA@228:U1, DRC@229, DRT@231, DCX@233, DCP@235, DCF@237, DCD@239, DCR@241, DSW@243, DUC@245, DPT@247, DUT@249, DPD@251, DVT@253, DVD@255, DVW@257, PFR@259, PFC@261, DDP@263, DDS@265, PNS@267, PTO0@269:U4, PTO1@273:U4, PTO2@277:U4, PTH0@281:U4, PTH1@285:U4, PTH2@289:U4, PTD@293:U4, PSA@297:U1, PSI@298:U4, PFM0@302:U1, PFM1@303:U1, PFM2@304:U1, PFT0@305, PFT1@307, PFT2@309, PFD0@311, PFD1@313, PFD2@315, PFV0@317, PFV1@319, PFV2@321, PDT@323, PSM@325:U1

- **Fw0A080004 D1**: PSD@3, PST@5, PSV@7, PSW@9, PSS@11:U1, ASN@12, ARR@14, APM@16, APS@18, APN@20, APF@22, ASI@24:U4, ASD@28, ADD@30, ADT@32, ARC@34, AHR@36, AHD@38, SFN@40, SDC@42:I2, SGT@44, SET@46, SBM@48:U1, SBT@49, SSA@51, SSI@53:U4, SDS@57, SAM@59:U1, SAX@60, SAN@62, SAO@64, SPT@66, SSP@68, PPC0@70:R4, PPC1@74:R4, PPC2@78:R4, PPC3@82:R4, PPE@86:U1
- **Fw0A080031 D0**: (params ver.)@8:U1, (CTD type)@9:U1, MSN@10, MPS@12:U1, MPC@13, MNM0@15, MNM1@17, MNM2@19, MNS0@21:U1, MNS1@22:U1, MNS2@23:U1, DTD0@24, DTD1@26, DTD2@28, MDS@30, APT@32:U1, BTM@33:U1, DDB@34:U1, XAS@35:U1, CPS@36:U1, CST@37:U1, CTF@38, CTL@40, CTU@42, CTC@44:U1, CSP@45:U4, CPB1@49, CPB2@51, CPB3@53, CPN1@55, CPN2@57, CPN3@59, CPP1@61, CPP2@63, CPP3@65, CRD@67, CRS@69, MTD@71, MTT@73:U4, MTS@77:U4, MAM@81:U1, MAT1@82, MAT2@84, MAP@86, ACA@88, ACG@90, ACR@92, PBL@94, PBH@96, PP0@98, PP1@100, PVL@102, PVH@104, PV0@106, PV5@108, PC0@110, PC1@112, ICM@114:U1, IMM@115, ICA0@117, ICA1@119, ICB0@121, ICB1@123, IMT0@125, IMT1@127, ILT0@129, ILT1@131, ICN@133, ICX@135, ICT@137:I2, ICS@139, ISS0@141:U1, ISS1@142:U1, ISM@143:U1, IST@144, ISO@146, ISP@148:U1, IUE0@149:U1, IUE1@150:U1, ICD0@151, ICD1@153, ICR0@155:U4, ICR1@159:U4, ICG0@163:U1, ICG1@164:U1, ISA@165:U1, ISD@166, IAT@168, ISV@170, ISW@172, ICO0@174, ICO1@176, ICF@178:U4, ICW@182:U4, DNS@186, DRA@188, DSD0@190, DSD1@192, DSD2@194, DST0@196, DST1@198, DST2@200, DDF@202, DDC@204, DSI@206:U4, DDD@210, DDT@212, DFT0@214, DFT1@216, DFT2@218, DCM@220:U1, DAL@221:U1, DCT0@222, DCT1@224, DCT2@226, DCA@228:U1, DRC@229, DRT@231, DCX@233, DCP@235, DCF@237, DCD@239, DCR@241, DSW@243, DUC@245, DPT@247, DUT@249, DPD@251, DVT@253, DVD@255, DVW@257, PFR@259, PFC@261, DDP@263, DDS@265, PNS@267, PTO0@269:U4, PTO1@273:U4, PTO2@277:U4, PTH0@281:U4, PTH1@285:U4, PTH2@289:U4, PTD@293:U4, PSA@297:U1, PSI@298:U4, PFM0@302:U1, PFM1@303:U1, PFM2@304:U1, PFT0@305, PFT1@307, PFT2@309, PFD0@311, PFD1@313, PFD2@315, PFV0@317, PFV1@319, PFV2@321, PDT@323, PSM@325:U1
- **Fw0A080031 D1**: PSD@3, PST@5, PSV@7, PSW@9, PSS@11:U1, ASN@12, ARR@14, APM@16, APS@18, APN@20, APF@22, ASI@24:U4, ASD@28, ADD@30, ADT@32, ARC@34, AHR@36, AHD@38, SFN@40, SDC@42:I2, SGT@44, SET@46, SBM@48:U1, SBT@49, SSA@51, SSI@53:U4, SDS@57, SAM@59:U1, SAX@60, SAN@62, SAO@64, SPT@66, SSP@68, WMT@70, PPC0@72:R4, PPC1@76:R4, PPC2@80:R4, PPC3@84:R4, PPE@88:U1